ACE MANUFACTURING SYSTEMS LTD

INDEX - MACHINE HISTORY

AMS — In Pursuit of Excellence

MACHINE NUMBER : M11-995

SL NO	DESCRIPTION	FORMAT NUMBER	ENCLOSED
1	MACHINE DISPATCH CLEARANCE	F-QA-FA -16	✓
2	PACKING LIST / DISPATCH DETAILS	F-TSG-18 & F-PAC-01	✓
3	MINUTES OF MEETING	F-TSG-17	— N A —
4	WORK ORDER	F-TSG-15	✓
5	MACHINE SEQUENCE RUNNING REPORT	F-QA-FA-12	✓
6	SPINDLE STABILISATION REPORT	F-QA-FA-11	✓
7	MACHINE HEALTH CHECK POINTS	F-QA-FA-09	✓
8	CHECK LIST - MECHANICAL	F-QA-FA-07	✓
9	MACHINE HISTORY - MECHANICAL	F-QA-FA-10	✓
10	CHECK LIST - ELECTRICAL	F-QA-FA-08	✓
11	MACHINE HISTORY - ELECTRICAL	F-QA-FA-14	✓
12	FINAL ASSEMBLY REPORT - MECHANICAL	F-QA-FA-05/15	✓
13	GEOMETRICAL TEST REPORT	DS-DSN-PROD-TC 01~ 06	✓
14	PRACTICAL TEST REPORT	F-TSG -34	✓
15	BALL BAR REPORT	F-QA-FA-02	✓
16	AXIS CALIBRATION REPORT	F-QA-FA-01	✓
17	SPINDLE BALANCING REPORT	F-QA-FA-04	✓
18	SPINDLE ASSEMBLY	F-QA-IPR-SPL- 01	✓
19	HYDRAULIC ASSEMBLY REPORT (OPTIONAL)	F-QA-IPR-HYD-01	— N.A —
20	GEAR BOX / ATC ASSEMBLY	F-QA-IPR-GBA- 01 / SUPPLIER REPORT	Not filed
21	BALL SCREW ASSEMBLY	F-QA-IPR-BSA- 01	✓
22	BASE BUILDING	F-QA-IPR-BBA- 01/ 02	✓

Date : 8/7/14 V. Gopal CHECKED BY APPROVED BY

ACE MANUFACTURING SYSTEMS LTD

AMS
In Pursuit of Excellence

MACHINE DISPATCH CLEARANCE

M/c No.: M 11 -995	CUSTOMER : M/S. KAANRA TECHNOLOGY INC
MODEL : MCY - 400	DATE : 14/07/14

DEVIATION DETAILS

ASSEMBLY INCHARGE | QUALITY INCHARGE | AUTHORISED SIGNATURE

CORRECTIVE ACTION (BY CSG)

RESPONSIBLE PERSON : SIGNATURE OF CSG HOD

Note : Customer Support Group will confirm the closure of deviation and Return to QA.

400-NB

ACE MANUFACTURING SYSTEMS LIMITED.

DISPATCH ADVICE

AMS® In Pursuit of Excellence

Sl No.	Description Of Dispatch Details	D.A No. : DNo#014899
1	Customer Name	KAANRA TECHNOLOGY INC [KAATE001]
2	Telephone No.	044 - 24995505 / 09500045211
3	Fax No.	044 - 24997281
4	E-Mail	kaanratechnologyinc@gmail.com
5	Customer's ECC NO.	AAMFK7916MEM001
6	Customer's Sales Tax No. & Date	CST : 1129122 30/01/2013 TIN : 33940762722 30/01/2013
7	Sales Order No.	MO10004399
8	Machine Model	MCV-400
9	Machine No.	M11-995
10	Invoice Address	KAANRA TECHNOLOGY INC [KAATE001] WORKS : NO. 35, 3RD CROSS STREET, , NEHRU NAGAR, , KOTTIVAKKAM, , CHENNAI , 600041
11	Place Of Delivery	Default Address
12	Purchase Order No. & Date	KTI/PO/AMS/14-007, Date : 04-JUN-14
13	P.O Value(Excluding Taxes & Duties)	Rs. 29,25,000.000
14	Transportation	Door delivery through any reliable carriers and freight paid basis
15	Insurance	To be arranged & borne by AMS. Dispatch details to be faxed to the customer.
16	Transportation Budget	23,000.00
17	Hypothecation	
18	Special Scheme	NONE
17	Scheduled Dispatch	12/07/2014
18	Payment	Refer payment collection detail sheet
19	Special Instructions	Keep the machine ready for dispatch. Contact HOD before dispatch. Shortage list to be made while preparing machine Invoice All pending points to be attended before machine dispatch.
20	**Special Remarks**	

For ACE MANUFACTURING SYSTEMS LTD.

Authorised Signatory

ACE MANUFACTURING SYSTEMS LIMITED

PACKING LIST

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Packing List No. :	DNo#014899	Date :	12/07/2014 1
Sales Order No. :	MO10004399	Date :	04/06/2014 1
Customer :	KAANRA TECHNOLOGY INC CHENNAI		
P.O. No. :	KTI/PO/AMS/14-007 Date : 04-JUN-14		

Sl No.	Item Id.	Model	Qty.
1	M11-995	MCV-400	1

Sl No.	Description	Short Supply?
1	CNC Vertical Machining Center Model "MCV-400"	
	Spindle nose taper 7/24 No.40	
	Spindle power (Fanuc) 15 Min./Cont. Rating - 7.5 / 5.5 KW	
	AC Servo axes drive (Fanuc)	
	20 tool magazine	
	Twin Arm (Random) type High Speed Auto tool changer	
	Tool shank BT-40	
	LM Guideways for all the three axes	
	CNC System Fanuc 0IMATE-MD with 8.4" Colour LCD	
	Ball screw stretching for X, Y & Z Axes	
	Panel cooler for electrical cabinet	
	Fail safe brake for Z-Axis	
	Standard work light	
	Coolant tank and chip tray	
	Three tier indication lamp	
	Flexible coupling for X, Y & Z Axes	
	Manual pulse generator	
	Rigid tapping	
	Total machine enclosure	
	Chip disposal to the rear side of the machine through two chutes on the sides	
	One set of manuals	
	Spindle speed of 60 - 6000 rpm (Indirect drive)	
	Rapid traverse of 30 / 30 / 30 m/min. for X, Y & Z Axes	
	Tool life management	
	Air drier unit (Small)	
2	Spindle with MAS - 403 type pull stud clamping system	
3	Flood coolant system around the spindle	
4	Chip flushing system on the base	
5	Increased day light area (NO increase in Z-stroke) - 125mm (Distance from spindle face to table top will be 300 - 800 mm	
6	UCAM CNC Rotary table - Model URH 201(With Pneumatic Brake) with Fanuc motor, drives & 4th Axis Feature	
7	Wider guarding	

For ACE MANUFACTURING SYSTEMS LTD.

Authorised Signatory

8	Hydraulic interfacing (only electrical wiring) with looping assembly for 4 lines on the machine	
9	Fixture interfacing with necessary M-codes	
10	Structure base (Suitable to hold cradle palte of Maximum length 580mm & Maximum swing of Dia 400mm)	
	with 2 Nos of L Brackets & UCAM Truniion Support with 4 Lines rotary joint	
11	Maximum Height of the VMC Machine while functional must be - 2950mm and during maintenance should be suitable for customer site condition	
	i.e. to the available height of 3000mm. AMS to ensure that equipment installation during erection/installation.	
12	: ACE BLUE & LIGHT GRAY COMBINATION	
13	MAS 403 Pull Studs - 20 Nos	
14	SERVO SPIN - 12 lubricating oil (1 liter)	
15	1 set of glass fuses(inside the cabinet)	
16	Electrical documentation manual & Fanuc system Warranty Card	
17	Electrical Back up CD (Inside the cabinet pouch)	
18	Geometrical test report	
19	NAS test report	
20	Laser Calibration test report	
21	Heavy cut & Rigid tapping test report	

For ACE MANUFACTURING SYSTEMS LTD.

Authorised Signatory

ACE MANUFACTURING SYSTEMS LIMITED			
PACKING LIST ANNEXTURE-II			
Doc No:	PAN#001970	**M/c No:**	**M11-995 / MCV-400**
Date:	12/07/2014	**So No:**	MO10004399
M/s	**KAANRA TECHNOLOGY INC CHENNAI**		

Sl. No	Description	Qty	Remarks	Part No
1	O'Ring ø8.1 x 1.6	6		S15-005-00
2	O'Ring ø24.5 x 3.0	6		S15-005-17
3	O'Ring ø57.6 x 2.4	6		S15-005-18
4	O'Ring ø19.1 x 1.6	2		S15-003-33
5	O'Ring ø11.1 x 1.6	4		S15-005-01
6	O'Ring ø51.6 x 2.4	2		S15-003-70
7	Cord of O'Ring ø3 X 2 Mtrs	1		S15-015-20
8	Spanner Double Ended 10 X 11	1		S31-001-01
9	Spanner Double Ended 12 X 13	1		S31-001-02
10	Spanner Double Ended 18 X 19	1		S31-001-03
11	Spanner Double Ended 30 X 32	1		S31-001-04
12	Spanner Double Ended 36 X 41	1		S31-001-11
13	Allen Key Set	1		S31-002-101
14	Extended Allen Key (A/F 8)	1		C99-5106-8904-00
15	Steel Ball Ø8	8		S09-015-02
16	Fuse Set (Fanuc Make)	1		S22-010-11
17	Glass Fuse Set (Kept in Electrical Cabinet)	1		
18	Lubrication Oil (Spin-12 / Process Oil-12)	1		S04-005-07-CAN
19	Pull Stud BT-40 (Mas-403)	20		C74-3102-8114-01
20	FANUC Series Oi Model D / Oi Mate-Model D USER`S MANUAL	1		D00-003-00-27
21	FANUC Series Oi Model D / Oi Mate-Model D MAINTENANCE MANUAL	1		D00-003-00-28
22	Fanuc AC SERVO & SPINDLE MOTOR Bis series /alpha series MAINTENANCE MANUAL	1		D00-003-00-03
23	Operating Instructions And Maintainance Manual	1		D00-003-00-05
24	Spare Parts Manual	1		D00-003-00-04
25	Air Cooler Manual	1		D00-003-00-06
26	ATC ARM Cranking Tool	1		C99-5109-8954-00

Packed by: .....

Printed Date & Time: 12/07/2014 4:18:27 PM

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ACE MANUFACTURING SYSTEMS LIMITED			
PACKING LIST ANNEXTURE-III			
Doc No:	PAN#001964	**M/c No:**	**M11-995 / MCV-400**
Date:	12/07/2014	**So No:**	MO10004399
M/s	**KAANRA TECHNOLOGY INC CHENNAI**		

Sl. No	Description	Qty	Remarks	Part No
1	Electrical Documentation Manual	1		D00-003-00-00-000-E0
2	Maintenance Service Note (Fanuc Warranty Card)	1		D00-003-00-23
3	MACHINE BACK UP CD (Kept Machine Electrical Cabinet Pouch)	1		D00-003-00-00-000-E3

Packed by: .....

Printed Date & Time: 12/07/2014 3:12:56 PM

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ACE MANUFACTURING STSTEMS LTD

PACKING CASE IDENTIFICATION LIST

AMS

Machine Number : MLL-995 Date : 14/7/2014

Sl. No.	Case - 1 Contents	Quantity
1	Machine Assembly	01NO
2	Standard Spares & Machine Manuals	01Box
3	Servo Spin - 12 Oil	01Lifter
4	Levelling Screws & Lock Nuts	—
5	Levelling Blocks - (100) / 180 / 220 / 300	08NOS
6	Cylinder - Hydraulic / Pneumatic	—
7	Machine Guards	—
8	Side Door Handles	—
9	Power Pack	—
10	Power Pack Stand	—
11	A P C Levelling Pads	—
12	~~Machine Lifting Ropes~~ Rotary Table Manuals	01Box
13	Fixture	01NO
14	Customer Components	—
15	Rotary Table	—
16		—
17		—

	Case - 2 Contents	
1	Coolant Tank / Conveyor Assembly	01NO
2	Coolant Pumps	03NOS
3	Mesh	08NOS
4	Chip Bin	01NO
5	Chip Bin Mesh	02NOS
6	Chucker Plates	01NO
7	Oil Bin	01NO
8	Chip Trolley	—
9	Conveyor Top Guard	—
10	Bag Filters - 50 / 100 Bar	—
11	Grund Fos Pump	—
12	Base Guard	—
13		
14		
15		
16		
17		

	Case - 3 Contents	
1		
2		
3		
4		
5		
6		
7		

Checked By : Jayanna

F-PAC-03-R0

ACE MANUFACTURING SYSTEMS LTD.

MACHINE DISPATCH CHECK POINTS

AMS — In Pursuit of Excellence

CUSTOMER'S NAME: Kaamra Technologies

M/C No.: MH-995 MCV-400 DATE: 14/7/014

S.No	DESCRIPTION	YES/ NO/NA	S.No.	DESCRIPTION	YES/ NO/NA
1	Packing list available	Yes	21	Chip Trolley cleaned	NA
2	Hydraulic / Pneumatic Counter Balance Cylinder removed	NA	22	Standard Spares & m/c manual available	Yes
3	Z- Motor Cover Removed & Packed	NO	23	Coolant tank assly. packed	Yes
4	Side door handles removed & Packed	NO	24	Index Table Cover packed properly	NA
5	Base guard removed & Packed	NA	25	Oil skimmer bin cleaned	NA
6	Coolant Gun tied properly	NA	26	Voltage Stabilizer packed	NA
7	Air gun tied properly	NA	27	Machine Platform cleaned	NA
8	Leveling Screws, Nuts removed & Packed	NO	28	Wanner Pump & cover cleaned & put back	NA
9	Leveling blocks (100)/180/220/300 Ht Qty: - 4/6/7/(8)/11/20	Yes	29	Customer components packed properly	NA
10	Automatic Tool Changer	Yes	30	Index Table packed properly	NA
11	APC Assembly packed Properly	NA	31	Chiller unit cleaned	NA
12	Pallet removed & Packed	NA	32	Rotary Table packed properly	NO
13	APC leveling screws - 1set	NA	33	Toolings packed properly	NA
14	Power pack stand	NA	34	Tail Stock packed properly	NA
15	Lub Tank oil drained - retain 2 ltrs:	Yes	35	Practical Test piece packed	NA
16	Machine oil Bin cleaned	Yes	36	Fixture/s packed properly	Yes
17	Coolant tank mesh cleaned & put back	Yes	37	Air removed from cover with Vaccum cleaner	NA
18	Chip Bin cleaned & put back	Yes	38	Desiccant Bags Hung	Yes
19	Chip Bin mesh cleaned & put back	Yes	39	Fumigation Certificate available	NA
20	TPM points stickers fixed properly	NA	40		

REMARKS / SHORTAGES IF ANY:

NIL

ASSEMBLY	STORES	FIXTURE	ELECTRICAL	DISPATCH	AUTHORISED SIGNATORY
			P.S.Ramesh	Jayanna	

ACE MANUFACTURING SYSTEMS LTD.

AMS
In Pursuit of Excellence

DISPATCH CHECK POINTS

CUSTOMER'S NAME : Kaanra Technologies.

MACHINE NUMBER : MU-995

DATE : 14/7/014

S.No	CHECK POINTS	STATUS - YES / NO	REMARKS
1	PACKING CASE IS IN GOOD CONDITION.	Yes	
2	CHIP CONVEYOR IS FREE FROM CHIPS & UNWANTED MATERIALS.	Yes	
3	PAINT SHOULD NOT BE PEELED OFF IN ANY PART OF THE MACHINE.	Yes	
4	ALL POCKETS IN ATC ARE EMPTY & CLEANED	Yes	
5	NO TOOL IN SPINDLE & CLEANED	Yes	
6	NO LOOSE PARTS KEPT ANYWHERE IN THE MACHINE.	Yes	
7	ENSURE RUST PREVENTIVE OIL IS APPLIED TO ALL MACHINED SURFACE.	Yes	
8	BOLT & NUT ARE FITTED AT ALL RELAVENT PLACES ON PACKING CASE.	Yes	
9	SPARE BOX KEPT INSIDE THE PACKING CASE.	Yes	
10	GEOMETRICAL, AXIS CALIBRATION, CUTTING TEST REPORTS ARE SENT WITH THE MACHINE (in side the cabinet)	Yes	
11	ELECTRICAL CABINET CLEANED.	Yes	
12	ENSURE MACHINE IS CLEANED & NO DUST ON ANY WHERE ON M/c	Yes	
13	SUPRABHA COVER IS SEALED PROPERLY.	Yes	
14	ENSURE PACKING BOX SHOULD HAVE MINIMUM 3" GAP ALL OVER	Yes	
15	COOLANT WATER FROM ALL PIPES REMOVED BY AIR JET.	Yes	
16	NO RUST OBSERVED ANYWHERE IN THE MACHINE & RUST PREVENTIVE SPRAYED TO ALL FASTENERS & BLACKEDNED PARTS	Yes	
17	THERE IS NO TOUCHUP PAINT DEPOSITED ON MATCHING PARTS.	— NO —	
18	MACHINE LIFTING PROCEDURE CHART FIXED ON THE PACKING CASE	Yes	
19	CABINET KEY SENT WITH MACHINE TIED TO THE AC UNIT EYE BOLT	Yes	
20	PANEL COOLER FILTER IS CLEANED	Yes	
21	CHECK ITEMS AS PER PACKING LIST	Yes	
22	WOODEN PACKING PROVIDED BELOW THE APC STATION.	— NO —	
23	WOODEN Support given below the Grundfoss Pump Mounting bracket	———	this proposal can issue 23/4/2014
24.	grundfoss pump Removed and packed seperately	Yes / NA / No	
25	MACHINE LIFTING PROCEDURE SKTCH PASTED ON THE GOURD	— Yes —	

Fayana — V. Gopal — KJ

ATTENDED BY — QUALITY ASSURANCE — DISPATCH - INCHARGE

F-PAC-02-RO

ACE MANUFACTURING SYSTEMS LIMITED

WORK ORDER

AMS In Pursuit of Excellence

Customer :
M/s. KAANRA TECHNOLOGY INC
WORKS : NO. 35, 3RD CROSS STREET,, NEHRU NAGAR,, KOTTIVAKKAM,, CHENNAI, 600041, TAMILNADU

SO.No MO10004399
PO.No KTI/PO/AMS/14-007
WO.Date : 26/06/2014 **PO.Date :** 04/06/2014
S.O.TYPE : Standard

Sl No.	Item Id.	Item Description	Unit	Qty.
1	VMC	CNC VERTICAL MACHINING CENTER - MCV-400	Nos	1

MACHINE COLOUR :

Machine Colour : ACE BLUE & LIGHT GRAY COMBINATION

WARRANTY :

a) One year (i.e. 12 months) from the date of commissioning of the machine or 15 months from the date of supply of the machine whichever is earlier against faulty material and workmanship. Guarantee is not applicable for part, which has normal wear & tear (like springs, balls etc.) and certain other items like Bulbs, Lamps, Indicator Lamps, Revolving Lamps, Seals and Glass Fuses.
b) For CNC System, motors and drives, it is Two Years from the date of dispatch. This warranty is applicable only for the first installation of the machine.

Scope of Supply, Technical Specs & CNC Features As Per **Annexure I**

Machine Inspection	Geometrical accuracies and machine acceptable as per test chart and NASA test
CustomerInspection Type	DIRECT
Customer Inspection Date	04/07/2014
Component Trials Required?	Not Applicable

For ACE MANUFACTURING SYSTEMS LTD.,

PIRAMADAS
MANAGING DIRECTOR

L.S.UMESH
CHIEF EXECUTIVE OFFICER

ACE MANUFACTURING SYSTEMS LTD.

AMS® In Pursuit of Excellence

PRODUCT SPECIFICATION SHEET

MODEL: MCV-400

Sl. No.	Description	Unit	MCV-400
8	**INSTALLATION DATA - STANDARD MACHINE**		
8.1	MACHINE WIDTH	mm	1900
8.2	MACHINE DEPTH	mm	3300
8.3	MACHINE HEIGHT	mm	2750
8.4	MACHINE WEIGHT - NET	Kgf	5100
8.5	POWER SUPPLY FOR BASIC MACHINE	Kva	14
8.6	CNC CONTROL (WITH 8.4" COLOUR LCD)		FANUC 0IMATE-MD
8.7	PNEUMATIC PRESSURE REQUIRED	Kgf / Cm2	5
8.8	PNEUMATIC SUPPLY REQUIRED AT ATMOSPHERIC PR.	Lpm	300
9	**TANK CAPACITY - STANDARD MACHINE**		
9.1	COOLANT TANK	litr	300 400
9.2	LUBRICATION TANK (USEFUL QUANTITY)	litr	5 (4)
10	**MACHINE FEATURES / ACCESSORIES - STANDARD**		
10.1	PANEL COOLER FOR ELECTRICAL CABINET		
10.2	FAIL SAFE BRAKE FOR Z - AXIS		
10.3	STANDARD WORK LIGHT		
10.4	COOLANT TANK & CHIP TRAY		
10.5	THREE TIER INDICATION LAMP TO INDICATE CYCLE RUNNING READY FOR NEXT CYCLE & MACHINE IN ALARM STATUS		
10.6	FLEXIBLE COUPLING FOR X, Y & Z AXES		
10.7	MANUAL PULSE GENERATOR		
10.8	RIGID TAPPING		
10.9	TOTAL MACHINE ENCLOSURE		
10.10	CHIP DISPOSAL TO THE REAR SIDE OF THE MACHINE THROUGH TWO CHUTES ON THE SIDES		
10.11	MANUALS - ONE SET		
10.12	BALL SCREW STRETCHING FOR X, Y & Z AXES		
10.13	TOOL LIFE MANAGEMENT		
11	**STANDARD CNC FEATURES OF FANUC-0IMATE-MD**		
11.1	THREE CONTROLLED AXES		
11.2	SIMULTANEOUS CONTROLLABLE AXES - 3 AXES		
11.3	FEED RATE / RAPID OVERRIDE FACILITY		
11.4	CIRCULAR INTERPOLATION - ANY TWO OUT OF THREE AXES		
11.5	JOG MODE		
11.6	BACKLASH COMPENSATION		
11.7	PITCH ERROR COMPENSATION		
11.8	MANUAL DATA INPUT		
11.9	DRY RUN		
11.10	SINGLE BLOCK		
11.11	PROGRAM PROTECT KEY		
11.12	SELF DIAGNOSTIC FUNCTION WITH SYSTEM & MACHINE ALARM STATUS DISPLAY		
11.13	EMERGENCY STOP		
11.14	MIRROR IMAGE (X & Y CO-ORDINATES)		
11.15	TOOL OFFSET MEMORY - 400 NOS.		
11.16	TOOL LENGTH COMPENSATION		

ACE MANUFACTURING SYSTEMS LTD.

AMS® In Pursuit of Excellence

PRODUCT SPECIFICATION SHEET

MODEL: MCV-400

Sl. No.	Description	Unit	MCV-400
11.17	PART PROGRAM MEMORY 1280 MTS (512K) BATTERY BACKUP		
11.18	CUTTER DIAMETER COMPENSATION		
11.19	CANNED CYCLES FOR DRILLING, TAPPING AND BORING		
11.20	PROGRAMMABLE DATA INPUT (G10)		
11.21	HELICAL INTERPOLATION		
11.22	RIGID TAPPING		
11.23	CLOCK FUNCTION		
11.24	LOOK AHEAD CONTROL		
11.25	READER PUNCHER INTERFACE (RS 232)		
11.26	WORK CO-ORDINATE SYSTEM (G53 TO G59)		
11.27	ANGLE CHAMFERING / CORNER RADIUS		
11.28	CUSTOM MACRO-B		
11.29	CO-ORDINATE SYSTEM ROTATION		
11.30	NUMBER OF REGISTERED PROGRAMS TOTAL 200		
11.31	BACK GROUND EDITING		
11.32	EXTENDED PART PROGRAM EDITING		
11.33	RUN HOUR PARTS COUNT DISPLAY		
11.34	MANUAL PULSE GENERATOR		
11.35	ALARM HISTORY & OPERATION HISTORY		
11.36	SPINDLE LOAD DISPLAY		
12	**MACHINE FEATURES / ACCESSORIES - ADDITIONALLY ORDERED BY M/S KAANRA TECHNOLOGY INC, CHENNAI.**		
12.1	FLOOD COOLANT SYSTEM AROUND THE SPINDLE		
12.2	CHIP FLUSHING SYSTEM ON THE BASE		
12.4	INCREASED DAY LIGHT AREA (NO INCREASE IN Z STROKE) 125MM		
12.5	UCAM CNC ROTARY TABLE MODEL URH 201(WITH PNEUMATIC BRAKE) WITH FANUC MOTOR, DRIVES & 4TH AXIS FEATURE		
12.6	WIDER GUARDING		
12.7	HYDRAULIC INTERFACING (ONLY ELECTRICAL WIRING) WITH LOOPING ASSEMBLY FOR 4 LINES ON THE MACHINE		
12.8	FIXTURE INTERFACING WITH NECESSARY M-CODES		
12.9	STRUCTURE BASE (SUITABLE TO HOLD CRADLE PALTE OF MAXIMUM LENGTH 580MM & MAXIMUM SWING OF DIA 400MM) WITH 2 NOS OF L BRACKETS & UCAM TRUNIION SUPPORT WITH 4 LINES ROTARY JOINT		
12.10	MAXIMUM HEIGHT OF THE VMC MACHINE WHILE FUNCTIONAL MUST BE 2950MM AND DURING MAINTENANCE SHOULD BE SUITABLE FOR CUSTOMER SITE CONDITION I.E. TO THE AVAILABLE HEIGHT OF 3000MM. AMS TO ENSURE THAT EQUIPMENT INSTALLATION DURING ERECTION/INSTALLATION.		
12.11	MACHINE COLOUR : ACE BLUE & LIGHT GRAY COMBINATION		

V.G 07/07/14

ACE MANUFACTURING SYSTEMS LTD

SPINDLE TEMPERATURE STABILIZATION REPORT

AMS — In Pursuit of Excellence

M/c. NUMBER : M11-995	MAXIMUM RPM : 6000	DRIVE: ID/DD/IDCTS/DDCTS
SPINDLE SL No. : 14-1458	PULLEY RATIO : 1:1 / 1:1.25 / 1:1.5 /1:1.75 / 1:2	

Note : Refer TM-QA-07 for test procedure. **Recommended Temp. Rise : 20°C max. above ambient**

% of current drawn at max RPM of the spindle : (Recommended max : DD : β3 = ≤ 15%, β6 / α6 & above = ≤10%. ID : β3 = ≤ 20%, β6 / α6 & above = ≤15%.	Measured = 10 %
Spindle noise at max RPM [Recommended : 80 dB(A) upto 10 kW, 85 dB(A) for > 10 kW]	Measured = 78 dB (A)

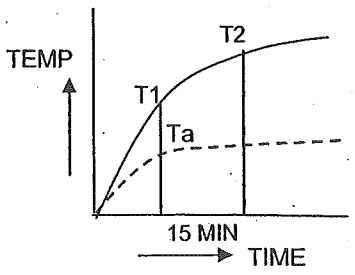
SPINDLE RPM	TIME	AMBIENT TEMPERATURE	SPINDLE TEMPERATURE
500	6 AM	24°C	—
6000	6:30	24°C	27°C
	7:00	24°C	30°C
	7:30	24°C	33°C
	8:00	25°C	35°C
	8:30	25°C	37°C
	9:00	26°C	38°C
	9:30	26°C	39°C
	10:00	26°C	40°C

1. The Stabilization point, Calculated as follows. Is the Temperature at which the rate of temperature rise over 15 minutes is less than 5%.

= [(T2 – T1) / (T1 –Ta)] X 100

= < 5%

T1 = Previous Temp., T2 = Temp. after 15 Min & Ta = Ambient Temp. at T1



2. RUN SPINDLE AFTER STABILIZATION FOR 3 HOURS.

8/7/14	V. G 2 dl
Date	Checked By

ACE MANUFACTURING SYSTEMS LTD	
MACHINE HEALTH CHECK POINTS	
MACHINE NUMBER : M11-995	DATE : 08/07/14

I	AXIS	
1	Ensure plugs are fitted for L M Guide ways.	✓
2	Ensure the lube timer is set 12 min for HG and 20 min for LM Guide ways.	✓
	In Grease Lubrication lube timer is set to 6 Hours.	N.A
3	Ensure leveling bolts are fixed with locknut.	✓
4	Check the Table tenon slot 14H8 / 18H8 (Tolerance 0 to +0.027 mm)	✓
5	Ensure butt reference function.	✓
6	Check all the axes reference arrows are matched.	✓
7	Check all Emergency push button function of Machine / APC / Double Hand	✓
8	Ensure marking of all screws with Yellow paint marker	✓
9	Ensure cover fixed to Index table, rotary joint & tool breakage sensor	N.A
10	Ensure smooth movement of sliding door without play and unhealthy noise	✓
11	Ensure effective function of Ball catch assembly on sliding doors.	✓
II	SPINDLE	
1	Ensure air coming through all air jet (4/8 holes) at oriented spindle position	✓
2	Ensure supply of air to spindle Air Purge system	N.A
3	Check the tightness of push stud.	✓
4	Ensure grub screws are fitted to spindle flange.	✓
5	Check spindle orientation position and note down the value. (Parameter No. : 4031 for Fanuc controller, ... for Siemens controller)	1283
6	Ensure spindle balancing is completed	✓
7	Ensure presence of gap between draw bar bush & ejection screw / roller (visually or by feel)	✓
8	Check the "Tool not in spindle" proximity message.	✓
III	ATC	
1	Ensure ATC / APC / Counter balance bracket pinning.	✓
2	Ensure oil level is max. in ATC gear box.	✓
3	Check the Tool Number display on panel board is same as in spindle	✓
4	Check tool change position limit switch for proper functioning & band width (Max. 2.5 mm on each side).	N.A
5	Ensure Proper working of Tool Ejector in HSK Magazine	N.A
6	Ensure ATC Auto door smooth function	N.A
7	Ensure Magazine rotates clockwise direction in manual mode	✓
8	Ensure recovery of ATC function during power failure / Emergency	✓
9	Ensure all Proximity switch & Pinch clamp mounting screws tightness to be checked and paint with Yellow marker	✓

V.Gopal	1...4
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ACE MANUFACTURING SYSTEMS LTD		
MACHINE HEALTH CHECK POINTS		
IV		COOLANT
1	✓	Ensure no coolant leakage around the machine & No presence of coolant in lubrication oil drain pipe
2	✓	Ensure proper working of float switch (by shifting the sensor upwards).
3	✓	Check function of Flood coolant (M7), Chip push (M182), Base flush (M181), CTS (M52) & Coolant gun (through selector switch).
4	NA	Ensure drum wash in conveyors (Optional)
5	NA	Check conveyor overload message & Direction (Optional)
6	NA	Check oil in conveyor gear box. (Optional)
7	NA	Ensure functioning of screw conveyor and chip conveyor in forward & reverse directions (through selector switch).
8	NA	Ensure CTS flow switch setting by opening the bag filter air vent ball valve (Optional)
9	NA	Ensure suction strainer & Check Valve in CTS tank & pr. gauge functioning (Optional)
10	NA	Ensure CTS pin locator hole is plugged (Optional)
11	NA	Ensure that twin arm plunger & CTS pin locator screw do not foul (Optional)
12	NA	Ensure bag filter clog message by closing bag filter outlet ball valve (Optional)
13	NA	Ensure High pressure air blast after CTS switched off (Optional)
V		APC
1	NA	Check pallet initialization
2	NA	Ensure supply of air jet to pallet rest pad cleaning.
3	NA	Ensure Pallet clamp seat check and arm seat check signal
4	NA	Ensure pallet change takes place smoothly with specified load (as per work order) without abnormal / ~~unhealthy noise~~
5	NA	Ensure Auto door smooth function
6	NA	Ensure Clamp line pressure relief valve
VI		HYDRAULIC, PNEUMATIC & LUBRICATION
1	✓	Ensure air pressure switch is working properly.
2	✓	Ensure proper setting of oil (1 drop / Two tool change) in FRL unit & Oil Level low message
3	✓	Check the FRL drain hose is fixed.
4	NA	Ensure identification name plates are fixed on power pack hoses.
5	NA	Check the Hyd.oil level low message (by adjusting sensor position) & Pr. gauge function
6	NA	Ensure hydraulic flow control valve grub locking in APC machine.
7	NA	Ensure no oil leakage in hydraulic valves & fittings
8	✓	Ensure lubrication pressure switch is set to 15 bar. \ 10 Bar for Grease lubrication
9	✓	Ensure pneumatic quick change coupling fixed to 4th axis provision
10	NA	Check Hydraulic specification is as per circuit diagram
VII		GENERAL
1		Check all the points of MOM are attended.
2	✓	Ensure coolant tank capacity engraved or marked in name plate
3		
4		

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MACHINE HEALTH CHECK POINTS

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MACHINE No. : M11-995 ASSEMBLY TECHNICIAN :

S.No	REMARKS OBSERVED	STATUS	SIGN
(1)	UCAM CNC rotary table mtg.	✓	
(2)	Looping hoses to be fixid.	✓	
(3)	Column rear guard to be fixid	✓	
(4)	Pending points as per internal check list to be completed.	✓	
(5)	ATC gear box, Spindle assy. Spindle balancing, Axies calibration, Ball bar practical test, Geomentrical test reports to be filed.	✓	

Date	ASSEMBLY IN CHARGE	CHECKED BY	PAGE No.
07/02/14		V. Gr 09	3...4

F-QA-FA-09-R2

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MACHINE HEALTH CHECK POINTS	In Pursuit of Excellence

MACHINE No. : ASSEMBLY TECHNICIAN :

S.No	REMARKS OBSERVED	STATUS	SIGN

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AMS — In Pursuit of Excellence

CHECK LIST - MECHANICAL

MACHINE NUMBER : M11-995 **DATE :** 07/07/14

I Vertical axis drop during power off : ——

' - ve' direction < 0.005 mm	'+ ve' <0.015mm direction
System Reading : 0.001	System Reading. : 0.007
Dial Reading. : 0.001	Dial Reading. : 0.007

II % of Axis current drawn (Recommended as per Data sheet DS -QA -02)

A	Axis	25 % Rapid Rate	50% Rapid Rate	100% Rapid Rate	600 mm/m Feed rate
	X	16%	18%	48%	14%
	Y	17%	20%	50%	15%
	Y / Z Up	54%	55%	63%	50%
	Y / Z Down	31%	29%	28%	33%
	B - Forward				
	B - Reverse				

B % of Axis holding current (Recommended as per Data sheet DS -QA -02)

Axis	100% Rapid Rate		2000 mm/min. Feed Rate	
	Motor end	Brake end	Motor end	Brake end
X	08%	10%	02%	02%
Y	14%	13%	05%	03%
Z	53%	33%	41%	42%

C Over travel against Axis dead stopper in mm (Recommended as per Drawing)

Axis	Brake end	Motor End
X	22.563	24.011
Y	16.436	13.242
Z	19.022	14.028

III **SPINDLE**

A	Push stud stroke length for Tool ejection (mm)	Recommended Values	Measured Value
	BT – 30/40 /50	Refer DS-DSN-PROD-01	0.8 mm
	HSK A50/63/100		——
B	Tool clamping force-Kgf	Refer DS-DSN-PROD-01	820 kgf

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C	Axial play in the spindle (Ref. TM-QA-10 for procedure) (Recommended max.except BT50 : 0.005 mm , for BT 50 : 0.008 mm)		
	1. Dial Reading. 'R1' (Ref. Step D)	0 ⟶ +0.032 ⟶ 0	Difference = 0.032 mm (R1~R2)
	2. Dial Reading. 'R2' (Ref. Step G)	0 ⟶ -0.024 ⟶ 0.023	
	1. Dial Reading. 'R1' (Ref. Step D)	0 ⟶ + ⟶	Difference = mm (R1~R2)
	2. Dial Reading. 'R2' (Ref. Step G)	0 ⟶ - ⟶	

IV	Turret type ATC	
A	Check the spring lever compression at :	
	1. After the magazine butts against the dead stopper (mm)	N.A
	2. Z – axis position below zero position (mm)	N.A
	3. Recommended : BT 40 - 2.5 to 3 mm , BT 30 - 1.5 to 2 mm (i.e. 1-2)	N.A
B	Tool picking position for Turret Type ATC Recommended : BT 40 - 60 to 80 mm , BT 30 - 50 to 65 mm	N.A

V	A.P.C	
A	Pressure setting for the pallet transfer cylinder in kg/cm^2 (As per circuit diagram)	N.A
B	Pressure setting for the pallet clamp-de clamp in kg/cm^2 (As per circuit diagram)	N.A

VI	Cycle Time in seconds (Refer Work order for recommended value)	
	1. Check the tool to tool time	2.00 Sec
	2. Check the chip to chip time with Z axis only	3.60 Sec
	3. Check the chip to chip time with all axes	4.40 Sec
	4. Check the pallet changing time	— N.A —

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CHECK LIST - MECHANICAL

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M/c. NUMBER : M11-995

DATE: 13/07/14

1. MACHINE LEVEL READING :

Along X-Axis

1/2 →	0	1/2 →
1/2 →	0	3/4 →
1/4 →	0	3/4 →

Along Y-Axis

1/2 ↑	1/4 ↑	1/4 ↑
1 ↑	|	3/4 ↑
1/2 ↑	3/4 ↑	1/2 ↑

2. TABLE TOP READINGS:

Along X-Axis

0	-5	-1
0	-6	0
0	-8	-5

Along Y-Axis

+4	+6	+9
+5	+7	+6
0	0	0

3. TENON SLOT READING :

-2	+4	0

PALLET - II

NA

PALLET - II

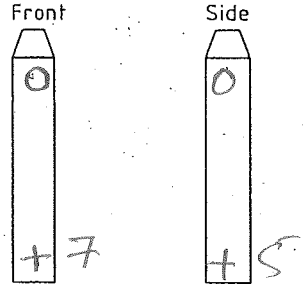
Along X-Axis

NA

Along Y-Axis

NA

4. MANDREL READINGS:



RUNOUT:

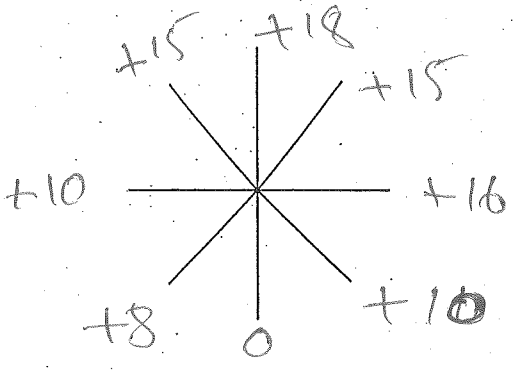
a. TOP : (0.010mm)

1. 0.001

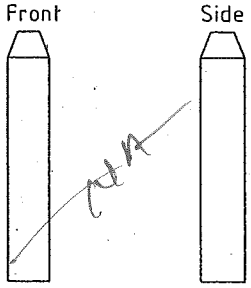
b. BOTTOM :(0.020mm)

1. 0.003
2. 0.007
3. 0.008
4. 0.004

5. TURN AROUND READING:



6. MANDREL READINGS:



RUNOUT:

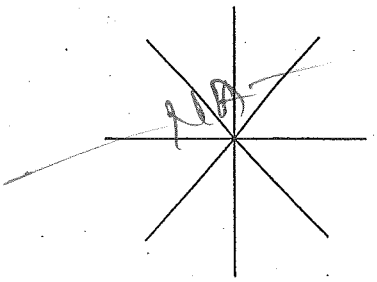
a. TOP : (0.010mm)

1. NA

b. BOTTOM :(0.020mm)

1.
2. NA
3.
4.

7. TURN AROUND READING:



NOTE: 1. For recommended value refer Design Data sheet DS-DSN-PROD-TC 01-06.

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CHECK LIST - MECHANICAL

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8.COLUMN READINGS :

FRONT

-6

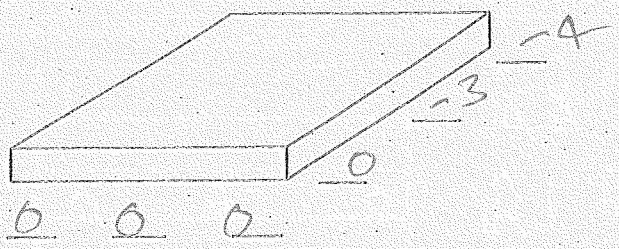
0

SIDE

+9

0

9. SQUARENESS READING (X & Y):



10. PALLET HEIGHT VARIATION:

Pallet-1: N/A

Pallet-2 : N/A

11. VARIATION IN HEIGHT OF GAUGE PLANE OF SPINDLES:
w.r.t TABLE/PALLET SURFACE (MEASURED ON MANDREL FACE)

N/A

12. PARALLELITY OF SPINDLES PLANE w.r.t X-AXIS SIDE MOVEMENT:

N/A

13.VARIATION IN CENTER DISTANCE BETWEEN SPINDLES:

N/A

10;NOTE: 1. For recommended value refer Design Data sheet DS-DSN-PROD-TC 01-06.

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MACHINE HISTORY - MECHANICAL

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	MACHINE NUMBER : M11-995	**DATE :** 07/07/14	
1	ATC Details : Model	ISO	4020V / 4024V / 4030V / 4020H / 4024H / 4030H
		HSK	A50 20V/A50 24V/A50 30V/A63 20V/A63 24V/A63 30V/A63 40V/A100 20V/A100 24V/A100 30V
1.1	ATC : Turret Type/Twin Arm	Sl.No	22187
1.2	ATC Torque Motor	TM 50 / 63 / 80 / 100 / 125 / Servo ATC	
2	**COOLANT SYSTEM DETAILS :**		
2.1	Make : Rajmane Industries	Model	Type
	Flood Coolant Pump - Crown/Flexinox/Disc Nozzle	RKM 306 - 206 / RV 160 - 270	Centrifugal Pump
2.2	Chip Pushing Coolant Pump	RKM 206 / RV 160 - 270	
2.3	Base Flushing Coolant Pump	RKM 206 / RV 160 - 270	
2.4	Coolant Gun & Drum Wash Pump N.A	RKM 206 / RV 160 - 270	
2.5	Arm Cleaning N.A		
2.6	Shower N.A		
2.7	Oil Skimmer - Make / Model N.A		
2.8	High Pressure Coolant Pump	Make	GF/ WP / BP
		Model	
2.9	Pressure Relief Valve N.A	Make	Polyhydron
		Model	DPRH-10T-50-11
2,10	Rotary Joint	Make	Deublin / RIX
		Model	
2.11	Chiller unit Details N.A	Make :	Sl. No:
3	**FLOAT SWITCH :**		
	Make	Filpro Electronics	
4	**CHIP CONVEYOR DETAILS :**		
4.1	Type of Conveyor	Scraper / Slat cum Scraper / Magnetic along X / Y / Z - axis	
4.2	Model	N.A	
4.3	Make	Universal Conveyor / SRV / Filcon	
4.4	Sl.No	N.A	
5	**CONVEYOR GEAR BOX DETAILS :**		
5.1	Model	N.A	
5.2	Make	Universal Conveyor / SRV / Rotomotive	
6	**FILTER SPECIFICATIONS :**		
6.1	Model / Working pressure	8 /15 / 50 bar / 100 bar	
6.2	Make	N.A Universal Conveyor	
6.3	Sl.No	N.A	
7	**PAPER BAND FILTER SPECIFICATIONS :**		
	Make	Sl No. N.A	
	Model		
8	**COIL / SCREW CONVEYOR DETAILS :** Qty : 1 / 2		
8.1	Make N.A	Universal Conveyor	
9	PRESSURE SWITCH FOR PNEUMATIC SYSTEM		
9.1	Operating Range	2 to 15 Kg /cm2	
9.2	Make	SUCO / SMC Digital	
9.3	Quantity	1 +1	

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MACHINE HISTORY - MECHANICAL

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10	**HYDRAULIC SYSTEM :**			
10.1	Make		Pascal / Hyd Tech / Rexorth / Yuken /Jai Chamundi /Hymech	
10.2	Product No		QC. No :	
10.3	Tank Capacity (Liters)		32 / 40 / 63 / 70 / 100	
10.4	Maximum Working Pressure		50 / 70 / 150 Bar	
10.5	HYDRAULIC PUMP	Make	Model	
		Rexroth / Yuken Corotech / Vickers		
	MOTOR	Rotomotive / Siemens	90L-4-1400 / 90S-4-1390 / 100LA-4	
11	**PRESSURE SWITCHES FOR HYDRAULIC SYSTEM :**			
11.1	Make		Polyhydron / Yuken	
11.2	Model		1PS 30-50 / 30-100 / 10 - 30 / 30-350	
11.3	Quantity			
12	**HYDRAULIC ACCUMULATOR :**			
12.1	Make	FOX / BOSCH / HYDAC	FOX / BOSCH / HYDAC	
12.2	Set Oil Pressure	Tool de clamp :	Counter balance:	
12.3	Gas Pressure	Tool de clamp :	Counter balance:	
12.4	Volume (Litres)	Tool de clamp : 1.4	Counter balance : 4 or 10	
13	**HYDRAULIC CYLINDERS :**			
	Make	Model	Function	Qty
13.1	Island Hydraulics / Preac	Ø 40 x 22 L - 125/152/180/1150	Pallet Forward / Reverse	
14	**FRL UNIT**			
	Make	Model	Operating Range	Grade of Oil
	Shavo / Festo / Norgan / Rexroth	MINI / MIDI	0-16 Kg/cm²	Servo Way Spin 12
15	DRYER	Make		Specification
		Trident / Norgan		VXD - 2
16	**TOOL DECLAMP SYSTEM :**		Mechanical / Pneumatic / Hydraulic / Hydro-Pneumatic	
17	**TURRET / INDEXING TABLE (OPTIONAL)**			
	Make	Model	Sl.No	No of Station
	Pragathi / AMS			8 /12 /
18	ROTARY TABLE	Make	Model	Sl.No
		NIKKEN / UCAM	MTO10750-82	URH-201-RP
19	TAIL STOCK / TRUNION SUPPORT	Make	Model	Sl.No
		UCAM	UFT-140-P	MTO.9176-181
20	**COUNTER BALANCING**			
	Pneumatic		Hydraulic	
20.1	Make	SMC / Rotex / Festo	Island Hydraulic	
20.2	Cylinder Diameter :	Ø 80 / 100 / 125	Ø 50 X 32 X 670	
20.3	Air Reservoir used	Yes / No		
	ADDITIONAL IF ANY			

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CHECK LIST - ELECTRICAL

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M/C NUMBER:- M11-995 | DATE:- 8/7/16

I.AXES

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the work order specification for the following		
	a)Servo motor	✓	
	b)Axis stroke	✓	
	c)Rapid rate & Feed rate	✓	
2	Check lub inch & Lubrication timers as required for the machine		
	LM RAIL: **20Min.** HARD.GUIDWAY: **12 Min.** Grease LUB: **6 Hours.**	✓	
4	Set the torque limit parameter wherever aplicable(P2060- **X / Y / Z**)	NA	
3	Check the connection of cable both at motor & drive end.	✓	

II.SPINDLE

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the connection of cable both at motor & drive end.	✓	
2	Specifications of spindle motor /speed /power is as per work order.	✓	
3	Check the spindle motor fan rotation.		
4	Check the following spindle parameters		
	a)Speed locking (P3735,3736)	✓	
	b)Power locking (P4028,4029)	✓	
	c)Rigid tapping (P4016#4,4044,4052,4085,5241,5261)	✓	
5	Check the spindle for following functions		
	a)Maximum & Minimum speed	✓	
	b)Manual Increment & decrement	✓	
	c)Manual stop, clockwise & counter clockwise.	✓	
	d)Star/Delta switching of Contactors for required speed	NA	
	(switching speed D340 =)		
	e)Spindle coasting	✓	
	f)Spindle orientation @ maximmum speed.	✓	

III.ATC

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the connection of cables.(cabinet, junction box, drive, etc.)	✓	
2	Check ATC for the following functions		
	a)Magzine motor direction	✓	
	b)Arm motor direction	✓	
	c)Tool display for tool no.1,2,4,8,16,32	✓	
	d)Tool declamp in manual mode	✓	
	e)Auto tool change	✓	
	f)Power failure conditions	✓	
3	Check for Test mode,Handheld MPG operation if **servo drive** is used.	NA	
4	Check for Test mode,parameter set / lock if **VFD** is used.	NA	

IV. APC

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the connection of cables.(cabinet, junction box, etc.)		
2	Check for pallet selection program(**O100**).		
4	Check APC for the following functions		
	a)MOP operations		
	b)Pallet set up sequence		
	c)Auto pallet change(M80) & related interlocks.	NA	
	d)Power failure conditions		
	e)Fixture interlock		

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V.PNUEMATIC

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the functions of following elements		
	a)Solenoids	✓	
	b)Pressure switches	✓	
	c)FRL unit & auto drain	NA	
2	Check for solenoid nut,screws & prsw caps.	✓	

VI.HYDRAULIC

SL.No	PARAMETERS	STATUS	REMARKS
1	Check for Hydraulic Motor / cooling fan direction.		
2	Check for cables/ hose / plugs termination & routing.	✓	
2	Check for solenoid/pressure sw mounting screws & gasket.	✓	
3	Check the functions of following elements	✓	
	a)Solenoids	✓	
	b)Pressure switches	✓	
	c)Power pack Float , Thermal , clog switch.	✓	

VIII.FIXTURE

SL.No	PARAMETERS	STATUS	REMARKS
1	Check for manual operation through push buttons.	NA	
2	Check for auto operation through M codes.	✓	
3	Check for timers,bypass bits, seat check,other related interlocks.	✓	

VII.COOLANT

SL.No	PARAMETERS	STATUS	REMARKS
1	Check for direction of all coolant / conveyor motors.	✓	
2	Check for cables/ hose / plugs termination & routing.	✓	
3	Check the functions of following elements		
	a)Manual mode operations.(SKB push buttos)	✓	
	b)MDI / Auto mode operations	✓	
	c)Float switch	✓	
	b)Chip/ Screw Conveyor operation(lsw, proximity,,fwd/ rev etc)	NA	
	c)HP coolant functions(clog,flow,air blast sol,relief valve etc.)	NA	
	d)Push/Flush sequence through M180 command.	✓	
	d) Coolant gun on/off through selector sw.	✓	

VIII.ROTARY/ INDEX TABLES

SL.No	PARAMETERS	STATUS	REMARKS
1	Cable/ hoses are routed properly at safe position.	✓	
2	Check for commanded position , table position & shortest path.	✓	
3	Check for tool display & table position in case of index table.	✓	

IX.CABINET & OPERATING PANEL

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the rating & spec.of following elements as per circuit diagram.		
	a)MPCB's	✓	
	b)Contactors	✓	
	c)MCB's	✓	
	d)Transformer	✓	
	e)PCB's,Relay's & relay boards	✓	
2	Set the MPCB's for 100% of rated current(refer IOM for torque motor)	✓	
3	Check the panel cooler with cab.door interlock.set for 35°C& lock par.	✓	
4	Ensure the rotation of drive heat sink fan & also other fans if any.	✓	

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CHECK LIST - ELECTRICAL

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5	Check for the following		
	a)Mounting of elements inside cabinet	✓	
	b)Mounting of elements on operating panel	✓	
	c)All Terminal tightness/ loose connections	✓	
	d)Input/ output short connection	✓	
	e)Ferruling of cable/wire(with adress if necessary)	✓	
	f)Identification stickers	✓	
	g)Moisture free beadings(both in cabinet & operating panel)	✓	

X.GENERAL

SL.No	PARAMETERS	STATUS	REMARKS
1	Check the functions of following		
	a)All emergency push buttons	✓	
	b)Emergency monitor	✓	
	c)Work lamp & m/c indication lamp(3 tier)	✓	
	d)Auto door & m/c door limit switches if provided	NA	
2	Check for - All earthing connections.	✓	(X)
3	Check for -Protective covers.	✓	
4	Check for -PVC/Metal Channel cover fixing.	✓	(X)
5	Check for -Name plate & tags on all elements.	✓	
6	Check for -Routing of cables & hoses.	✓	
7	Check for- Model/Sl.No./ rating in M/C No. plate	✓	

INTERNAL CHECKLIST POINTS

- ~~Product top plate~~
- → 16-Pin Hyd. dummy conn to be put.
- → Cabinet Door alignment & lock lever alignment
- ~~→~~ New macro title
- → ~~CSK~~ New circuit wiring
- →

XI. PMC DATA

Instruction: Please note all D-bits which is enabled at the time of shipping .
Also give the details if it is used for special function

D0	7	6	5	(4)	3	2	(1)	(0)	D16	7	6	5	4	3	2	1	0
D1	7	6	5	4	3	2	1	0	D17	7	6	5	4	3	2	1	0
D2	7	6	5	4	3	2	1	0	D18	7	6	5	4	3	2	1	0
D3	7	(6)	5	4	3	2	1	0	D19	7	6	5	4	3	2	1	(0)
D4	7	6	(5)	4	3	2	1	0	D20	7	6	5	4	3	2	1	0
D5	7	6	5	4	3	2	1	0	D21	7	6	5	4	3	2	1	0
D6	(7)	6	5	4	3	2	1	0	D22	7	6	5	4	3	2	1	0
D7	7	6	5	4	(3)	2	1	0	D23	7	6	5	4	3	2	1	0
D8	7	6	5	4	3	(2)	1	(0)	D24	7	6	5	4	(3)	(2)	1	0
D9	7	6	5	4	3	2	1	(0)	D25	7	6	5	4	3	2	1	0
D10	7	6	5	4	3	2	1	0	D26	7	6	5	4	3	2	1	0
D11	7	6	5	4	3	2	1	0	D27	7	6	5	4	3	2	1	0
D12	7	6	5	4	3	2	1	0	D28	7	6	5	4	3	2	1	0
D13	7	6	5	4	3	2	1	0	D29	7	6	5	4	3	2	1	0
D14	7	6	5	4	3	2	1	0	D30	7	6	5	4	3	2	1	0
D15	7	6	5	4	3	2	1	0									

CHECKED BY:- K.V.

VERIFIED BY:-

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CHECK LIST - ELECTRICAL

D110	00000000	1	1
D000	00000000	3	30

XII.PROGRAM/ LOGIC VERSION

Instruction: Please ckeck for latest version before making entry.

Logic version No.:	Remarks:
TMA01-B2-11-M00	
Macro version No.:	**Remarks:**
VER-2.11.09. OCTOBER 2010	

Progrm version Nos.:

O9001	A2.2.03	O9010			
O9002	B2.2.02	O9011			
O9003					

XIII.CNC PARAMETERS

REFERENCE POSITION	X	Y	Z
ATC 2nd Ref. 1241	-600	0	-4.73
APC 3rd Ref. 1242	NA		
APC 4th Ref. 1243	NA		

RAPID TIME CONSTANT	X	Y	Z
1620	96	112	120
1621	88	96	96

Check the following with system:		'1'	'0'	
No.of Ladder steps**	Dgn:1141 # 6	8000 ✓	5000	**N/A for 0 iMD
4th axis option**	1279 # 4 OR 1148 # 7	Yes ✓	No	**N/A for 0 iMD
Macro executer option	Dgn:1236 # 1	Yes ✓	No	
MPG option for PMM	Dgn:1292 # 6	Yes	No ✓	
AICC1 option**	Dgn:1273 # 0	Yes	No ✓	
AICC2 option**	Dgn:1148 # 0	Yes	No ✓	
3111#7 = 0 (NPA) - Message display	✓			

XIV.SHIPPING

Instruction: Please perform following activities before shipping .
Also give the details if any special activity is done

Sl.No	PARAMETERS	REMARKS
1	Ensure the servo guide back up	
2	Check the various communication like RS232C,PCMCIA	✓
3	Edit title,& set time, date other original factory set values	
4	Take all programs back up &Lock 8000 & 9000 series programs (P3202)	✓
5	Take the back up of following :	
	a) CNC parameters	
	b) Pitch error compensation	
	c) PMC parameters	
	d) ID info	
	e) M-info	
	f) Ladder	
	g) Macro	
	h) S-RAM	
6	Make I/O channel '0' for RS232C communication	
7	Keep spare fuse inside the cabinet	
8	Ensure any updation/correction in electrical circuit diagrm	
9	Ensure all the elements are in good condition before shipping	
10	Compare CNC parametrers with standard data & store in backup file	
11	Ensure password protection to edit ladder diagram.	

CHECKED BY:- JOY K.V.

VERIFIED BY:

ACE MANUFACTURING SYSTEMS LTD

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CABINET ENERGIZING REPORT

Route card No.: H59	Date: 13/06/14
Sales order No.: 4399	M/C No.: M11-995

WIRING OPTIONS

DESCRIPTION		CHECKPOINTS
STANDARD MACHINE	☑	STD CLNT,
CHIP CONVEYORS	☐	SCRAPER TYPE,SLAT TYPE , SCREW/COIL TYPE & HC CONNECTORS,
APC	☐	I/O CARD, RELAY BOARD, EI/O3CABLE,TB1 TERMINATION,
ROTARY TABLE	☐	SYSTEM OPTION, DRIVE INTERFACE, MS CONNECTORS,
INDEX TABLE	☐	MPCB, CONTACTOR,SSR, ACR3,
FIXTURE	☐	I/O CARD, RELAYBOARD, EI/O2 CABLE,TB2 TERMINATION,
SPECIAL COOLANT	☐	
ADD I/O CARDS 1 / 2 / 3	☐	
	☐	
	☐	

SHORT CIRCUIT TEST

Test to be done with the help of continuity tester/ multimeter/ megger

1)BETWEEN PHASE & EARTH -- ok	2)BETWEEN PHASE & NEUTRAL -- ok
3)BETWEEN TWO PHASES -- ok	4)BETWEEN 24V & EARTH -- ok

VOLTAGE TEST

Max. allowed tolerance For 415V ac -±15V, For 200 - 230V ac - ±10V, For 24V dc- ±1V,

1) INCOMING MAINS: 395.9V	2) NEUTRAL TO EARTH: 2.09V
3) TFR 1 SECONDARY(220 V / 3 Ph): 210.6V	4) TFR 1 SECONDARY(220 V / 1 Ph): 214.3V
5) TFR 1 SECONDARY(230 V / 1 Ph): 224.3V	
5)REG.POWER SUPPLY(PS1)O/P : 24.30VDC	5)REG.POWER SUPPLY(PS2)O/P : 24.31VDC

AXIS CURRENT TEST

Check with MPG feed, Check for noise& current less than 10%

1)X -AXIS - 3.1	2)Y -AXIS - 3.1
3)Z -AXIS - 4.1	

SYSTEM LOADING DATA

Please refer data sheet for running version Nos.

1)LOGIC VER No.: TMA01-02-07-M00	2)MACRO VER No.: 2.T1.01
3)O9001 VER No.: A2.2.03	4)O9010 / O9013 VER No.:
5)O9003 VER No.: C1.A.02	6)O9011 / O9012 VER No.:
7)O9002 VER No.: B2.2.02	8)O8020 VER No.:

ASHOK

ACE MANUFACTURING SYSTEMS LTD

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CABINET ENERGIZING REPORT

CHECKPOINTS

SL No.	Description	status
1	Cabinet colour - As per work order	yes
2	Panel layout - As per layout diagram	yes
3	Ferruling- As per circuit diagram	yes
4	Identification stickers /name plates/ tags - As per circuit diagram	yes
5	Lug Crimping - Physical check	yes
6	Terminal tightness - Physical check for loose connections	yes
7	Rating of main transformer& related contactor,mcb - As per DATA SHEET	yes
8	Rating of other control mcb's - As per circuit diagram / DATASHHET	yes
9	Rating of mpcb's& contactors - As per circuit diagram	yes
10	Cabinet door setting- check for lsw setting, Beading, locking,allignment	yes
11	Health check of elements- For physical damage, enrgization, mounting/ fixing	yes
12	Cabinet mounting - Check for Screw tightness, rubber beading if required	yes

ADDITIONAL CONTENTS WITH CABINET

1)MACHINE HISTORY	☑	2)PRE WORK ORDER	☑
3)M/C No. PLATE	☐	4)PVC CHANNEL COVER	☑
5)PENDENT BEADINGS	☑	6)SPLIT PLATE	☑
7)SPINDLE POWER,FAN,F/B CABLE	☑	8)	☐
	☐		☐

SHORTAGE/ PENDING WORK DECLARATION

No.	DESCRIPTION	ACTION PLAN
1	S/a option bit Required	✓
2		
3		
4		
5		
6		
7		

CHECKED BY:- Ashwa

VERIFIED BY:-

53734

得意先名:
CUSTOMER: FANUC INDIA PRIVATE LIMITED
CONTRACT NO.: YD06819- 000 DATE OF MANUFACTURE:2013-12
MODEL NO.: A02B-0321-B500 DATE OF TEST :2013/12/25
SERIAL NO.: E13ZH1623 CHECKED BY :A. Watanabe
SETTING OF INPUT: APPROVED BY : K.Miura

CNC IDENTIFICATION NUMBER

3C7B5D01
2391A24F
5E34C389
7892C491

EDITION OF PRINTED BOARD & ROM

NAME	EDIT	NAME	EDIT	NAME	EDIT
A20B-8201-0088	03A	A02B-0319-H500#60W3	05		
		A02B-0319-H580#40B0	11		
		A02B-0319-J541#60VM	07		
		A02B-0319-J571#658E	07		
		A02B-0319-J572#658F	06		
		A02B-0319-J575#659B	03		
		A02B-0319-J579#BW50	03		
		A02B-0320-J578#DX50	03		
		A02B-0320-J578#DX51	03		
		A02B-0322-H501#D5F1	32		
		A06B-6057-H507#90C5	06		
		A06B-6057-H507#90C8	06		

LIST OF CARD , DIMM & BACK BOARD

A20B-3300-0768	1		
A20B-3900-0286	1		
A20B-8002-0703	1		
A20B-8200-0560	1		

本データシートはファナック工場出荷時のCNC装置の装置情報です
Information contained in this Data Sheet is system configuration at the time of FANUC factory shipment.

NAME	NAME	SPECIFICATION	ORDR
BASIC CONTROL UNIT	ﾍﾞｰｼｯｸ ﾕﾆｯﾄ	A02B-0321-B500	1
AXIS CONTROL CARD A2 FROM/SRAM MEMORY A1 POWER SUPPLAY UNIT MAIN BOARD DISPLAY UNIT NAME PLATE	ｼﾞｸ ｾｲｷﾞｮ ｶｰﾄﾞA2 FROM/SRAMﾒﾓﾘ A1 ﾃﾞﾝｹﾞﾝ ﾕﾆｯﾄ ﾒｲﾝﾎﾞｰﾄﾞ ﾋｮｳｼﾞ ﾕﾆｯﾄ ﾒｲﾊﾞﾝ	A02B-0319-H032 A02B-0319-H051 A02B-0319-H111 A02B-0319-H125 A02B-0319-H164#M A02B-0322-H190 A02B-0322-Rw21K	1 1 1 1 1 1
PACKAGE 3 CUSTOM SOFTWARE SIZE 512K BYTE MACRO EXECUTOR CONTROLLABLE AXES EXPANSION	ﾊﾟｯｹｰｼﾞ 3 ｶｽﾀﾑｿﾌﾄ ﾖｳﾘｮｳ 512Kbyte ﾏｸﾛ ｴｸﾞｾﾞｷｭｰﾀ ｾｲｷﾞｮｼﾞｸｽｳ ｶｸﾁｮｳ	A02B-0322-H532 A02B-0322-J738#512K A02B-0322-J888 A02B-0322-R689	1 1 1 1
I/O UNIT FOR 0i *MANUAL PULSE GENERATOR	I/O UNIT *ｼｭﾄﾞｳ ﾊﾟﾙｽ ﾊｯｾｲ	A02B-0309-C001 A860-0203-T001	1 1
DIGITAL SERVO UNIT DISCHARGE RESISTOR AC LINE FILTER	ﾃﾞｼﾞﾀﾙｻｰﾎﾞﾕﾆｯﾄ ｶｲｾｲﾃｲｺｳﾕﾆｯﾄ AC ﾗｲﾝ ﾌｨﾈﾀｰ	A06B-6130-H002 A06B-6130-H402 A81L-0001-0083#3C	1 1 1
	SPINDLE AMP.	A06B-6164-H343#H580	1
	BIS4/4000 BIS12/3000 BIS12/3000 BIS22B/3000 SPB6/10000I	A06B-0063-B003 A06B-0078-B103 A06B-0078-B103#0100 A06B-0082-B403 A06B-1445-B103	1 1 1 1 1

الطبعة الأولى ١٤٢٤هـ - ٢٠٠٣م

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

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M/C NO.: m11-995	DATE: 13/06/14
MODEL: MCV-400.	CUSTOMER: KAANRA TECHNOLOGY PVT.

M/C BACKUP (PATH) : \\arya3\E-Backup-II\ELECTRICAL MACHINE DATA\AMS DISPATCH MACHINE BACKUPS (20014-15)\FANUC

CNC SYSTEM

0i-MD ☐ 0i-MATE-MD ☑ ☐	PART No A02B-0321-B500
	Sl.No.: ~~E13X400102~~ E13ZH1623

SYSTEM OPTIONS:

β SERVO DRIVE SVPM

#H580

7.5i (20/20/40) ☐ 15i (40/40/80) ☑	PART No A06B-6164-H343
	Sl.No.: V13921758

α SERVO DRIVE PSM

11i ☐ 15i ☐ 26i ☐ ☐	PART No A06B-____-____
	Sl.No.:

α SERVO DRIVE SPM

11i ☐ 15i ☐ 22i ☐ ☐	PART No A06B-____-____
	Sl.No.:

α SERVO DRIVE SVM2

40/40i ☐ 80/80i ☐ 160/160i ☐	PART No A06B-____-____
	Sl.No.:

α SERVO DRIVE SVM1(1'st)

40i ☐ 80i ☐ 160i ☐	PART No A06B-____-____
	Sl.No.:

α SERVO DRIVE SVM1(2'nd)

40i ☐ 80i ☐ 160i ☐	PART No A06B-____-____
	Sl.No.:

FANUC I/O MODULE / I/O CARD

I/O MODULE WITH MPG ☑	I/O CARD WITH MPG ☐	I/O CARD WITHOUT MPG ☑	
PARTNo.: A02B-0309-C001	PART No.: A20B-2002-0520	PART No.: A20B-2002-0521	
Sl No. E13YD1002	Sl No.	Sl No.	QTY: 01 Nos

AC REACTOR1

0.14mH/62A ☑ 0.07mH/110A ☐ ☐	PART No A81L-0001-0156
	Sl.No.: RT1352362

CHECKED BY:- Ashok	VERIFIED BY:-

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

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SERVO MOTOR - X AXIS

β8/3K	β12/3K	β22/2K		
☐	☑	☐	☐	

PART No.: A06B-0078-B103 #100

Sl.No.: C13XW0HC9

SERVO MOTOR - Y AXIS

β8/3K	β12/3K	β22/2K		
☐	☑	☐	☐	

PART No.: A06B-0078-B103

Sl.No.: C139W1GB2

SERVO MOTOR - Z AXIS

β22/2K	β22/3K	β40/2K		
☐	☑	☐	☐	

PART No.: A06B-0082-B403

Sl.No.: C13XA5262

B AXIS DETAILS

ROTARY TABLE ☑ INDEX TABLE ☐ ☐

DRIVE:	βiSV-20	βiSV-40	
	☑	☐	

PART No.: A06B-6130-H002

Sl.No.: V1A52039?

MOTOR:	β4/4K	β8/3K	
	☑	☐	

PART No.: A06B-0063-B001

Sl.No.: ✓ C137F01134

REACTOR: A81L-0001-0083#3C (0.15mH/24A) ☐

Sl No.: RT13N0115

DISCHARGE RESISTOR: 16Ohms ☐ 30Ohms ☑

PMM AXIS DETAILS NIL

SERVO ARM ☐ SERVO MAG ☐ TILTING TABLE ☐

DRIVE:	βiSV-20	βiSV-40	
	☐	☐	

PART No.: A06B-____-H002

Sl.No.:

MOTOR:	β4/4K	β8/3K	
	☐	☐	

PART No.: A06B-0____-____

Sl.No.:

REACTOR: A81L-0001-0083#3C (0.15mH/24A) ☐

Sl No.:

DISCHARGE RESISTOR: 16Ohms ☐ 30Ohms ☐

SPECIAL INFORMATIONS:

CHECKED BY:- AAhob

VERIFIED BY:-

PRINTED IN INDIA

ACE MANUFACTURING SYSTEMS LTD

MACHINE HISTORY - ELECTRICAL

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ISOLATTION SWITCH (SW1) — AC 3 Ph 415V ☑

40A	63A	100A	MAKE: SALZER ☑	ABB ☐	☐
☑	☐	☐	TYPE:		

ISOLATTION TRANSFORMER (TFR1) PRIMARY- 415V/3Ph SECONDARY - 220V/3Ph, 220V/1Ph, 230V/1Ph

11.5KVA	14.3KVA	18 KVA	22.5KVA	MAKE: SALZER ☐	T.S INTERNATIONAL ☑
☐	☑	☐	☐	SL.No: 1103/6/14	

PRIMARY MCB (F15) — AC 3 Ph 415V / 3POLE ☑

25A	32A	40A	50A	MAKE: SCHNEIDER ☑	MERLIN GERIN ☐
☐	☑	☐	☐	TYPE:	

SECONDARY MCB (F19) — AC 3 Ph 220V / 3POLE ☑

40A	50A	63A	80A	MAKE: SCHNEIDER ☑	ABB ☐
☐	☑	☐	☐	TYPE:	

DRIVE CONTROL MCB (F20) 2A — AC 3 Ph 220V / 2POLE ☐

1A	3A			MAKE: SCHNEIDER ☐	MERLIN GERIN ☐
☐	☐	☐	☐	TYPE:	

SPINDLE MOTOR FAN MPCB (F56) — AC 3 Ph 220V / 3POLE ☑

0.1-0.16A	0.25-0.4A		MAKE: ABB ☑	SCHNEIDER ☐
☑	☐	☐	MPCB SETTING: 0.1A / 0.13A / 0.28A	

PSM CONTACTOR (K19) — 2 30V AC ☑

CA3-30-10	A40-30-10	LC1E80		MAKE: ABB ☑	SCHNEIDER ☐
☐	☑	☐	☐	TYPE:	

REGULATED POWER SUPPLY DETAILS

PS1 ☑	PS2 ☑	PS3 ☐	OUTPUT: 24V DC 10A ☑	24V DC 20A ☐
MAKE:	ABB (CP-E 24/10.0) ☐	OMRON(S8VS 24024) ☑		KB ELECTORNICS ☐
Sl. No.: PS1:	PS2:	PS3:		

ISOLATTION TRANSFORMER (TFR4) PRIMARY- 415V/1Ph SECONDARY - 200V/1Ph, 230V/1Ph

900VA	450VA		MAKE: SALZER ☐	T.S INTERNATIONAL ☐
☐	☐	☐	SL.No:	

TFR4 PRIMARY MCB (F16) : 3A/2P ☐	☐	MAKE:SCHNEIDER ☐	MERLIN GERIN ☐
TFR1 -220V-MCB (F58) : 6A/2P ☑	☐	MAKE:SCHNEIDER ☑	MERLIN GERIN ☐
TFR1 -230V- MCB (F59) : 10A/2P ☑	☐	MAKE:SCHNEIDER ☑	MERLIN GERIN ☐

SPECIAL INFORMATIONS:

CHECKED BY:-	VERIFIED BY:-

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MACHINE HISTORY - ELECTRICAL

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CONTROL CIRCUIT

SEALED KEY BOARD/MCP: Sl.No.: 88

TYPE: (YSL-AMS-SKB-2-REV1-10/09) ☑ YSL-SPARK-MCP-5-REV0-03/10 ☑

ATC BRAKE RELAY (PCB 1) :

TYPE: 1 SSR RELAY -YSL-SSRP-BR-3-REV3-08/11 ☐ 2 SSR RELAY - YSL-S26-009-77-REV6-08/11 ☑

I/O DISTRIBUTON PCB(PCB2) (PCB104 /105 /106 /107 /156 /157 /256 /257) : QTY: 05 nos

TYPE: YSL-S26-009-56-REV0-04/08 ☐ YSL IFB-F50-WT-P3-02-0313 ☐ UL-FLKM50M-42 R2 ☑

CNC ON/OFF RELAY (PCB 3) :

TYPE: YSL -SSR-CNC ON / OFF-REV8-08/10 ☑

Z AXIS BRAKE RELAY (PCB 5) :

TYPE: YSL-SSRP-BR3-REV3-08/11 ☑

Z AXIS BRAKE CURRENT SENSINGRELAY (CSR1) :

TYPE: YSL ELECTRONICS (CSR1-2) ☑

AC RELAYS: Z BRAKE/EMG (ACR1) ☐ ATC/CAB. THERMAL (ACR2) ☑ I. TABLE THERMAL(ACR3) ☐

MAKE: OMRON ☑ ELESTA ☐

RELAY BOARD(STD-X20) 14 1RELAY(ELECTRO MAGNETIC-DC) ☐

TYPE: YSL-UMK-1RM-24V- DC-CN-FI-2-REV1-12/09 ☐

RELAY BOARD(STD-X31) 1RELAY(ELECTRO MAGNETIC-DC/AC) ☑

TYPE: YSL-S26-009-082-REV1-06/10 ☑

RELAY BOARD(APC-X25) 8 RELAY (SSR TYPE-DC) ☐

TYPE: YSL-S26-009-064-REV4-09/11 ☐

RELAY BOARD(FIXTURE-X26) 2 / 4 / 8 RELAY (SSR TYPE-DC) ☑

TYPE: YSL-S26-009-062-REV4-09/11 ☐ S26-009-063-REV4-09/11 ☑ S26-009-064-REV4-09/11 ☐

RELAY BOARD(TOOL DECLAMP - X33) nil 1 RELAY (SSR TYPE-DC) ☐

TYPE: YSL-S26-009-061-REV4-07/12 ☐

RELAY BOARD(- X) 1 RELAY (SSR TYPE-DC) ☐

TYPE: YSL-S26-009-061-REV4-07/12 ☐

RELAY BOARD(- X) 2 RELAY (SSR TYPE-DC) ☐

TYPE: YSL-S26-009-062-REV4-07/12 ☐

CHECKED BY:- Ansb VERIFIED BY:-

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MACHINE HISTORY - ELECTRICAL

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EMERGENCY CONTACTOR (K22)			24V DC ☑
KC6-40E ☑	CA3KN-40BD ☐	MAKE: ABB ☑	SCHNEIDER ☐
		TYPE:	
RELAY BOARD(- X)			1 RELAY (SSR TYPE-DC) ☐
TYPE: YSL-S26-009-061-REV4-07/12 ☐			

SINGLE PHASE POWER SUPPLY CIRCUIT

MCB (F18)			AC 1 Ph 230V / 1POLE ☐
10A ☐ ☐ ☐		MAKE: SCHNEIDER ☐	MERLIN GERIN ☐
		TYPE:	
PANEL COOLER:		POWER: 500W ☐	1000W ☑
MAKE: ADVANCE (TURBO1000R33) ☑	ADVANCE (TURBO500R50) ☐		ACE (IGLOO - 750) ☑
Sl.No.: 1063			
UTILITIES (CABINET):			
CABINET LAMP: PEARL(CFL-11W) ☑	PHILIPS(CFL-11W) ☐		☐
DOMESTIC CONNETION BOX: MURR ELEKTRONIK-MODILINK MSDD basic ☑			
DOOR INTERLOCK: LIMIT SW (HONEYWELL-ZLD A01D) ☑			
MCB FOR CAB. LAMP, 5 PIN SOCKET (F66-3A/1PMAKE: SCHNEIDER) ☑			
UTILITIES (SWITCHGEARS): NIL			
ADD ON BLOCK(PCB5/PSM(K19) INTERLOCK):	SCHNEIDER(LA1KN11) ☐		ABB(CA5-10) ☐
ADD ON BLOCK(STAR DELTA INTERLOCK):	SCHNEIDER(LA1KN11) ☐		ABB(CA5-10 & 01-02 each) ☐
ADD ON BLOCK(HYDRAULIC FAN POWER):	SCHNEIDER(LA1KN11 ☐		ABB(CAF6-20-M) ☑
ADD ON BLOCK(AUTO DRAIN SOL):	SCHNEIDER(LA1KN11) ☐		ABB(CAF6-20-M) ☑

SPINDLE STAR / DELTA CIRCUIT NIL

CONTACTORS (K20 / K21)			230V AC ☐
A75-30 ☐	CA7 - 85 ☐ ☐	MAKE: ABB ☐	SPRECHER &SCHU ☐
		TYPE:	
RELAY BOARD(X32)			2 RELAYS (SSR TYPE) ☐
TYPE: S26-009-80-CN--F-24DC-230AC ☐			

SPECIAL INFORMATIONS:

CHECKED BY:- Ashok	VERIFIED BY:-

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MACHINE HISTORY - ELECTRICAL

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MAGAZINE MOTOR POWER

MPCB (F1)	1-1.6A ☑	0.63-1A ☐		CONTACTOR (K1/K2)	LP2K09-01BD ☐	VBC6-30-01 ☐	
MPCB SETTING:	1A ☑	1.2A ☑		MOTOR ID:	SA9715 ☐	SA4067 ☑	SA4060 ☐ TM80 ☐
SOLID STATE RELAY (SSR1):	MAKE: YSL-ACSSC-2P-4 ☑	ERI ELECTRONICS- SJDA - 482524(DSPST ☐					

ARM MOTOR POWER

MPCB (F2)	1-1.6A ☑	MCB (F81-4A / 3P) ☐		CONTACTOR (K3/K76)	LP2K09-01BD ☐	VBC6-30-01 ☐	
MPCB SETTING:	1.2A ☑	1.4A ☐		MOTOR ID:	SAG909 ☐	SA5907 ☑	SA4412 ☐ ☐
VFD(HITACHI-WJ200-1.5 KW) ☑	SPEED(50/ 30/ 15HZ) ☐			REACTOR/CHOKE:	2HP/ 6.5mH /3.4A(TSI) ☐		
SOLID STATE RELAY (SSR3):	MAKE: YSL-ACSSC-2P-4 ☑	ERI ELECTRONICS- SJDA - 482524(DSPST) ☐					

LUBRICATION MOTOR POWER

MCB (F3)	1A/3P ☑	MPCB(0.63-1A) ☐		CONTACTOR (K4)	LP1K09-10BD ☐	BC6-30-01 ☑	
GREASE LUBRICAION SYSTEM (LUBE) ☑				MOTOR POWER:	0.09KW/0.5A ☐	0.12KW/0.6A ☑	

FLOOD COOLANT MOTOR POWER

MPCB (F4)	0.63-1A ☑	2.5-4A ☐		CONTACTOR (K5)	LP1K09-10BD ☐	BC6-30-01 ☑	
MPCB SETTING:	0.9A ☑	3.4A ☐		MOTOR POWER:	0.3KW/0.9A ☑	1.5KW/3.4A ☐	
SURGE SUPRESSOR(RC4):					YSL-NIR-RC3-1/022-575VAC ☑		

CHIP PUSH COOLANT MOTOR POWER

MPCB (F5)	0.63-1A ☐	2.5-4A ☑		CONTACTOR (K6)	LP1K09-10BD ☐	BC6-30-01 ☑	
MPCB SETTING:	0.9A ☐	3.4A ☑		MOTOR POWER:	0.3KW/0.9A ☐	1.5KW/3.4A ☑	
SURGE SUPRESSOR(RC5): YSL-NIR-RC3-1/022-575VAC ☑				COMMON CONTACOR KIT - PUSH/FLUSH(ABB-VBC6 ☐			

CHIP FLUSH COOLANT MOTOR POWER

MPCB (F6)	0.63-1A ☐	2.5-4A ☑		CONTACTOR (K7)	LP1K09-10BD ☐	BC6-30-01 ☑	
MPCB SETTING:	0.9A ☐	3.4A ☑		MOTOR POWER:	0.3KW/0.9A ☐	1.5KW/3.4A ☑	
SURGE SUPRESSOR(RC6): YSL-NIR-RC3-1/022-575VAC ☑				COMMON MPCB WITH CHIP PUSH ☐			

COOLANT GUN / CONVEYOR DRUM WASH MOTOR POWER

MPCB (F7)	0.63-1A ☐	2.5-4A ☐		CONTACTOR (K8)	LP1K09-10BD ☐	BC6-30-01 ☐	
MPCB SETTING:	0.9A ☐	3.4A ☐		MOTOR POWER:	0.3KW/0.9A ☐	1.5KW/3.4A ☐	
SURGE SUPRESSOR(RC7):					YSL-NIR-RC3-1/022-575VAC ☐		

HIGH PRESSURE COOLANT MOTOR POWER

MPCB (F9)	4-6.3A ☐	2.5-4A ☐		CONTACTOR (K10)	LP1K09-10BD ☐	BC7-30-01 ☐	
MPCB SETTING:	4.9A ☐	5.2A ☐		MOTOR POWER:	2.2KW/4.9A ☐	2.2KW/5.2A ☐	
SURGE SUPRESSOR(RC9):					YSL-NIR-RC3-1/022-575VAC ☐		

HYDRAULIC 1 MOTOR POWER

MPCB (F8)	4-6.3A ☐	2.5-4A ☑		CONTACTOR (K9)	LP1K09-10BD ☐	BC6-30-01 ☑	
MPCB SETTING:	2.7A ☐	3.5A ☑		MOTOR POWER:	1.1KW/2.7A ☐	1.5KW/3.5A ☐	
SURGE SUPRESSOR(RC8):					YSL-NIR-RC3-1/022-575VAC ☑		

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INDEX TABLE MOTOR POWER N/A

MPCB (F12)	1-1.6A ☐	☐	CONTACTOR(K15/K16)	LP1K09-10BD ☐	VBC6-30-01 ☐
MPCB SETTING:	1.2A ☐	1.4A ☐	MOTOR POWER:	1.1KVA/1.5A ☐	2.3KVA/3.2A ☐
SOLID STATE RELAY (SSR5):	MAKE: YSL-ACSSC-2P-4 ☐	ERI ELECTRONICS- SJDA - 482524(DSPST) ☐			

CHIP CONVEYOR 1 MOTOR POWER N/A

MPCB (F10)	0.63-1A ☐	1-1.6A ☐	CONTACTOR(K11/K12)	LP1K09-10BD ☐	VBC6-30-01 ☐
MPCB SETTING:	0.65A ☐	☐	MOTORPOWER:	0.18KW/0.65A ☐	☐
SURGE SUPRESSOR(RC10):			YSL-NIR-RC3-1/022-575VAC ☐		

CHIP CONVEYOR 2 MOTOR POWER N/A

MPCB (F11)	0.63-1A ☐	1-1.6A ☐	CONTACTOR(K13/K14)	LP1K09-10BD ☐	VBC6-30-01 ☐
MPCB SETTING:	0.65A ☐	☐	MOTORPOWER:	0.18KW/0.65A ☐	☐
SURGE SUPRESSOR(RC11):			YSL-NIR-RC3-1/022-575VAC ☐		

COIL / SCREW CONVEYOR 1 MOTOR POWER N/A

MPCB (F13)	0.63-1A ☐	1-1.6A ☐	CONTACTOR(K17/K18)	LP1K09-10BD ☐	VBC6-30-01 ☐
MPCB SETTING:	0.65A ☐	1.1A ☐	MOTORPOWER:	0.18KW/0.65A ☐	0.37KW/1.1A ☐
SURGE SUPRESSOR(RC13):			YSL-NIR-RC3-1/022-575VAC ☐		

COIL / SCREW CONVEYOR 2 MOTOR POWER N/A

MPCB (F30)	0.63-1A ☐	1-1.6A ☐	CONTACTOR(K28/K29)	LP1K09-10BD ☐	VBC6-30-01 ☐
MPCB SETTING:	0.65A ☐	1.1A ☐	MOTORPOWER:	0.18KW/0.65A ☐	0.37KW/1.1A ☐
SURGE SUPRESSOR(RC30):			YSL-NIR-RC3-1/022-575VAC ☐		

OIL SKIMMER MOTOR POWER N/A

MPCB (F21)	0.25-0.4A ☐	☐	CONTACTOR (K23)	LP1K09-10BD ☐	BC6-30-01 ☐
MPCB SETTING:	0.25A ☐	☐	MOTOR POWER:	40W/0.22A ☐	☐
SURGE SUPRESSOR(RC21):			YSL-NIR-RC3-1/022-575VAC ☐		

MOTOR POWER (OPTIONAL-1) N/A

MPCB (F)	☐	☐	CONTACTOR (K	)LP1K09-10BD ☐	BC6-30-01 ☐
MPCB SETTING:	☐	☐	MOTOR POWER:	☐	☐
SURGE SUPRESSOR(RC):			YSL-NIR-RC3-1/022-575VAC ☐		

MOTOR POWER (OPTIONAL - 2) N/A

MPCB (F)	☐	☐	CONTACTOR (K	)LP1K09-10BD ☐	BC6-30-01 ☐
MPCB SETTING:	☐	☐	MOTOR POWER:	☐	☐
SURGE SUPRESSOR(RC):			YSL-NIR-RC3-1/022-575VAC ☐		

SPECIAL INFORMATIONS: MPCB'S- SCHNEIDER(EasyPactTVSGZ1E08) ☐ ABB (MS 116) ☑

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ATC & SPINDLE DETAILS

SPINDLE MOTOR:	MODEL:	β3 ☐	β6 ☑	β12 ☐	☐
	MOTOR RPM:	6000 ☐	8000 ☐	10000 ☑	☐
	MOTOR POWER:	5.5/7.5 KW			
	PART No:	A06B-1445-B103			
	Sl. No:				
	GEAR RATIO:	1:1 ☐	1:1.25 ☐	1:1.5 ☑	DIRECT DRIVE ☐
	POWER LIMIT:	IN KW: 5.5/7.5			
	SPEED LIMIT:	IN RPM: 60 - 6000			

PROXIMITY SENSORS-SPINDLE

MAKE; TEKNIC ☑ TYPE: (24V DC - PNP) ☐

SPINDLE ORIENTATION (PXS1) ☑	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☐	☐	☑
TOOL CLAMP(PXS3) ☑	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☐	☐	☑
TOOL DECLAMP(PXS4) ☑	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☐	☐	☑
NO TOOL IN SPINDLE(PXS32) ☑	M08X50	M08X68	M12X73	
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☑	☐	☐
ONE ROTATION(PXS39) ☑	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☐	☐	☑

PROXIMITY SENSORS-ATC(DISC TYPE) N.V.D

MAKE; TEKNIC ☑ TYPE: (24V DC - PNP) ☐

MAGAZINE IN POSITION (PXS2) ☑	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☑ NON CONN. ☐	☐	☐	☐	☑
MAGAZINE REFERENCE (PXS46) ☑	M12X73	FG2X05APO24-5000LFLAT		
TYPE: CONNECTOR ☑ NON CONN. ☐	☑	☐		☐

PROXIMITY SENSORS-ATC(TWIN ARM TYPE)

MAKE; TEKNIC ☑ TYPE:EGT 12 X 02 AP - 024 SEM4L(24V DC - PNP) ☑

MAGAZINE IN POSITION PROXIMITY (PXS2) ☑	MAG REFERENCE PROXIMITY (PXS46) ☑
POCKET UP PROXIMITY (PXS5) ☑	POCKET DOWN PROXIMITY (PXS6) ☑
ARM POSITION 1 PROXIMITY-PS1 (PXS7) ☑	☐
ARM POSITION 2 PROXIMITY-PS2 (PXS58) ☑	ARM POSITION 3 PROXIMITY-PS3 (PXS59) ☑

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PNUEMATIC ELEMENTS (SPINDLE + ATC)

MAKE; SUCO ☑	TYPE:(24V DC - NO CONTACT) ☐
MAIN AIR PRESSURE SWITCH (PRS 2) ☑	SPINDLE AIR PRESSURE SWITCH (PRS 101) ☑
AIR PURGE PRESSURE SWITCH (PRS 102) ☐	☐
MAKE; FESTO ☑ SMC ☐	TYPE:(24V DC -COIL) ☐
SPINDLE AIR JET SOLENOID (Y1) ☑	SPI AIR PURGE SOLENOID (Y115) ☐
POCKET UP SOLENOID (Y2) ☑	POCKET DOWN SOLENOID (Y3) ☑
ATC DOOR CLOSE SOLENOID (Y56) ☐	ATC DOOR OPEN SOLENOID (Y57) ☐
AIR THROUGH NOZZLE SOLENOID (Y99) ☐	☐

TOOL CLAMP/DECLAMP DETAILS

MAKE; YUKEN ☐ SMC ☐ FESTO ☑	TYPE:HYDRAULIC(DC COIL) ☐ PNUEMATIC(DCCOIL) ☐
TOOL CLAMP SOLENOID (Y51) ☐	TOOL DECLAMP SOLENOID (Y4) ☑
TOOL DECLAMP PUSH BUTTON :	ILLUMINATED GREEN ☑ ☐

UTILITIES(SPINDLE + ATC) N/A

TOOL CHANGE POSITION LIMIT SWITCH- (LS3) :	TEKNIC SG-01-R-610 ☐		☐
ATC / I.TABLE DISPLAY SEL SWITCH :	2 POSITION ☐	ATC DOOR CLOSE READ SW	☐
ARM/MAG TEST MODE KEY SWITCH (KSW7) :	2 POSITION ☐	ATC DOOR OPEN READ SW	☐
POCKET UP/DOWN PB ☐ ARM FWD/REV SEL SW ☐	☐	ATC DOOR OPEN/CLOSE PB	☐

SPECIAL INFORMATIONS:

AUTO DOOR & CYCLE START DETAILS N/A

MAKE; FESTO ☐	TYPE:(24V DC -COIL) ☐
AUTO DOOR OPEN SOLENOID (Y6) ☐	AUTO DOOR CLOSE SOLENOID (Y7) ☐
AUTO DOOR SAFETYSOLENOID (Y8) ☐	☐
DOOR SWITCHES	TYPE: LS5 - NG1 RS 510 (24V DC -NO CONTACT) ☐
MAKE; TEKNIC ☐ FESTO ☐	READ SWITCHES (RSW30& RSW31) ☐
CYCLE START SWITCH	TYPE: LS45 - XCE-154 (24V DC -NO CONTACT) ☐
MAKE; ☐ TELEMECHANIC ☐	☐
DOUBLE HAND CYCLE START BOX	CYCLE START PB (MUSHROOM HEAD GREEN) (02 Nos) ☐
MAKE; TEKNIC ☐ ☐	EMERGENCY PB (MUSHROOM HEAD SPRING RETURN) ☐
AUTO DOOR OPERATION	AUTO DOOR OPEN/CLOSE PB (GREEN) ☐
MAKE; TEKNIC ☐ ☐	AUTO DOOR INTERLOCK KEY SW (2 POSITION) ☐

SPECIAL INFORMATIONS:

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COOLANT TANK / CONVEYORS DETAIL

UTILITIES (STD COOLANT)

COOLANT LEVEL FLOAT SWITCHES	LOW-FILPRO(LIM 18PF SC)	☑	HIGH-FILPRO(LIM 18PR SC)	☐
COOLANT GUN SEL SW :	2 POSITION	☐		☐
COOLANT VALVES: PUSH(Y125) /FLUSH(Y126) /ARM FLUSH(Y127) /FLUSH2(Y219) :	MAKE:AVCON / SMC	☐		
COOLANT PLUG (HC1) :	16 PIN	☑	24 PIN	☐
LOCATION(FROM M/C REAR) :	LHS	☐	RHS	☑

UTILITIES (HP COOLANT): NA

FLOW SWITCH (FLW1) :	IFM (SH0 500)	☐		☐
CLOG SWITCH (FCLG1) :	HIRLEKAR	☐	SALVIN	☐
AIR BLAST SOLENOID (Y5) :	FESTO	☐	ROTEX RELIEF VALVE (Y235)	☐
M/C DOOR CLOSE LIMIT SW (LS5) :	TEKNIC(NG1 RS 510)	☐		☐

UTILITIES (CHIP CONVEYOR): NA

CHIP CONVEYOR 1 O/L LIMIT SW (LS12) :	TEKNIC(NG1 RS 510)	☐		☐
CHIP CONVEYOR 2 O/L LIMIT SW (LS13) :	TEKNIC(NG1 RS 510)	☐		☐
CHIP CONVEYOR FWD/REV SEL SW :	3 POSITION(SPRING RETURN)	☐		☐
CHIP CONVEYOR PLUG (HC3) :	10 PIN	☐	24 PIN	☐
LOCATION(FROM M/C REAR) :	LHS	☐	RHS	☐

UTILITIES (COIL/ SCREW CONVEYOR): NA

SCREW CONVEYOR 1 O/L PROXIMITY (PXS20) :	M12 X 52MM (CONN. TYPE)	☐	M12 X 72MM (CONN. TYPE)	☐
SCREW CONVEYOR 2 O/L PROXIMITY(PXS104) :	M12 X 52MM (CONN. TYPE)	☐	M12 X 72MM (CONN. TYPE)	☐
SCREW CONVEYOR FWD/REV SEL SW :	3 POSITION(SPRING RETURN)	☐	COMMON WITH CHIP.CNYR	☐

SPECIAL INFORMATIONS:

HYDRAULIC TANK DETAIL

OIL LEVEL FLOAT SWITCH (FL4) :	SHRIDHAN	☐		☐
OIL CLOG SWITCH (CLG2) :	HYDAC	☐	REXROTH	☐
HYDRAULIC THERMAL SWITCH WITH CONTROLLER (TH4) :	SRI ELECTRINICS	☐		☐
HYDRAULIC PLUG (HC2) :	16 PIN	☑	24 PIN	☐
MAKE;INDO ELECTRIC ☐ LOCATION(FROM M/C REAR) :	LHS	☑	RHS	☐

SPECIAL INFORMATIONS:

Hydraulic fixtures - Feeding interface only.

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LUBRICATION TANK / FRL DETAIL

OIL LEVEL FLOAT SWITCH **(FL1)** :	CENLUB	☑	GREASE-LUBE	☐	
LUB PRESSURE SWITCH **(PRS1)** :	SUCO	☑	GREASE-LUBE	☐	
AUTO DRAIN SOLENOID **(Y44)** :	ROTEX	☐	FLOAT SW**(FL3)**	☐	

MACHINE LAMPS

MAKE:			PATANKAR	☐	AISHWARYA	☑
MACHINE LAMP1 **(ML1)** :	11W	☐	9W+9W	☐	36W	☑
MACHINE LAMP2 **(ML2)** :	11W	☐	9W+9W	☐	36W	☑
MACHINE INDICATION LAMP(PL5) :	1 TIER	☐	3 TIER	☑	3 TIER WITH HOOTER	☐
MAKE:			PATANKAR	☑	LAXMI(VANTECH)	☐

UTILITIES(MACHINE)

EMERGENCY MONITOR (PCB 12)		YSL-EMG-PCB-4-REV3-12/11		☑
HAND HELD MPG:		FANUC ☐	FUTURE(TAIWAN)	☐
	Sl. No.:			
TPM TRACK :		PIONEER CT ☐		☐
TOOL BREAKGE SENSOR :		METROL ☐		☐
SERVO VOLTAGE STABILIZER:		15KVA ☐	25KVA	☐
INPUT: **OUTPUT:**		**MAKE:** SERVOMAX ☐		☐
	Sl. No.:			
COOLANT / SPINDLE CHILLER UNIT:		**MAKE:** POINT ☐		☐
POWER:	**MCB(F31):**10A / 3P	**Sl. No.:**		
CONNECTOR :	INDO ELECTRIC (10 PIN) ☐	**LOCATION:** LHS ☐	RHS	☐

FIXTURE DETAIL

MAKE; REXROTH / YUKEN / SUN HYD. ☐		**PNEUMATIC** ☐	**HYDRAULIC** ☐
STATION 1 CLAMP SOLENOID **(Y70)** ☐		STATION 1 DECLAMP SOLENOID **(Y71)** ☐	
STATION 2 CLAMP SOLENOID **(Y72)** ☐		STATION 2 DECLAMP SOLENOID **(Y73)** ☐	
STATION 3 CLAMP SOLENOID **(Y84)** ☐		STATION 3 DECLAMP SOLENOID **(Y85)** ☐	
STATION 4 CLAMP SOLENOID **(Y86)** ☐		STATION 4 DECLAMP SOLENOID **(Y87)** ☐	
MAKE; POLYHYDRON / ☐		**PNEUMATIC** ☐	**HYDRAULIC** ☐
STATION 1 CLAMP PRESSURE SW **(PRS20)** ☐		STATION 2 CLAMP PRESSURE SW **(PRS22)** ☐	
STATION 3 CLAMP PRESSURE SW **(PRS31)** ☐		STATION 4 CLAMP PRESSURE SW **(PRS32)** ☐	

SPECIAL INFORMATIONS: CLAMP PB'S (GREEN) -

DECLAMP PB'S (RED) -

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MACHINE HISTORY - ELECTRICAL

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APC (LINEAR TYPE) DETAIL

MANUAL OPERATION PANEL :	YSL-AMS-APC-PCB-REV2-12/11 ☐
PALLET SET UP PB :	LUMINOUS GREEN ☐
EMERGENCY PB :	MUSHROOM HEAD RED ☐

PROXIMITY SENSORS:

MAKE; TEKNIC ☐	TYPE: (24V DC - PNP) ☐

PALLET RECEIVED PROXIMITY(PXS11) ☐	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐
ARM 1 IN PROXIMITY (PXS13) ☐	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐
ARM 1 OUT PROXIMITY (PXS14) ☐	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐
ARM 2 IN PROXIMITY (PXS15) ☐	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐
ARM 2 OUT PROXIMITY (PXS16) ☐	M12X40	M12X52	M12X68	M12X73
TYPE: CONNECTOR ☐ NON CONN. ☐	☐	☐	☐	☐

PNEUMATIC SOLENOIDS:

MAKE; FESTO ☐	TYPE: 24V DC -COIL ☐
APC DOOR OPEN SOLENOID (Y12) ☐	APC DOOR CLOSE SOLENOID (Y13) ☐
PALLET PAD CLEAN SOLENOID (Y18) ☐	☐

HYDRAULIC SOLENOIDS:

MAKE; YUKEN ☐ REXROTH ☐	TYPE: 24V DC -COIL ☐
PALLET CLAMP SOLENOID (Y10) ☐	PALLET DECLAMP SOLENOID (Y11) ☐
ARM 1 IN SOLENOID (Y14) ☐	ARM 1 OUT SOLENOID (Y15) ☐
ARM 2 IN SOLENOID (Y16) ☐	ARM 2 OUT SOLENOID (Y17) ☐

HYDRAULIC PRESSURE SWITCHES:

MAKE; POLYHYDRON ☐	TYPE: ☐
PALLET CLAMP PRSW (PRS3) ☐	PALLET DECLAMP PRSW (PRS60) ☐

LIMIT SWITCHES:

MAKE; TEKNIC(NG1VS 510) ☐	JAI BALAJI (JLSSPR) ☐
PALLET 1 ON STATION LIMIT SW (LS10) ☐	PALLET 2 ON STATION LIMIT SW (LS11) ☐

READ SWITCHES:

MAKE; FESTO ☐	APC DOOR INTERMEDIATE READ SW (RSW 65) ☐
APC DOOR OPEN READ SW (RSW 58) ☐	APC DOOR CLOSE READ SW (RSW 59) ☐

SPECIAL INFORMATIONS:

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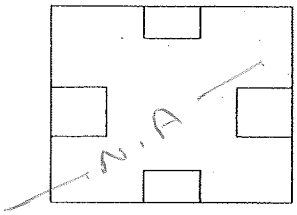
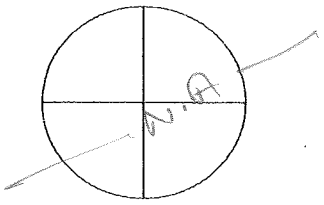
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CHECK LIST - FINAL ASSEMBLY (VMC / CMV) - MECHANICAL

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MACHINE No. : M11-995	CUSTOMER : M/s KANRA TECHNOLOGY

I. Motor Mounting Bracket Face Out and CONCENTRICITY for Direct Drive

	FACE OUT (Recommended ≤ 0.015 mm)	CONCENTRICITY (Recommended ≤ 0.010 mm)
		

S.No.	DESCRIPTION	RECOMMENDED	ACTUAL
1	Spindle Connector Radial Run Out	≤ 0.010 mm	
2	Motor Current at Maximum RPM	β3 < 15%, β6 / α6 & ABOVE <10%	
3	CTS Rod Guide Radial Run Out (DDCTS)	≤ 0.010 mm	N.A
4	Spindle Clamping Force	As per DS-DSN-PROD-01	
5	Spindle Motor Specification	As per Work Order	
II	**INDIRECT DRIVE**		
1	Motor Pulley Radial Run Out	≤ 0.050 mm	0.03 mm
2	Spindle Pulley Radial Run Out	≤ 0.050 mm	0.02 mm
3 a)	Belt Tension	As per DS-DSN-PROD-02	308 N/m.
b)	Belt Specification	Length : 770 mm Width : 40 mm	Quantity : 01 No
4	Motor Current at Maximum RPM	β3 < 20%, β6 / α6 & ABOVE <15%	20 %
5	Rotary Joint Radial Run Out (ID CTS)	≤ 0.040 mm	N.A
6	Spindle Clamping Force	As per DS-DSN-PROD-01	840 kg
7	Spindle Motor Specification	As per Work Order	126
8	Pulley Ratio	As per Work Order	1 : 1.5
III	Axis Dead Stopper Value / Butt Referencing Value (As per Drawing)		
	Axis	MOTOR END	BRAKE END
1	X	24.011	-22.563
2	Y	-13.242	16.436
3	Z	14.028	-19.022
IV	% Of axis CURRENT Drawn at 100 % Rapid Rate (Recommended as per DS-QA-02)		
	X - Axis	48 %	
	Y- Axis	51 %	
	Z - Axis Up	64 %	
	Z - Axis Down	27 %	
V	Accumulator Oil Pressure	As per Circuit Diagram	N.A
	Accumulator Gas Pressure	As per Circuit Diagram	N.A
NAME & SIGN	Madhu Sudhan CHECKED BY	APPROVED BY	1 ... 7 Page No.

ACE MANUFACTURING SYSTEMS LTD

CHECK LIST - FINAL ASSEMBLY (VMC / CMV) - MECHANICAL

AMS — In Pursuit of Excellence

S.No.	DESCRIPTION	RECOMMENDED	ACTUAL
VI	**ATC ALIGNMENT (TURRET TYPE)**		
1	Parallelism of gripper Arm face wrto. X - Axis	≤ 0.050 mm	
2	Tool Ejection Stroke	As per DS - DSN-PROD-01	N.A
3	Swing Lever Gap	As per Drawing	
4	Side Cam Alignment	0.020 mm Maximum	
5	Gap between Draw Bar bush & Roller/Screw	1 ± 0.1 mm	
VII	**ATC ALIGNMENT (TWIN ARM)**		
1	X - Axis	≤ 0.2 mm	0.05mm
2	Y - Axis	≤ 0.2 mm	0.03 m.m
3	Z - Axis	≤ 0.05 mm	0.02 m.m
4	Twin Arm bottom face parallelism in Y-Axis	≤ 0.1 mm	0.05 m.m
5	Gap between Draw Bar Bush & Roller	1 ± 0.1 mm	1 m.m
6	Tool Ejection stroke	As per DS - DSN-PROD-01	0.8 m.m
7	Gap between deflection Pads	0.02 TO 0.050 mm	N.A
VIII	**APC ALIGNMENT - LINEAR TYPE**		
1	Height difference of rest pads	≤ 0.010 mm	N.A
2	Pallet change Time	As per Work Order	N.A
IX	**APC ALIGNMENT- ROTARY TYPE**		
1	APC Base parallelism w.r.to. X - Axis	≤ 0.1 mm	
	APC Base parallelism w.r.to. Y - Axis	≤ 0.1 mm	
2	Arm Top parallelism w.r.to. X - Axis	≤ 0.05 mm	
	Arm Top parallelism w.r.to. Y - Axis	≤ 0.05 mm	N.A
3	Arm 1 Alignment ' FACE ' w.r.to X- Axis	≤ 0.02 mm	
	Arm 1 Alignment ' Taper ' w.r.to X - Axis	≤ 0.02 mm	
	Arm 2 Alignment ' FACE ' w.r.to X - Axis	≤ 0.02 mm	
	Arm 2 Alignment ' Taper ' w.r.to X - Axis	≤ 0.02 mm	N.A
4	Offset between Arm 1 & Arm 2 - Face	≤ 0.030 mm	
	Offset between Arm 1 & Arm 2 - Taper	≤ 0.030 mm	
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AMS *In Pursuit of Excellence*

CHECK LIST - FINAL ASSEMBLY (VMC / CMV) - MECHANICAL

S.No.	DESCRIPTION	RECOMMENDED	ACTUAL
5	Pallet - 1 Wedge Alignment along X - Axis	≤ 0.020 mm	
	Pallet - 2 Wedge Alignment along X - Axis	≤ 0.020 mm	
6	Gap between Pallet -1 side & Arm along X - Axis	Minimum 0.3 mm	
	Gap between Pallet -1 side & Arm along Z - Axis	0.1 ± 0.02 mm	N.A
	Gap between Pallet -2 side & Arm along X - Axis	Minimum 0.3 mm	
	Gap between Pallet -2 side & Arm along Z - Axis	0.1 ± 0.02 mm	
7	Pallet Sagging - Pallet -1 'W.r.to. X & Y - Axis	≤ 1 mm	
	Pallet Sagging - Pallet -2 'W.r.to. X & Y - Axis	≤ 1 mm	
8	Pallet Height Variation	As per TEST CHART	N.A
9	Backlash between Rack & Pinion at distance of 500 mm	≤ 1 mm	
10	Pallet Change Time	As per Work Order	

S.No.	CHECK POINTS	REMARKS
A	**AXIS**	
1	Ensure geometrical accuracies as per test chart.	OK
2	Ensure plugs are fitted for LM guide ways.	Ok
3	Ensure butt reference function.	Ok
4	Ensure effective wiping of saddle cover wipers.	N.A
5	Ensure lube timer is set 12 min for HG and 20 min for LM Guide ways.	OK
6	Ensure leveling bolt locknuts are tighten after machine leveling.	Ok
7	Check the all axes reference arrows are matched & Direction name plate fixed.	Ok
8	Ensure tightness & marking of all screws with Yellow paint marker	Ok
9	Ensure proper setting of Limit switches.	Ok
B	**ATC**	
1	Check oil level is maximum in ATC gear box.	Ok
2	Check tool change position limit switch for proper functioning & band width. (Max. 1 mm on each side)	N.A
3	Ensure smooth function of Tool change without abnormal/unhealthy noise.	OK
4	Ensure ATC bracket & Magazine Dowelling.	OK

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ACE MANUFACTURING SYSTEMS LTD CHECK LIST - FINAL ASSEMBLY (VMC / CMV) - MECHANICAL		AMS In Pursuit of Excellence
S.No.	CHECK POINTS	REMARKS
C	SPINDLE	
1	Check the tightness of push stud.	OK
2	Ensure air jet through all the (4 / 8) holes at oriented spindle position	OK
3	Ensure supply of air to spindle Air Purge system (If provided)	N.A
4	Ensure proper setting of proximities.	OK
5	Ensure smooth running of spindle without abnormal/unhealthy noise	OK
6	Ensure dummy plate fixing on CTS slot in Non CTS machine.	N.A
7	Check the Tool not in spindle proximity message.	OK
D	COOLANT	
1	Ensure no coolant leakage around the machine & presence of coolant in lubrication oil drain pipe	OK
2	Ensure proper working of float switch (by shifting the sensor upwards).	OK
3	Ensure all the hoses, fittings tightened and routed properly.	OK
4	Check function of Flood coolant (M7), Chip push (M182), Base flush (M181), CTS (M52) & Coolant gun (through selector switch).	OK
5	Ensure CTS flow sw.setting by opening the bag filter air vent ball valve. (Optional)	
6	Ensure suction strainer & Check Valve in CTS tank & pressure gauge working. (If provided)	
7	Ensure CTS pin locator hole is plugged. (If provided)	N.A
8	Ensure that twin arm plunger & CTS pin locator screw do not foul (if provided)	N.A
9	Ensure bag filter clog message by closing bag filter outlet ball valve (Optional)	
10	Ensure High pressure air blast after CTS switched off (If provided)	
11	Ensure drum wash in conveyors (If provided)	
12	Check conveyor overload message & Direction (If provided)	N.A
13	Check oil in conveyor gear box. (if provided)	
14	Ensure functioning of screw conveyor in forward & reverse directions. (through selector switch)	
15	Ensure splash guard fixed to conveyor.	
E	APC	
1	Ensure proper setting of proximities.	
2	Ensure supply of air jet to pallet rest pad cleaning.	N.A
3	Ensure pallet change takes place smoothly with specified load (as per work order) without abnormal / unhealthy noise	
4	Ensure APC base Dowelling.	
5	Check pallet initialization in APC Machine	
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ACE MANUFACTURING SYSTEMS LTD		AMS In Pursuit of Excellence
CHECK LIST - FINAL ASSEMBLY (VMC / CMV) - MECHANICAL		
S.No.	CHECK POINTS	
F	HYDRAULIC, PNEUMATIC & LUBRICATION	
1	Ensure pneumatic main line pressure switch setting (4 bar) & error message are satisfactory	OK
2	Ensure Spindle air jet pressure switch setting (3 bar) & error message are satisfactory	OK
3	Ensure Spindle air purge pressure switch setting (1.2 bar) & error message are satisfactory	N.A
4	Ensure proper setting of oil flow (1 drop/tool change) in FRL unit. & Oil Level low message	OK
5	Ensure proper routing of FRL drain hose.	OK
6	Ensure identification name plates are fixed on power pack hoses.	OK
7	Ensure Ball valve fixed to FRL Unit inlet	OK
8	Ensure pneumatic quick change coupling fixed to 4th axis provision	N.A
9	Ensure no oil leakage in hydraulic valves & fittings	N.A
10	Ensure hydraulic flow control valve grub locking in APC machine.	N.A
11	Ensure pneumatic counter balance bracket Dowelling.	N.A
G	GUARDING ASSEMBLY	
1	Ensure all internal guards are assembled.	OK
2	Ensure smooth movement of sliding door without play and unhealthy noise	N.A
3	Ensure Nylock nuts are fixed to door rollers.	OK
4	Ensure ATC door LM rail plugs fixed.	N.A
5	Ensure Proper tuning of Front auto door / ATC auto door cylinder.	N.A
H	ANY OTHER POINTS	
*	A.T.C Mtg bracket dowelly to be done	
*	Column Rear Cover to be fixed — Shunkyo	
*	Conduit Cover to be fixed	
NAME & SIGN	CHECKED BY: Madhu APPROVED BY:	5... 7 Page No.

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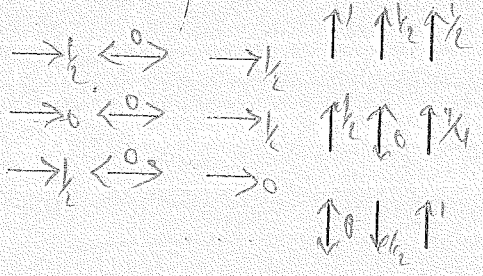
GEOMETRICAL TEST CHART(VMC / CMV)

AMS — In Pursuit of Excellence

PERMISSIBLE TOLERANCE AS PER TEST CHART (DS-DSN-PROD-TC-01)

1. MACHINE LEVEL READING:

Along X-Axis / Along Y-Axis



2. TABLE TOP READINGS:

Along X-Axis

0	-3	0
0	-5	0
0	-6	-4

Along Y-Axis

0	0	0
0	-1	-2
-4	-3	-7

3. T SLOT READING:

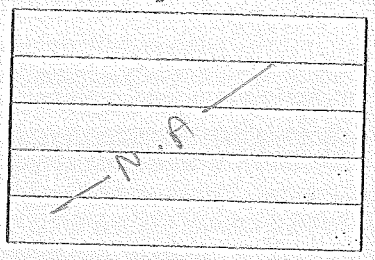
0	-6	0

PALLET - II

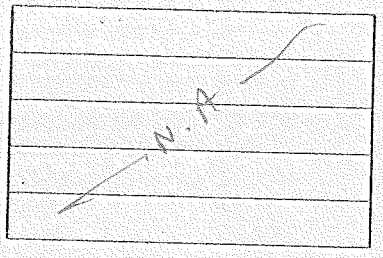
—	N.A	—

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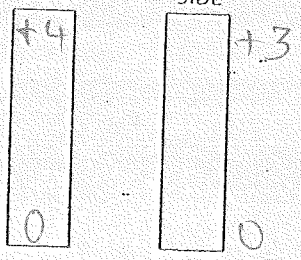
Along X-Axis



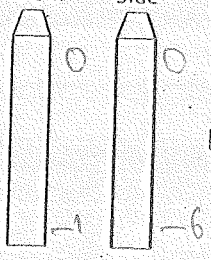
Along Y-Axis



4. COLUMN READINGS :



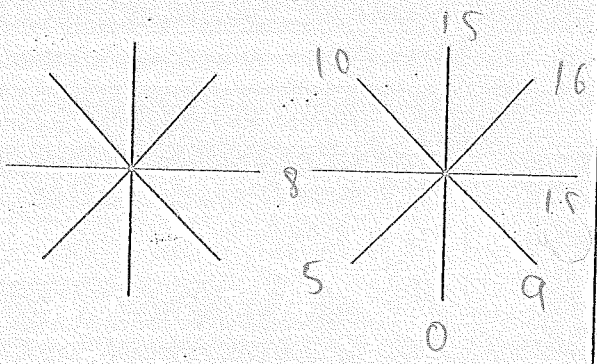
5. MANDREL READINGS:



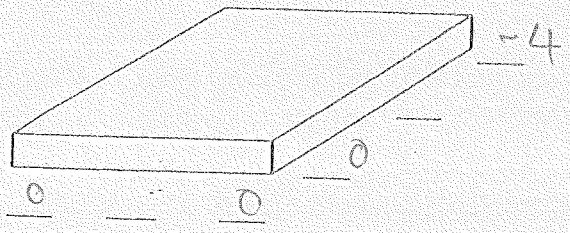
RUNOUT:

a. TOP :
   1. 0.001
b. BOTTOM :
   1. 0.007
   2. 0.008
   3. 0.007
   4. 0.01

6. TURN AROUND READING:



7. SQUARENESS READING (X&Y):



8. PALLET HEIGHT VARIATION:

Pallet-1: N.A

Pallet-2 : —

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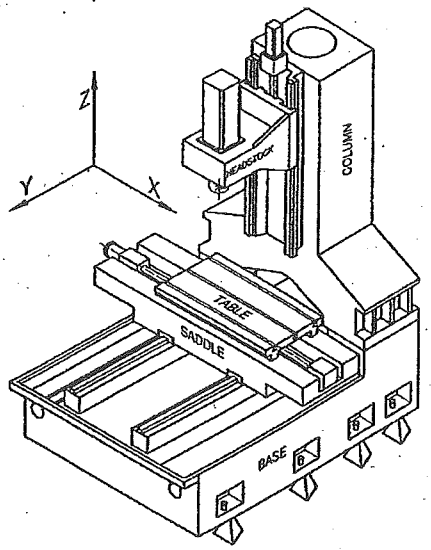
ACE MANUFACTURING SYSTEMS LTD

AMS
In Pursuit of Excellence

DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINIG CENTER

PRODUCT	:	VERTICAL MACHINING CENTER (FIXED COLUMN)
MODEL NAME	:	MCV-400
MACHINE NUMBER	:	M11-995
DATE OF TESTING	:	09-07-2014



(SANTOSH. R)

CHECKED BY
QUALITY ASSURANCE

CHECKED BY
ASSEMBLY - MECHANICAL

SIGN & DATE	BSR	IG	RP	07-02-2012	1 OF 5	DS-DSN-TC-PROD-01-R3
	PREPARED	CHECKED	APPROVED	ISSUE DATE	PAGE No.	DOCUMENT No.

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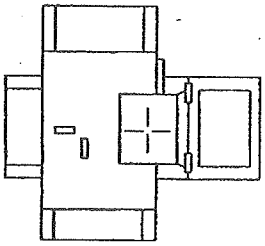
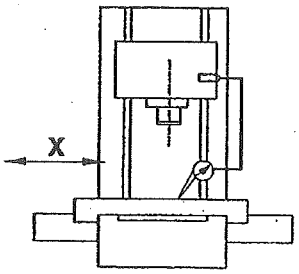
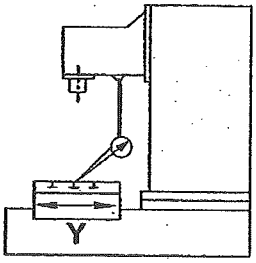
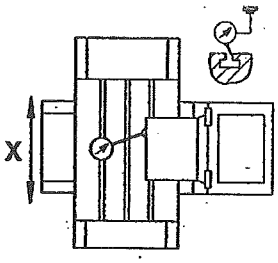
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DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINIG CENTER

All measurements in mm

SL. NO.	Description of the test	Figure	Tolerance	
			Permissible	Actual
G1	LEVELLING OF MACHINE X & Y AXES AT THEIR MIDDLE OF STROKE		0.03 / m ALONG X & Y AXES	LEVEL ALONG X-AXIS : 0.005 LEVEL ALONG Y-AXIS : 0.005
G2	PARALLELISM OF TABLE / PALLET SURFACE WITH RESPECT TO X-AXIS MOVEMENT		L ≤ 400 = 0.015 400 < L ≤ 630 = 0.025 630 < L ≤ 1000 = 0.040 1000 < L ≤ 2000 = 0.060 L= MOVEMENT OF THE TABLE / PALLET	TABLE : 0.011 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G3	PARALLELISM OF TABLE / PALLET SURFACE WITH RESPECT TO Y-AXIS MOVEMENT		L ≤ 400 = 0.015 400 < L ≤ 630 = 0.025 630 < L ≤ 1000 = 0.040 1000 < L ≤ 2000 = 0.060 L= MOVEMENT OF THE TABLE / PALLET	TABLE : 0.012 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G4	PARALLELISM OF REFERENCE T-SLOT TO X-AXIS MOVEMENT		L ≤ 400 = 0.015 400 < L ≤ 630 = 0.025 630 < L ≤ 1000 = 0.040 1000 < L ≤ 2000 = 0.060 L= MOVEMENT OF THE TABLE / PALLET	TABLE : 0.010 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)

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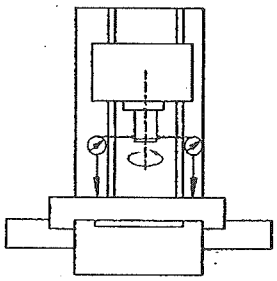
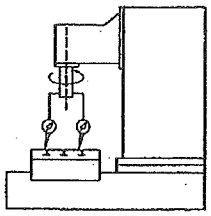
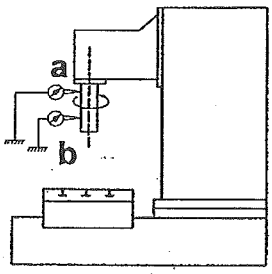
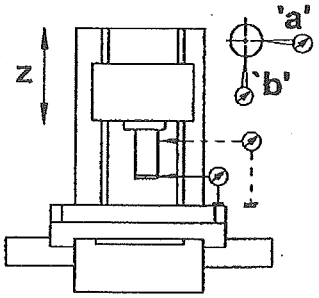
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DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINIG CENTER

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance Permissible	Actual
G5	SQUARENESS OF SPINDLE AXIS TO TABLE SURFACE IN XZ PLANE. X & Y AXES AT THEIR MID STROKE		0.020 FOR 300mm SWING DIAMETER	TABLE : 0.011 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G6	SQUARENESS OF SPINDLE AXIS TO TABLE SURFACE IN YZ PLANE. X & Y AXES AT THEIR MID STROKE		0.020 FOR 300mm SWING DIAMETER	TABLE : 0.012 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G7	RUN OUT OF SPINDLE TAPER a) NEAR SPINDLE NOSE b) AT A DISTANCE OF 300 FROM SPINDLE NOSE		a. 0.01 NEAR SPINDLE NOSE b. 0.02 AT A DISTANCE OF 300mm	a. 0.002 b. 0.012
G8	PARALLELISM OF SPINDLE AXIS & Z-AXIS IN a) XZ PLANE b) YZ PLANE TO BE CHECKED AT MEAN POINT OF RUNOUT		a. 0.02 / 300 b. 0.02 / 300	a. 0.010 b. 0.012

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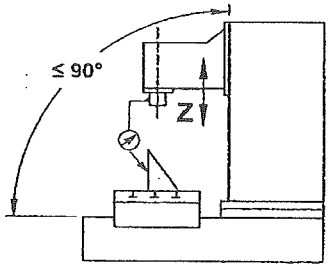
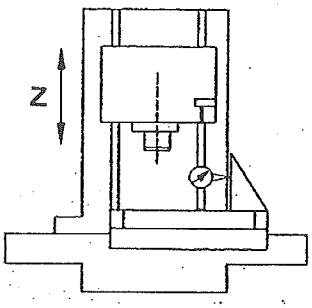
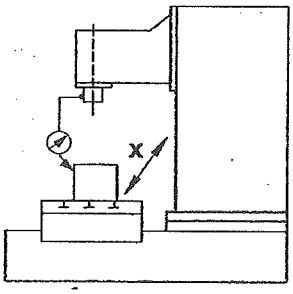
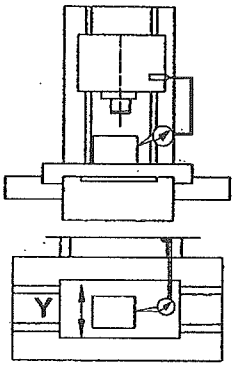
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DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINIG CENTER

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance: Permissible	Actual
G9	SQUARENESS BETWEEN TABLE SURFACE & Z-AXIS MOVEMENT IN YZ PLANE		0.015 / 300	TABLE : 0.010 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G10	SQUARENESS BETWEEN TABLE SURFACE & Z-AXIS MOVEMENT IN XZ PLANE		0.015 / 300	TABLE : 0.011 PALLET 1 : PALLET 2 : (FOR APC MACHINE ONLY)
G11	STRAIGHTNESS OF X-AXIS MOVEMENT IN XY - PLANE		0.010 / 300	0.002
G12	STRAIGHTNESS OF Y-AXIS MOVEMENT IN XY - PLANE		0.010 / 300	0.002

SIGN & DATE	BSR	IG	RP	07-02-2012	4 OF 5	DS-DSN-TC-PROD-01-R3
	PREPARED	CHECKED	APPROVED	ISSUE DATE	PAGE No.	DOCUMENT No.

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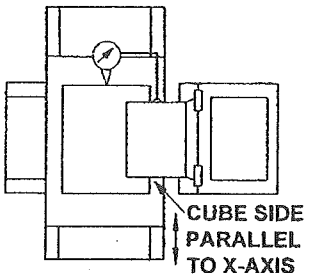
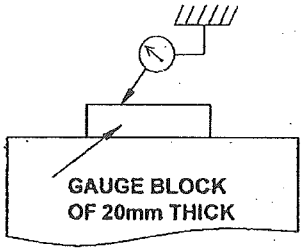
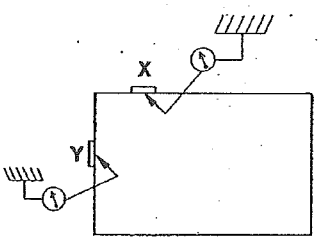
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DESIGN VERIFICATION / VALIDATION PLAN CHART

TITLE: GEOMETRICAL ACCURACY TEST CHART FOR VERTICAL MACHINIG CENTER

SL. NO.	Description of the test	Figure	All measurements in mm Tolerance: Permissible	Actual
G13	SQUARENESS OF Y - AXIS MOVEMENT WITH RESPECT TO X - AXIS MOVEMENT	MEASURE READING ALONG 'Y' 	0.015 / 300	0.010
G14	VARIATION BETWEEN THE PALLET HEIGHT (FOR APC MACHINE ONLY)		0.030	NA
G15	PALLET REPEATABILITY IN X & Y AXES WHEN LOADED ON THE MACHINE FROM THE PALLET STATION FOR SAME PALLET (FOR APC MACHINE ONLY)		0.020 IN X & Y AXES DIRECTION	PALLET 1 : NA PALLET 2 : (FOR APC MACHINE ONLY)

SIGN & DATE	BSR	IG	RP	07-02-2012	5 OF 5	DS-DSN-TC-PROD-01-R3
	PREPARED	CHECKED	APPROVED	ISSUE DATE	PAGE No.	DOCUMENT No.

ACE MANUFACTURING SYSTEMS LTD.

STANDARD TEST REPORT

1) HEAVY CUT REPORT :

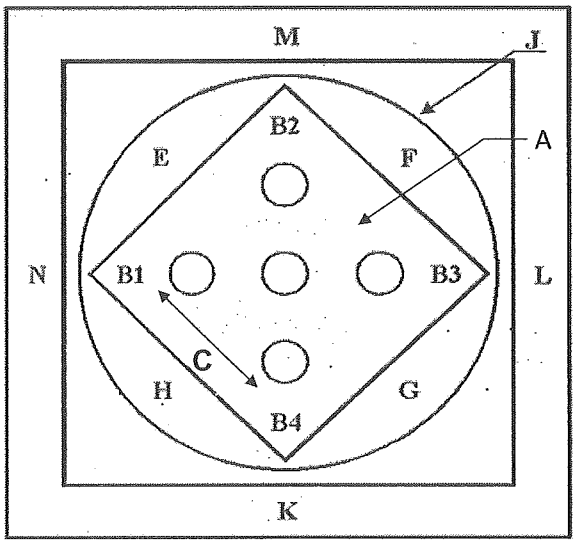
MODEL	: MCV-400	BASE SPEED OF MOTOR	:1500	rpm	MATERIAL	: CI (Gr-25)	
M/C NO.	: M11-995	SPINDLE SPEED RANGE	:60-6000	rpm	CUTTER Ø	: 80	
TYPE OF MOTOR	: βi 6/10000	MOTOR SPEED RANGE	:90-9000	rpm	WIDTH OF CUT	: 55mm	
PULLEY RATIO	: 1:1.5	MOTOR CAPACITY	:7.5/5.5	kW	C.S(V)	: 251.36	

Spindle speed in rpm	Feed rate mm/ins.	Feed rate mm/rev.	Feed rate mm/min.	Depth Of Cut in mm	M R R cc/min.	'X' Axis % as per CRT	Spl.load % as per CRT	Remarks
1000	0.15	0.90	900	2.5	124	22	48	SMOOTH CUTTING
1000	0.15	0.90	900	4.5	225	36	99	SMOOTH CUTTING

RIGID TAPPING : M8×1.25 RIGID TAPPING TEST CONDUCTED, CHECKED WITH GAUGE & FOUND OK.

2) NAS TEST REPORT :

No.	Description of Test	Test Applied to	Permissible deviation	Actual Readings in mm
1	Flatness of Surface	A	0.015	0.007
2	Parallelism of Surfaces	N-L	0.020	0.006
		M-K	0.020	0.012
		H-F	0.030	0.003
		G-E	0.030	0.010
3	Squareness of side faces	K-L	0.020	0.007
		L-M	0.020	0.014
		M-N	0.020	0.005
		N-K	0.020	0.008
		E-F	0.030	0.011
		F-G	0.030	0.007
		G-H	0.030	0.008
		H-E	0.030	0.005
4	Circularity of Machining	J	0.040	0.011
5	Linear pitch in ' X ' axis	B1-B3 (100)	0.025	-0.001
	Linear pitch in ' Y ' axis	B2-B4 (100)	0.025	-0.006
	Diagonal pitch	C(70.71)	0.035	0.001
6	Bore size variation		0.020	
	Hole No.	Max.	Min.	Ovality
	B1=	28.548	28.543	0.005
	B2=	28.547	28.544	0.003
	B3=	28.547	28.543	0.004
	B4=	28.546	28.542	0.004
7	Straightness	K	0.010	0.008
		L	0.010	0.006
		M	0.010	0.008
		N	0.010	0.003
		E	0.015	0.008
		F	0.015	0.007
		G	0.015	0.008
		H	0.015	0.003
8	Step difference in axis change over		0.010	0.003



SINGLE AXIS CONTROL K=L=M=N =	248.10 mm
DOUBLE AXIS CONTROL E=F=G=H =	157.00 mm
Ø OF CIRCLE " J " =	241.40 mm

Heavy cut Done By	NAS TEST DONE BY	Department incharge	QC / CMM
For KESHAV MOGER	MANJUNATH SV	SHAMAN P	
Date: 28/6/2014	Date: 7/7/2014	Date: 8/7/2014	Date: 09/07/14

Ballbar diagnostics table

M11-995-XY-B

RENISHAW

Operator: Shivu.
Date: 2014-Jul-06 10:55:54

Machine: Quick check
QC20-W: 153J45, Last calibrated: 2013-12-05

Error	Magnitude		Independent circularity		Ranking
Backlash X	▸0.0	◂0.2µm	0.2µm	(1%)	(12)
Backlash Y	▴0.2	▾1.6µm	1.6µm	(10%)	(4)
Reversal spikes X	▸3.9	◂3.3µm	3.9µm	(23%)	(1)
Reversal spikes Y	▴-3.0	▾-3.6µm	3.6µm	(22%)	(2)
Lateral play X	▸-0.8	◂0.5µm	0.4µm	(3%)	(10)
Lateral play Y	▴-0.1	▾0.2µm	0.1µm	(1%)	(13)
Cyclic error X	↑0.5	↓0.5µm	0.5µm	(3%)	(9)
Cyclic error Y	↑0.8	↓0.6µm	0.8µm	(5%)	(8)
Servo mismatch	0.05ms		0.8µm	(5%)	(7)
Squareness	2.7µm/m		0.4µm	(2%)	(11)
Straightness X	-2.2µm		1.1µm	(7%)	(5)
Straightness Y	1.8µm		0.9µm	(5%)	(6)
Scaling mismatch	-4.8µm		2.4µm	(14%)	(3)
Scaling error X	43.3ppm				
Scaling error Y	59.4ppm				
Cyclic pitch X	18.0000mm				
Cyclic pitch Y	30.0000mm				
Calculated feedrate	1000.8mm/min				
Centre offset X	-2.7µm				
Centre offset Y	-0.9µm				
Positional tolerance	44.7µm				
Best fit radius	150.0077mm				
Circularity	13.0µm				

Test parameters

Radius	150.0000mm
Feedrate	1000.0mm/min
Start/End/Overshoot	0°/360°/180°
Run sequence	CW CCW

ISO 230-4:2005(E)

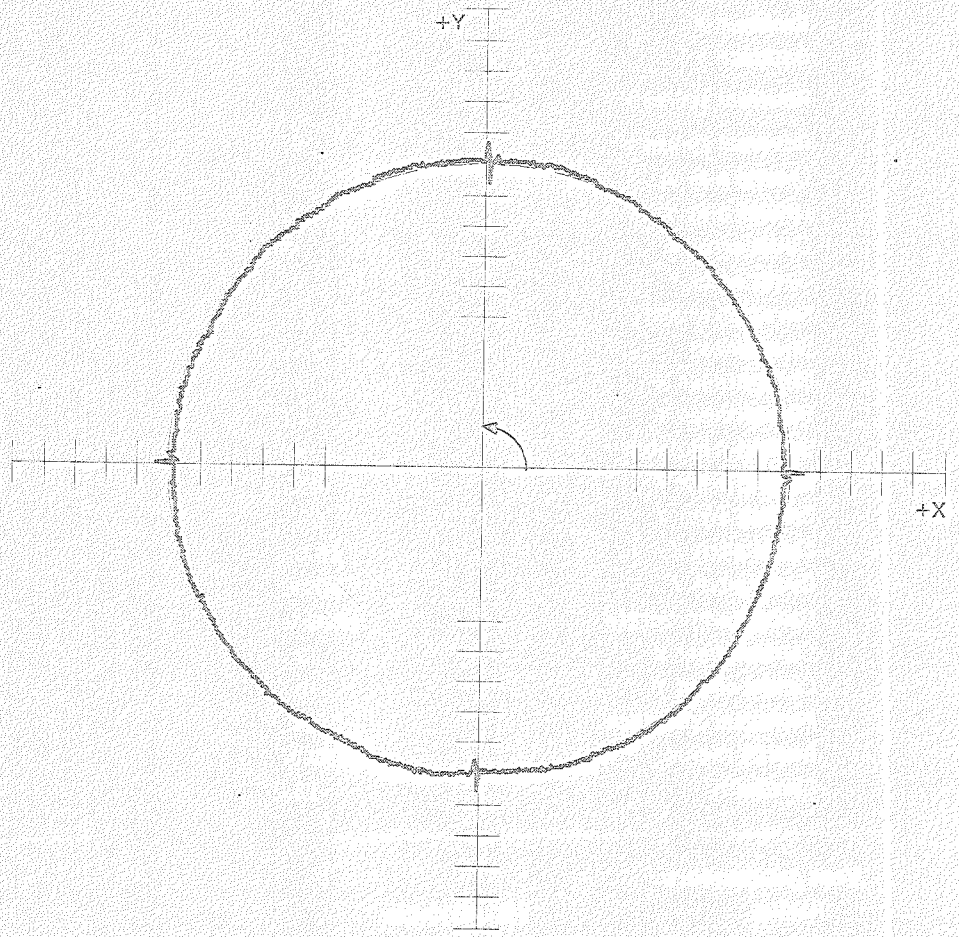
Circular deviation
M11-995-XY-B

Operator: Shivu.
Date: 2014-Jul-06 10:55:54

RENISHAW

Machine: Quick check
QC20-W: 153J45, Last calibrated: 2013-12-05

Circular deviation (CW)	
Value	12.8µm
Test parameters	
Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CW
Plane under test	XY
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 1

10.0µm/div

ISO 230-4:2005(E)

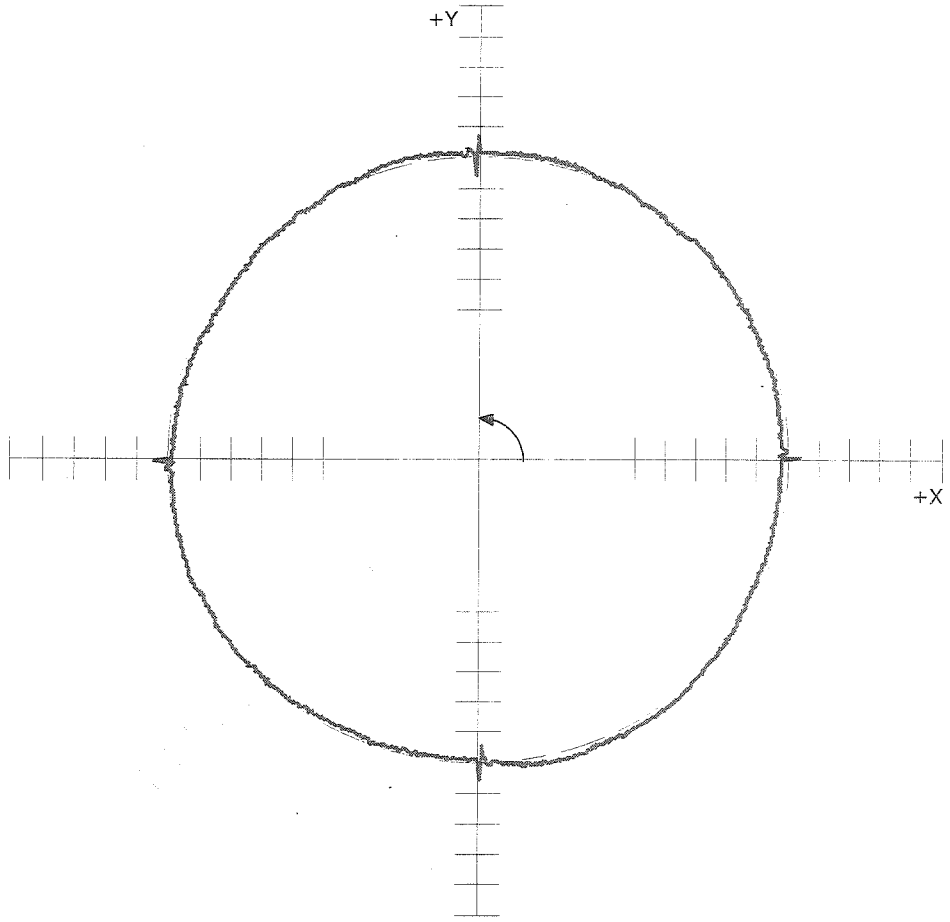
RENISHAW

Circular deviation
M11-995-XY-B

Operator: Shivu.
Date: 2014-Jul-06 10:55:54

Machine: Quick check
QC20-W: 153J45, Last calibrated: 2013-12-05

Circular deviation (CCW)	
Value	13.0µm
Test parameters	
Radius	150.0000mm
Sample rate	26.316Hz.
Feedrate	1000.0mm/min
Run direction	CCW
Plane under test	XY
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 2

10.0µm/div

Ballbar diagnostics table

M11-995-XZ-B

Operator: Shivu.
Date: 2014-Jul-06 11:15:23

RENISHAW

Machine: Quick check
QC20-W: 153345, Last calibrated: 2013-12-05

Error	Magnitude		Independent circularity		Ranking
Backlash Z	▵0.2	▿-0.1µm	0.2µm	(1%)	(12)
Backlash X	‹0.0	›0.1µm	0.1µm	(1%)	(13)
Reversal spikes Z	▵-1.9	▿-0.2µm	1.9µm	(12%)	(3)
Reversal spikes X	‹3.2	›2.3µm	3.2µm	(20%)	(2)
Lateral play Z	▵0.3	▿0.3µm	0.3µm	(2%)	(11)
Lateral play X	‹-1.0	›0.5µm	0.5µm	(3%)	(8)
Cyclic error Z	↑0.9	↓0.9µm	0.9µm	(6%)	(6)
Cyclic error X	↑1.0	↓1.1µm	1.0µm	(6%)	(5)
Servo mismatch	-0.07ms		1.1µm	(7%)	(4)
Squareness	5.7µm/m		0.9µm	(5%)	(7)
Straightness Z	0.6µm		0.3µm	(2%)	(10)
Straightness X	9.8µm		4.9µm	(31%)	(1)
Scaling mismatch	-1.0µm		0.5µm	(3%)	(9)
Scaling error Z	74.3ppm				
Scaling error X	77.5ppm				
Cyclic pitch Z	12.0000mm				
Cyclic pitch X	38.1000mm				
Calculated feedrate	1000.8mm/min				
Centre offset Z	-2.3µm				
Centre offset X	-0.3µm				
Positional tolerance	67.5µm				
Best fit radius	150.0114mm				
Circularity	10.5µm				

Test parameters

Radius	150.0000mm
Feedrate	1000.0mm/min
Start/End/Overshoot	90°/90°/180°
Run sequence	CW CCW

ISO 230-4:2005(E)

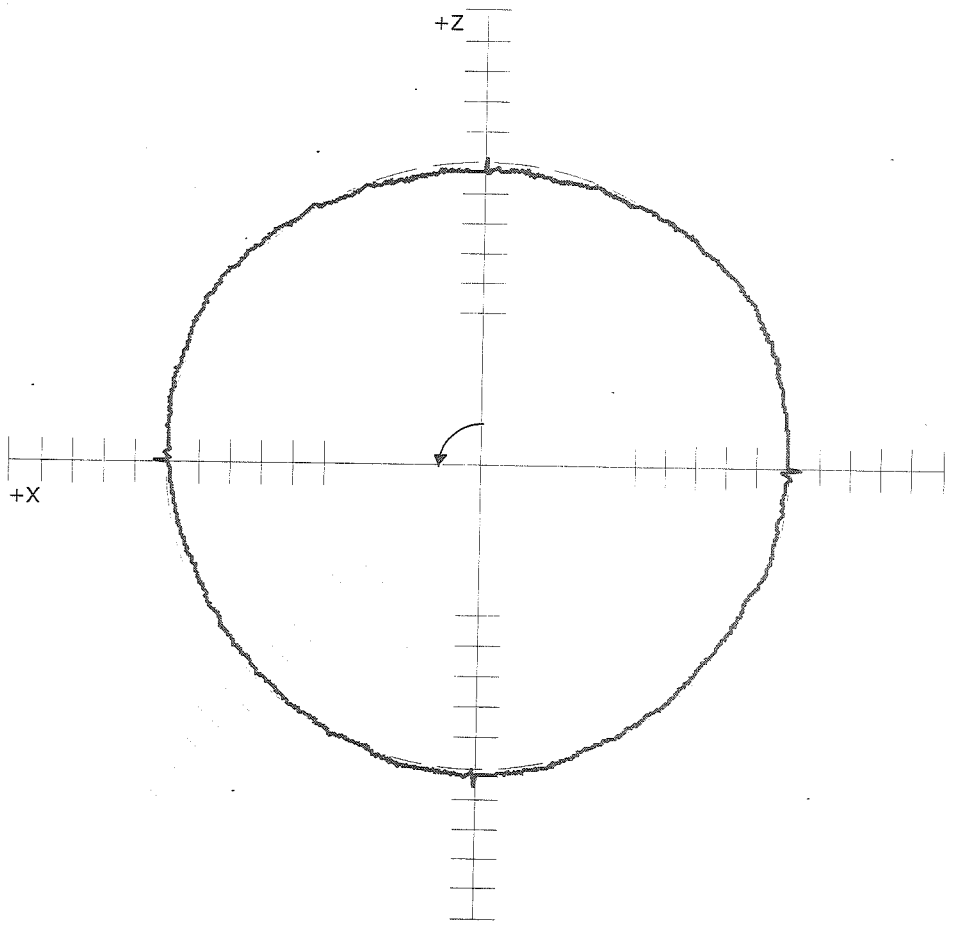
Circular deviation
M11-995-XZ-B

Operator: Shivu.
Date: 2014-Jul-06 11:15:23

RENISHAW

Machine: Quick check
QC20-W: 153J45, Last calibrated: 2013-12-05

Circular deviation (CW)	
Value	9.4µm
Test parameters	
Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CW
Plane under test	ZX
Test position	
Start angle	90°
End angle	90°
Overshoot angle	180°



Run 1

10.0µm/div

ISO 230-4:2005(E)

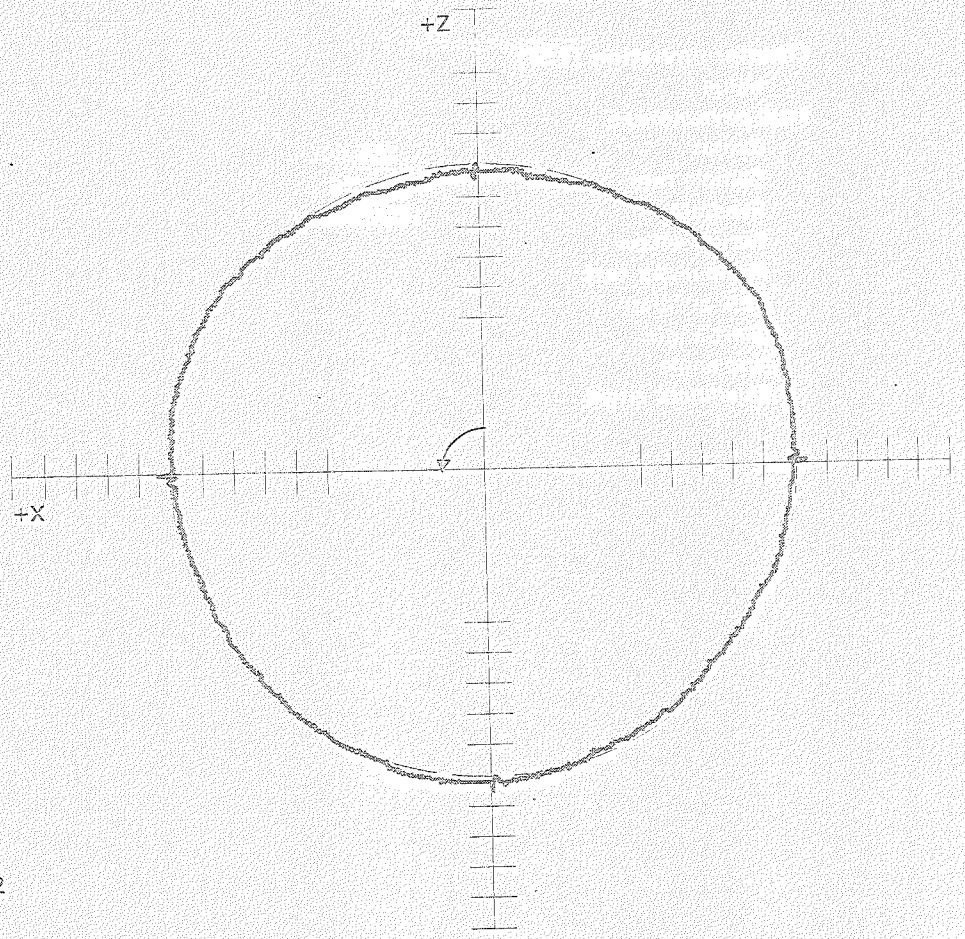
Circular deviation
M11-995-XZ-B

Operator: Shivu.
Date: 2014-Jul-06 11:15:23

RENISHAW

Machine: Quick check
QC20-W: 153J45, Last calibrated: 2013-12-05

Circular deviation (CCW)	
Value	9.4µm
Test parameters	
Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CCW
Plane under test	ZX
Test position	
Start angle	90°
End angle	90°
Overshoot angle	180°



Run 2

10.0µm/div

Ballbar diagnostics table

M11-995-YZ-A

Operator: Shivu.

Date: 2014-Jul-06 11:23:56

RENISHAW

Machine: Quick check

QC20-W: 153J45, Last calibrated: 2013-12-05

Error	Magnitude		Independent circularity		Ranking
Backlash Y	▸0.3	◂1.9µm	1.9µm	(9%)	(4)
Backlash Z	▴-0.1	▾-0.4µm	0.4µm	(2%)	(10)
Reversal spikes Y	▸-5.0	◂-5.2µm	5.2µm	(25%)	(1)
Reversal spikes Z	▴-1.5	▾1.4µm	1.5µm	(8%)	(5)
Lateral play Y	▸0.1	◂0.5µm	0.4µm	(2%)	(11)
Lateral play Z	▴-0.4	▾1.9µm	1.1µm	(5%)	(7)
Cyclic error Y	↑1.2	↓0.9µm	1.2µm	(6%)	(6)
Cyclic error Z	↑0.9	↓0.9µm	0.8µm	(4%)	(9)
Servo mismatch	0.02ms		0.3µm	(2%)	(12)
Squareness	16.6µm/m		2.5µm	(12%)	(3)
Straightness Y	8.4µm		4.2µm	(21%)	(2)
Straightness Z	-0.2µm		0.1µm	(0%)	(13)
Scaling mismatch	1.7µm		0.8µm	(4%)	(8)
Scaling error Y	81.4ppm				
Scaling error Z	75.9ppm				
Cyclic pitch Y	24.0000mm				
Cyclic pitch Z	31.7500mm				
Calculated feedrate	1000.8mm/min				
Centre offset Y	2.1µm				
Centre offset Z	15.1µm				
Positional tolerance	69.8µm				
Best fit radius	150.0118mm				
Circularity	19.4µm				

Test parameters

Radius	150.0000mm
Feedrate	1000.0mm/min
Start/End/Overshoot	0°/360°/180°
Run sequence	CW CCW

ISO 230-4:2005(E)

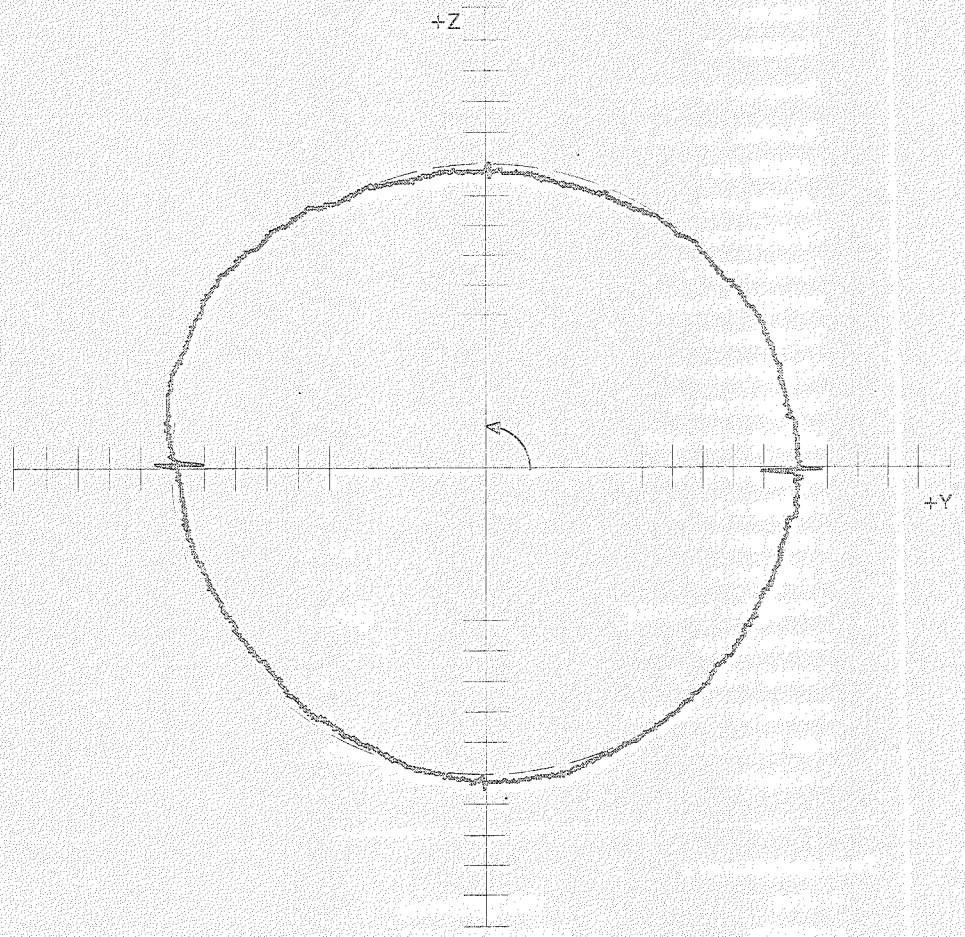
RENISHAW

Circular deviation
M11-995-YZ-A

Operator: Shivu.
Date: 2014-Jul-06 11:23:56

Machine: Quick check
QC20-W: 153J45, Last calibrated: 2013-12-05

Circular deviation (CW)	
Value	18.9µm
Test parameters	
Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CW
Plane under test	YZ
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 1

10.0µm/div

ISO 230-4:2005(E)

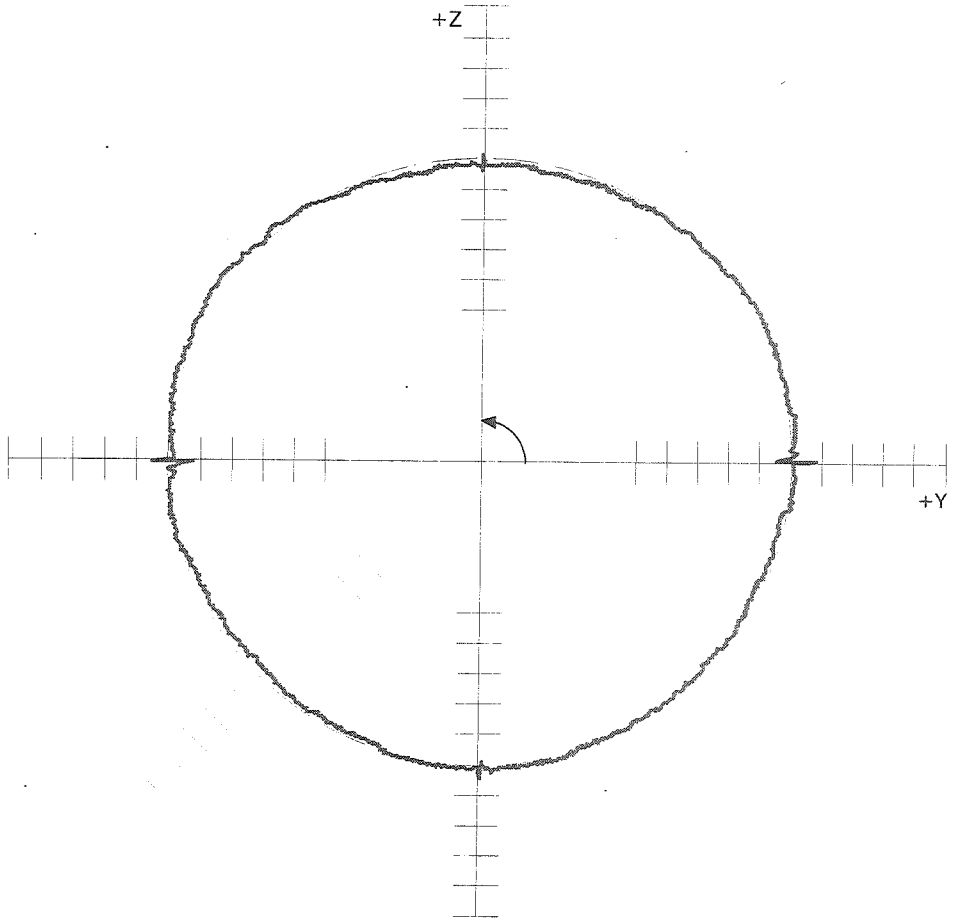
Circular deviation
M11-995-YZ-A

Operator: Shivu.
Date: 2014-Jul-06 11:23:56

RENISHAW

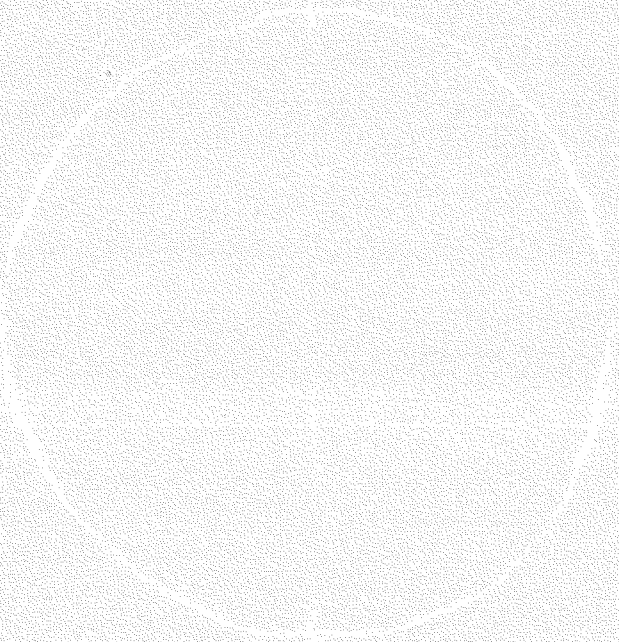
Machine: Quick check
QC20-W: 153J45, Last calibrated: 2013-12-05

Circular deviation (CCW)	
Value	16.1µm
Test parameters	
Radius	150.0000mm
Sample rate	26.316Hz
Feedrate	1000.0mm/min
Run direction	CCW
Plane under test	YZ
Test position	
Start angle	0°
End angle	360°
Overshoot angle	180°



Run 2

10.0µm/div



ACE MANUFACTURING SYSTEM LTD.

Machine No :-	M11-995.	Date :-	10:30 Jul 03 2014
Machine Modle :-	MCV-400.	Cond by:-	QAS/SUR.

AXIS CALIBRATION PITCH ERROR REPORT

X				Y				Z			
SL NO				SLNO				SL NO			
100 (X)	0			200 (Y)	0			300 (Z)	0		
101	1	141	0	201	0	241	2	301	0	341	1
102	0	142	0	202	0	242	1	302	0	342	1
103	0	143	0	203	0	243	1	303	0	343	0
104	0	144	0	204	-1	244	1	304	0	344	0
105	0	145	0	205	0	245	2	305	1	345	1
106	1	146	1	206	0	246	0	306	1	346	2
107	0	147	0	207	0	247	0	307	0	347	1
108	0	148	0	208	0	248	0	308	0	348	0
109	0	149	0	209	1	249	0	309	1	349	0
110	1	150	0	210	0	250	0	310	0	350	1
111	0	151	0	211	0	251	0	311	0	351	0
112	0	152	0	212	1	252	0	312	1	352	0
113	1	153	0	213	0	253	0	313	1	353	0
114	1	154	0	214	0	254	0	314	0	354	0
115	0	155	0	215	1	255	0	315	1	355	0
116	0	156	0	216	0	256	0	316	1	356	0
117	1	157	0	217	0	257	0	317	0	357	0
118	0	158	0	218	0	258	0	318	1	358	0
119	0	159	0	219	0	259	0	319	1	359	0
120	0	160	1	220	0	260	0	320	1	360	0
121	0	161	0	221	0	261	0	321	1	361	0
122	1	162	0	222	0	262	0	322	0	362	0
123	0	163	0	223	0	263	0	323	2	363	0
124	0	164	0	224	0	264	0	324	1	364	0
125	0	165	0	225	0	265	0	325	0	365	0
126	0	166	0	226	1	266	0	326	1	366	0
127	0	167	0	227	0	267	0	327	0	367	0
128	0	168	0	228	0	268	0	328	1	368	0
129	-1	169	0	229	1	269	0	329	2	369	0
130	0	170	0	230	1	270	0	330	1	370	0
131	0	171	0	231	1	271	0	331	1	371	0
132	0	172	0	232	0	272	0	332	0	372	0
133	0	173	0	233	1	273	0	333	0	373	0
134	0	174	0	234	0	274	0	334	1	374	0
135	-1	175	0	235	1	275	0	335	1	375	0
136	0	176	0	236	1	276	0	336	0	376	0
137	0	177	0	237	1	277	0	337	1	377	0
138	0	178	0	238	2	278	0	338	1	378	0
139	0	179	0	239	0	279	0	339	1	379	0
140	0	180	0	240	0	280	0	340	1	380	0

SL NO	PARARMETER NO	DISCRIPTION	X	Y	Z
1	3620	P COMP REF NO	160	245	350
2	3621	P COMP MIN NO	100	200	300
3	3622	P COMP MAX NO	165	250	355
4	3623	PCOMP MULTIPLICATION	10	10	10
5	3624	P COMP INTERVAL	10	10	10

ACE MANUFACTURING SYSTEM LTD.

AMS
In Pursuit of Excellence

M/c NO :-	M11-995.	COND BY:- QAS/SUR.
Machine Modle :-	MCV-400.	DATE:- 06-Jul-14

AXIS CALIBRATION BALL BAR PARAMETERS

SERVO PARAMETERS RELATED TO CIRCULARITY FOR FANUC SYSTEM

SL No.	PARAMETER No.	TYPICAL VALUE	DESCRIPTION	X	Y	Z	REMARKS
1	1851	###	BACKLASH IN FEED	30	65	20	
2	1852	###	BACKLASH IN RAPID	20	45	10	
3	1825	6000	SERVO LOOP GAIN	5000	5000	5000	
4	1826	500	IN POSITION WIDTH	500	500	500	
5	2021	256	LOAD INERTIA	192	128	161	******
6	2048	600 (###)	BACK ACCEL COMPENSATION VALUE	25	50	25	
7	2094	600 (###)	BACK ACCEL COMPENSATION VALUE	25	50	25	
8	2066	10 TO -20	250 MICRO SEC ACC FEEDBACK	-10	-10	-10	******
9	2068	70 TO700	FEED FWD CO- EFFICIENT	100	100	100	
10	2069	50 TO 200	VELOCITY FEED FWD CO- EFFICIENT FOR RAPID	100	100	100	
11	2071	50 TO 200	TIME DURING WHICH BACK ACCELERATION IS EFFECTIVE	100	100	100	
12	2145	100	VELOCITY FEED FWD CO- EFFICIENT FOR CUTTING	100	100	100	SHOULD NOT BE CHANGED
13	2144	9900	LOOK AHEAD FFW COEFFICIENT FOR CUTTING	9900	9900	9900	
14	2092	9200	LOOK AHEAD FFW COEFFICIENT FOR RAPID	9200	9200	9200	
15	2082	100	STAGE 2 START / END PARAMETER	100	100	100	
16	2039	100	STAGE 2 ACCELERATION AMOUNT	300	300	300	
17	1622	NA	CUT TIME CONSTANT	64	64	64	Same value for 3 axis

NOTE:-

******	DON'T CHANGE THE VALUE
###	VARIES FOR ALL MACHINES

RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

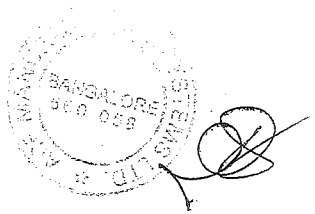
File: M11-995-X.RTL

Machine:MCV-400.
Date:11:23 Jul 03 2014
Axis:X.

Serial No:M11-995.
By:QAS/SUR.
Location:AMS.

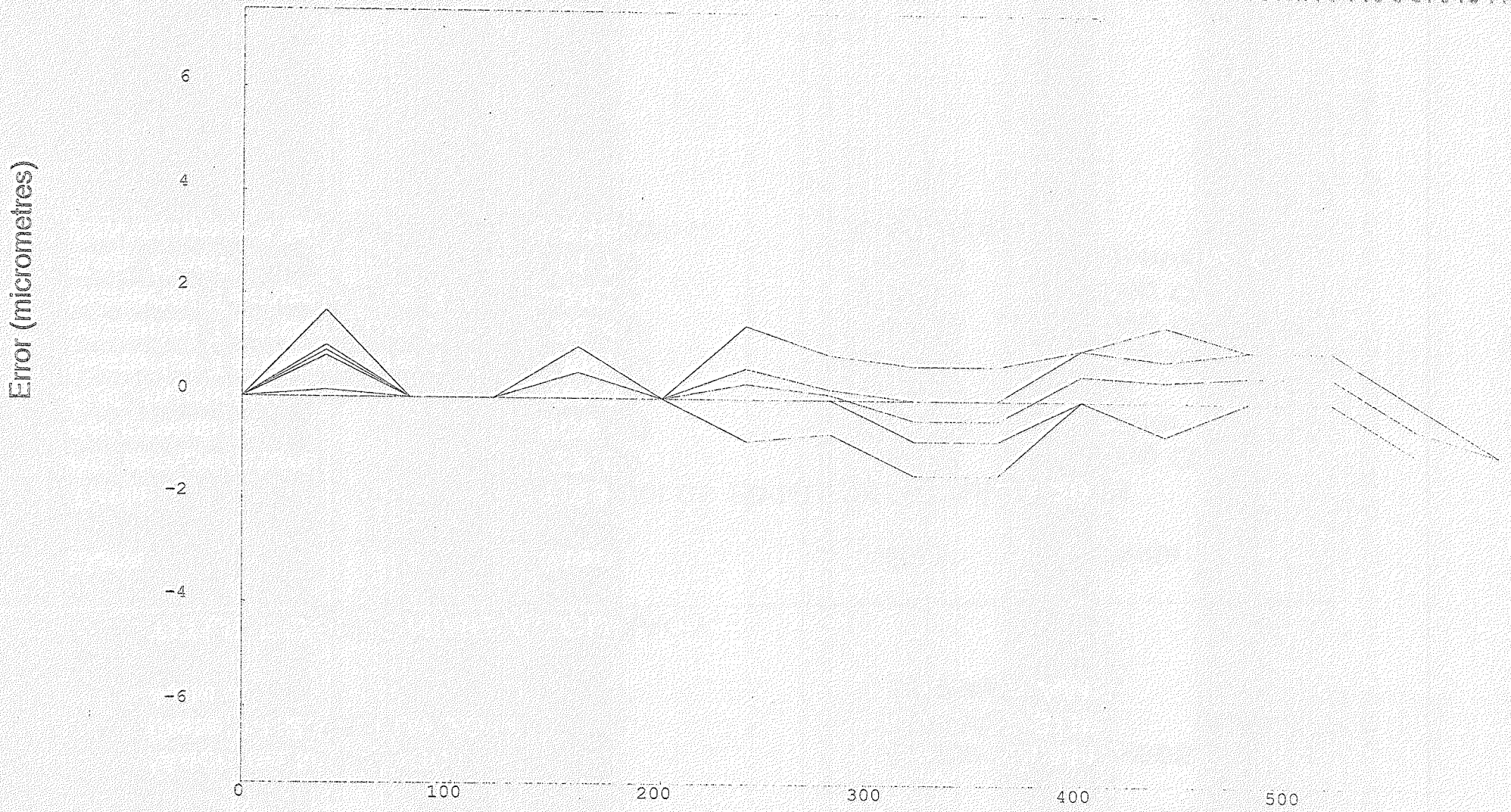
TITLE:POSITIONING ACCURACY.

Data	Start	End	Units
Time :	11:23 Jul 03 2014	11:33 Jul 03 2014	
Air temperature :	27.50	27.51	deg C
Air pressure :	908.00	908.00	mbar
Rel. humidity :	64.51	64.05	%
Material Temperature 1 :	--NC--	--NC--	deg C
Material Temperature 2 :	25.40	25.48	deg C
Material Temperature 3 :	--NC--	--NC--	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31640048	0.31640016	



VDI 3441

POSITIONING ACCURACY.



VDI 3441 - Linear

Machine:MCV-400.	Axis:X.		U max :	1.000
Serial No:M11-995.	Location:AMS.		P :	3.142
Date:11:23 Jul 03 2014	U mean :	0.525	Ps max :	1.643
By:QAS/SUR.	Ps mean :	0.522	Pa :	1.900

RENISHAW CALIBRATION INTERFEROMETER SYSTEM
ALL DATA

Machine:MCV-400. Serial No:M11-995.
Date:11:23 Jul 03 2014 By:QAS/SUR.
Axis:X. Location:AMS.
TITLE:POSITIONING ACCURACY. Filename: M11-995-X.RTL

Units of data values are micrometres

Target 1 (+) (0.0000 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 1 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 2 (+) (39.9990 millimetres)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 0.0
Target 2 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 3 (+) (79.9980 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 3 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 4 (+) (119.9970 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 4 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 5 (+) (159.9960 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 5 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 6 (+) (199.9950 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 6 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 7 (+) (239.9940 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 7 (-)
1) 1.0 2) 1.0 3) 1.0 4) 0.0
5) 0.0

Target 8 (+) (279.9930 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 8 (-)
1) 0.0 2) 0.0 3) 1.0 4) 0.0
5) 0.0

Target 9 (+) (319.9920 millimetres)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 9 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 10 (+) (359.9910 millimetres)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 10 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 11 (+) (399.9900 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 11 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 12 (+) (439.9890 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 12 ( )
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 0.0

Target 13 (+) (479.9880 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 13 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 14 (+) (519.9870 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 14 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 15 (+) (559.9860 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 15 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 16 (+) (599.9850 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 16 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

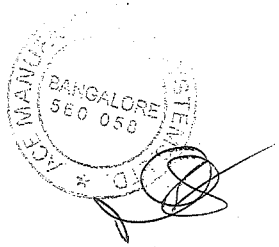
File: M11-995-Y.RTL

Machine:MCV-400.
Date:12:28 Jul 03 2014
Axis:Y.

Serial No:M11-995.
By:QAS/SUR.
Location:AMS.

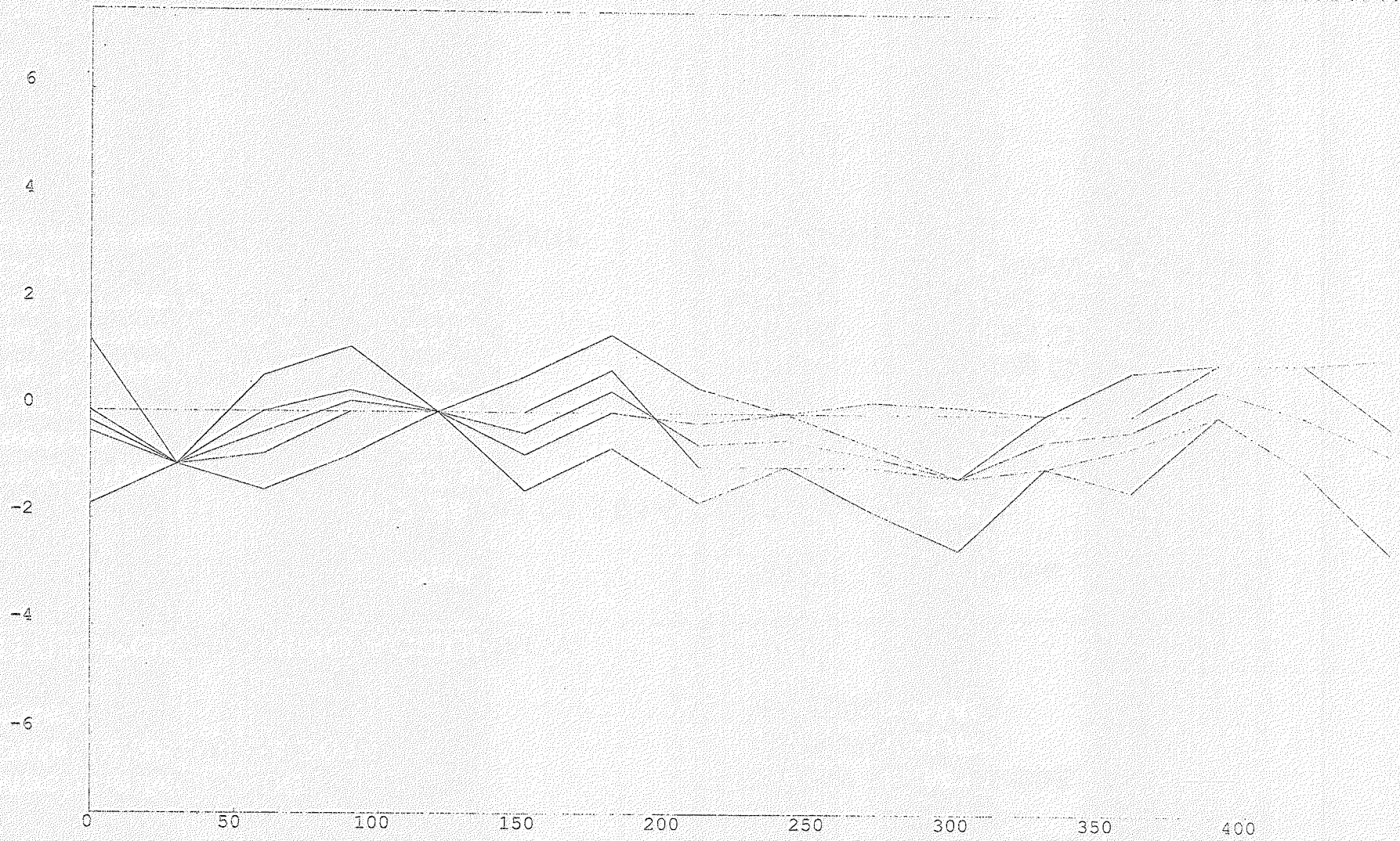
TITLE:POSITIONING ACCURACY.

Data	Start	End	Units
Time :	12:28 Jul 03 2014	13:13 Jul 03 2014	
Air temperature :	27.95	28.55	deg C
Air pressure :	907.20	906.60	mbar
Rel. humidity :	62.42	60.27	%
Material Temperature 1 :	--NC--	--NC--	deg C
Material Temperature 2 :	25.86	26.18	deg C
Material Temperature 3 :	--NC--	--NC--	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31639895	0.31639796	



VDI 3441

POSITIONING ACCURACY.



VDI 3441 - Linear

Machine:MCV-400.	Axis:Y.	U max : 2.000
Serial No:M11-995.	Location:AMS.	P : 4.012
Date:12:28 Jul 03 2014	U mean : 0.688	Ps max : 2.683
By:QAS/SUR.	Ps mean : 1.147	Pa : 1.700

RENISHAW CALIBRATION INTERFEROMETER SYSTEM
ALL DATA

Machine:MCV-400. Serial No:M11-995.
Date:12:28 Jul 03 2014 By:QAS/SUR.
Axis:Y. Location:AMS.
TITLE:POSITIONING ACCURACY. Filename: M11-995-Y.RTL

Units of data values are micrometres

Target 1 (+) (0.0000 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 1 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) -2.0

Target 2 (+) (29.9990 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 2 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 3 (+) (59.9980 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) 0.0
5) -1.0
Target 3 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 4 (+) (89.9970 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 4 (-)
1) 1.0 2) 0.0 3) 0.0 4) 1.0
5) 0.0

Target 5 (+) (119.9960 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 5 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 6 (+) (149.9950 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) 0.0
5) -1.0
Target 6 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 7 (+) (179.9940 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 7 (-)
1) 1.0 2) 1.0 3) 0.0 4) 1.0
5) 1.0

Target 8 (+) (209.9930 millimetres)
1) -1.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 8 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 9 (+) (239.9920 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 9 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 10 (+) (269.9910 millimetres)
1) -1.0 2) 0.0 3) 0.0 4) -1.0
5) -1.0
Target 10 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 11 (+) (299.9900 millimetres)
1) -2.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 11 (-)
1) -2.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 12 (+) (329.9890 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 12 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 13 (+) (359.9880 millimetres)
1) -1.0 2) -1.0 3) 0.0 4) 0.0
5) -1.0
Target 13 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 14 (+) (389.9870 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 14 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 15 (+) (419.9860 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 15 (-)
1) 1.0 2) 1.0 3) 1.0 4) 1.0
5) 1.0

Target 16 (+) (449.9850 millimetres)
1) -2.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 16 (-)
1) -1.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

RENISHAW CALIBRATION INTERFEROMETER SYSTEM

ENVIRONMENT DATA PRINT-OUT

Mode : Linear displacement

File: M11-995-Z.RTL

Machine:MCV-400.
Date:14:18 Jul 03 2014
Axis:Z.

Serial No:M11-995.
By:QAS/SUR.
Location:AMS.

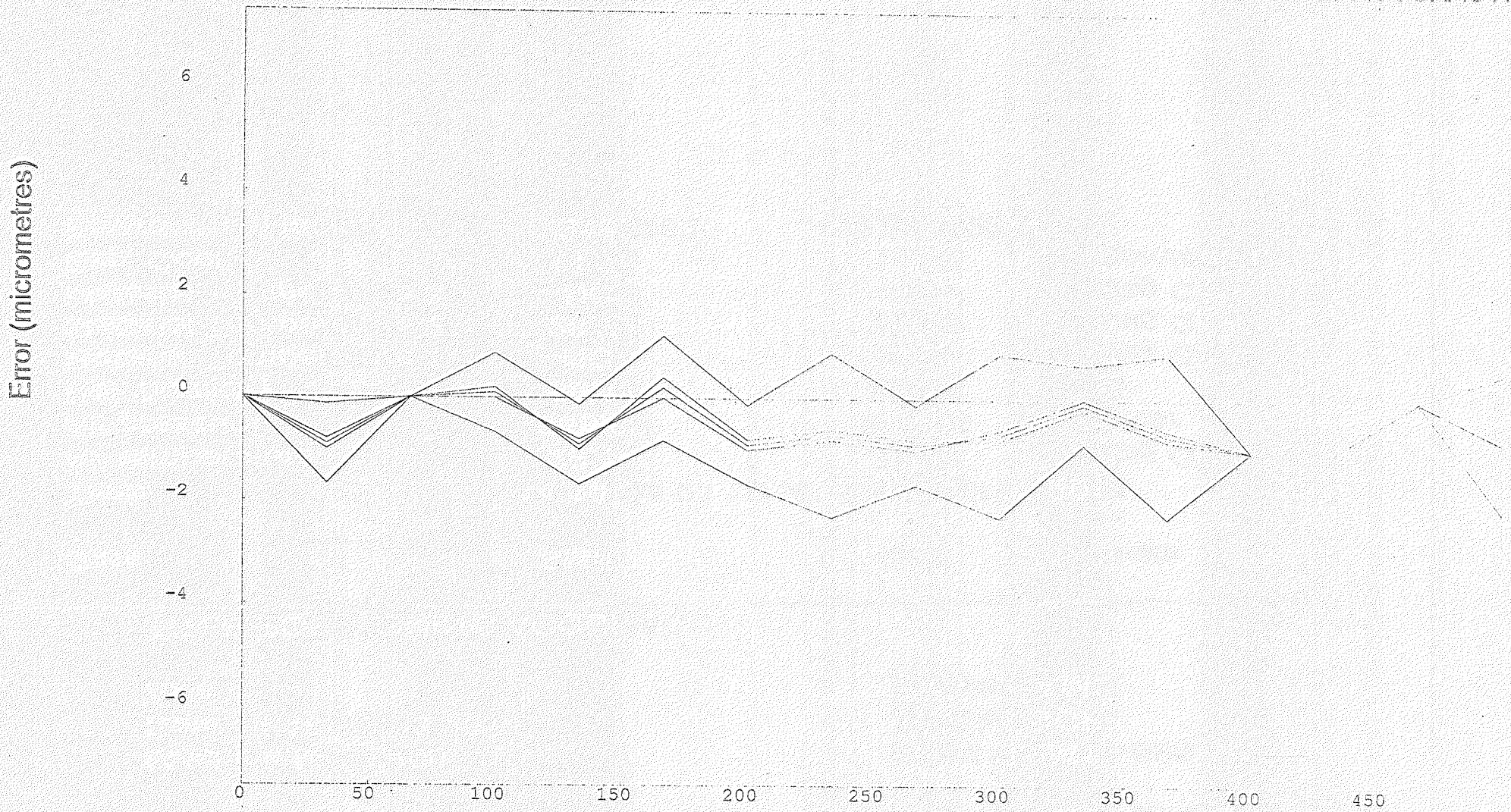
TITLE:POSITIONING ACCURACY.

Data	Start	End	Units
Time :	14:19 Jul 03 2014	14:29 Jul 03 2014	
Air temperature :	29.14	29.27	deg C
Air pressure :	905.60	905.60	mbar
Rel. humidity :	59.11	58.10	%
Material Temperature 1 :	--NC--	--NC--	deg C
Material Temperature 2 :	26.72	26.80	deg C
Material Temperature 3 :	--NC--	--NC--	deg C
Exp. Coeff :	11.70	11.70	ppm/deg C
Environment factor :	0.31639620	0.31639594	



VDI 3441

POSITIONING ACCURACY.



VDI 3441 - Linear

Machine:MCV-400.	Axis:Z.		U max :	0.400
Serial No:M11-995.	Location:AMS.		P :	3.514
Date:14:18 Jul 03 2014	U mean :	0.137	Ps max :	2.985
By:QAS/SUR.	Ps mean :	1.333	Pa :	1.200

RENISHAW CALIBRATION INTERFEROMETER SYSTEM
ALL DATA

Machine:MCV-400. Serial No:M11-995.
Date:14:18 Jul 03 2014 By:QAS/SUR.
Axis:Z. Location:AMS.
TITLE:POSITIONING ACCURACY. Filename: M11-995-Z.RTL

Units of data values are micrometres

Target 1 (+) (0.0000 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 1 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 2 (+) (33.3330 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 2 (-)
1) -1.0 2) 0.0 3) 1.0 4) -1.0
5) -1.0

Target 3 (+) (66.6660 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 3 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 4 (+) (99.9990 millimetres)
1) 1.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 4 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 5 (+) (133.3320 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 5 (-)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 6 (+) (166.6650 millimetres)
1) 1.0 2) 1.0 3) 0.0 4) 0.0
5) 0.0
Target 6 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 7 (+) (199.9980 millimetres)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 7 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 8 (+) (233.3310 millimetres)
1) 0.0 2) -1.0 3) 0.0 4) -1.0
5) -1.0

Target 8 (-)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 9 (+) (266.6640 millimetres)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 9 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 10 (+) (299.9970 millimetres)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 10 (-)
1) 0.0 2) 0.0 3) -1.0 4) -1.0
5) -1.0

Target 11 (+) (333.3300 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) -1.0
Target 11 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 12 (+) (366.6630 millimetres)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 12 (-)
1) 0.0 2) 0.0 3) -1.0 4) -1.0
5) -1.0

Target 13 (+) (399.9960 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 13 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 14 (+) (433.3290 millimetres)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 14 (-)
1) -1.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

Target 15 (+) (466.6620 millimetres)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0
Target 15 (-)
1) 0.0 2) 0.0 3) 0.0 4) 0.0
5) 0.0

Target 16 (+) (499.9950 millimetres)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0
Target 16 (-)
1) 0.0 2) -1.0 3) -1.0 4) -1.0
5) -1.0

ACE MANUFACTURING SYSTEMS LTD

SPINDLE BALANCING REPORT

AMS® In Pursuit of Excellence

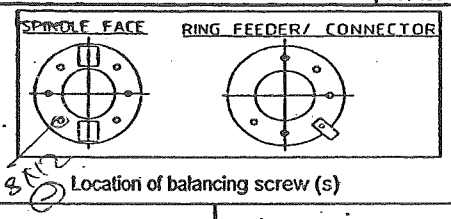
Machine Number : M11-995

Spindle Serial Number : 141458

Spindle Type : BT40

Spindle Speed (Max.) : 6000 rpm

Note : Refer TM-QA-09 for measurement procedure.



Sl No.	Speed in RPM	Velocity of vibration (mm/sec) before Balancing Along X-axis	After Balancing along X-axis <8000 rpm : 0.8 mm/sec max. > 8000 rpm : 1 mm/sec max.	After Balancing along Y-axis <8000 rpm : 0.8 mm/sec max. > 8000 rpm : 1 mm/sec max.
1	500			
2	1000	0.004	0.003	0.02
3	1500	0.015	0.006	
4	2000	0.028	0.023	0.007
5	2500	0.053	0.026	
6	3000	0.120	0.049	0.011
7	3500	0.109	0.043	
8	4000	0.136	0.078	0.045
9	4500	0.221	0.109	
10	5000	0.751 -190°	0.651	0.022
11	5500	0.981 -152°	0.483	
12	6000	0.898 -98°	0.600	0.067
13	6500			
14	7000			
15	7500			
16	8000			
17	8500			
18	9000			
19	9500			
20	10000			
21	10500			
22	11000			
23	11500			
24	12000			
25	12500			
26	13000			
27	13500			
28	14000			
29	14500			
30	15000			

Spindle max. vibration mm / sec :

Correction in gms :

REMARK IF ANY: Nil

Date: 02/07/14

Checked By

APPROVED BY

ACE MANUFACTURING SYSTEMS LTD

SPINDLE ASSEMBLY - INSPECTION REPORT

AMS — In Pursuit of Excellence

Machine Number : MH-995	Spindle Unit Number: 14-1458
Spindle Drive : ~~ID~~ / ID-CTS / DD / DD-CTS	Spindle Diameter : 70 mm
Pull stud type : DIN ~~MAS~~	Max. Spindle Speed : 6000 RPM
Bearing Make & Specification : FAG B7014E.TVPP4-UL	Draw bar Sl. No : SD70
Spindle Number: ACW-452	Cartridge Number : 217
Spindle Nose Taper : ISO 30 ~~40~~ 50	HSK A50 A63 A100

Sl. No.	Description	Recommended	Actual
1	Locknut Tightening Torque	Ref.DS-PDN-MEC-SA-08	115Nm
2	Spindle Bearing Preload	Ref.DS-PDN-MEC-SA-07	-0.014mm
3	Spindle Run out on taper	0.002 mm (*)	0.002mm
4	Mandrel Run out near the spindle nose	0.003 mm (*)	0.002mm
5	Mandrel Run out at 4 positions @ 90° at a distance of 300 mm (ISO 40/50, HSK 63/100), 250mm (ISO 30, HSK 50) from spindle nose	10 micron (*)	1. 5 2. 5 3. 4 4. 4
6	Spindle stabilization temperature on Test rig	Up to 15° C above ambient	33°C-23°C=10°C
7	Check for abnormal / unhealthy noise	Abnormal / unhealthy noise not acceptable	OK
8	Pulley ratio (For ID machines)	Refer Work order	1:1.5 Ratio
9	Spindle pulley Run out (For ID machines)	0.050 mm (*)	0.050mm
10	Spindle connector Run out (For DD machines)	0.010 mm (*)	—
11	Disc spring specification (Single / double stack)	Ref.DS-DSN-PROD-01	φ40x φ0.8x3mm
12	Number of disc springs used	Ref.DS-DSN-PROD-01	6 Only
13	Disc spring Compression, Free length & Compressed length	Ref.DS-DSN-PROD-01	9.0 - 6.9 = 2.3mm
14	Tool clamping force	Ref.DS-DSN-PROD-01	Showing
15	ID-CTS Deublin Run out in Assembly condition (For CTS Spindle)	0.030 mm	Deublin Sl. No. : — Run out :

(*) To be measured after temperature stabilization of spindle at max. speed.

REMARK IF ANY:

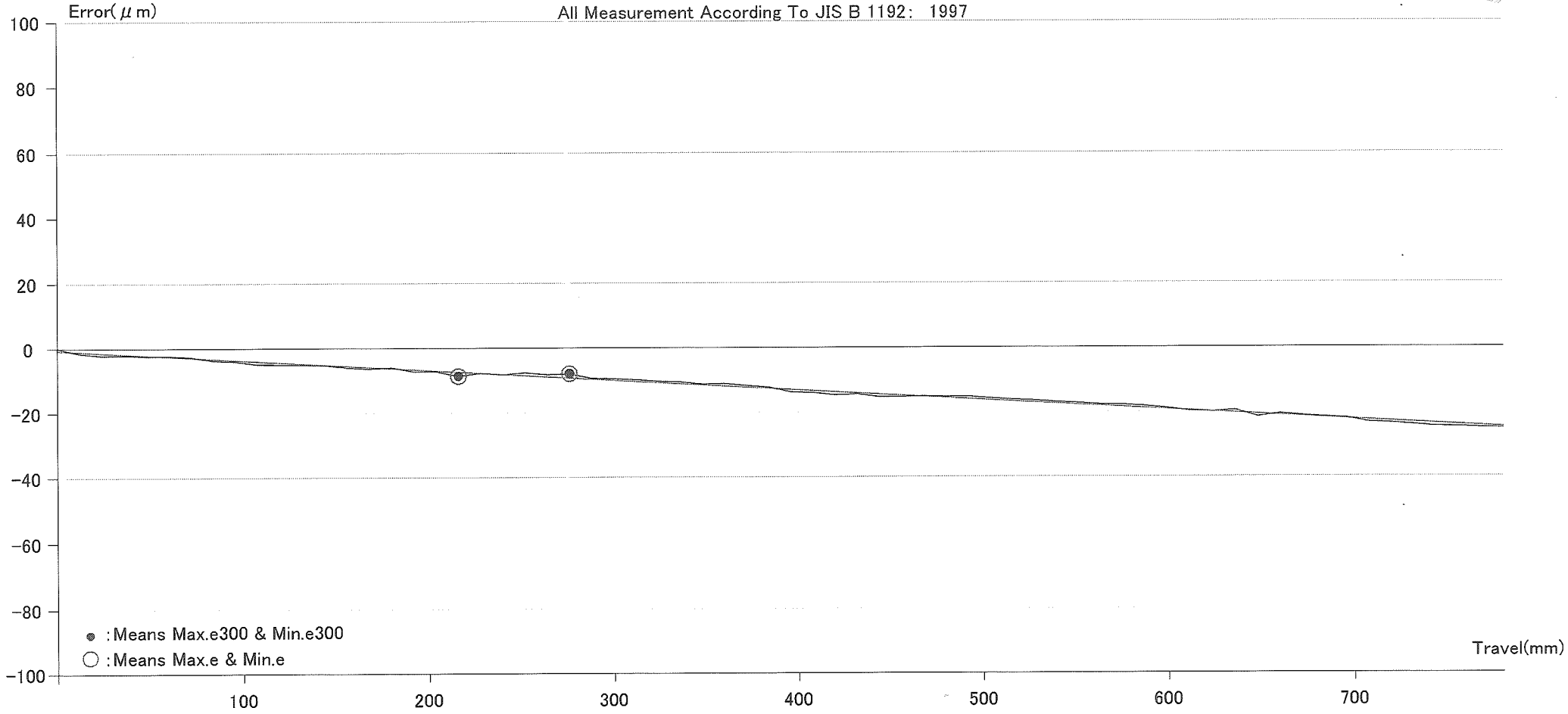
TEST ID:- 0809

Date: 1/7/24	Checked By	Approved By

PMI BALLSCREWS

BALLSCREW INSPECTION CERTIFICATE

Inspected By Agilent Laser Measuring System
All Measurement According To JIS B 1192: 1997



Specifications:	
·Customer	:AMS
·Order No.	:S3-130097891004
·Drawing No.	:BA300510002-0
·Part No.	: *:S12-016-95
·Specification	:R40-12C1-FDWE-846-1110-0.008
·Nominal Diameter (mm)	:40
·Nominal Pitch (mm)	:12
·Travel Length (mm)	:780
·E(μm)	:±18
·T(μm)	:-25

Measuring Result:	Permissible Error	Actual Measuring Error
·T+E(μm)	-43~-7	-24.22
·e(μm)	13	2.55
·e300(μm)	8	2.55
·Tq(kgf - cm)	4.16~7.28	5.333~6.74P
·Accuracy Grade	C3	C0

Manufactured By
PRECISION MOTION IND., INC.
www.pmi-amt.com

Measuring Environment:
·Refernce Temperatrue : 20℃ ± 1.0℃
·Relative Humidity : 50% ± 10%
Measuring Device:
·Air Bearing guided Measuring Machine
·Agilent Laser Interferometer Type 5527A

Inspector: JAY R

QC
2014/01/21
PASS

M11-995 X, オネガ

F-P010-01-04-E

1007

ACE MANUFACTURING SYSTEMS LTD

BALL SCREW ASSEMBLY REPORT

AMS — In Pursuit of Excellence

Machine Number : M11 - 995 | Ball screw Sl No: 831300978910004 | Axis : X

I. Ball screw Make & Specification : PMI (R40-12C1-FDWC-8H1-1110-0.008)

II. Ball Scerw Bearing Make & Specification : NACHI (7S1AB0003/GNM)

Sl No	Description	Recommended	Motor End	Brake End
1	Ball screw rotational Torque (Forward / Reverse)	As per DS - PDN - MEC - SA - 02	1.30 N/m	1.40 N/m
2	Bearing housing Bore diameter	As per drawing	6 to 6.1	9 to 9.1
3	Bearing housing centre height difference	0.01 mm	0.01mm	-
4	Bearing housing bore parallelism with respect to pads	0.01 mm	0.005mm	0.005mm

Motor End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	-	T1	-
2	Bearing width B1	-	T2	-
3	Bearing width B2	NACHI	T3	14.940
4	Bearing width B3	"	T4	14.940
5	Spacer cylindrical width B4	-	T5	-
6	Flange cover (2) spigot depth	C24-275-8123	T6	-
7	Housing Bearing depth	C37-103-2123	T7	43.500

IV. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm (103) .618

T6 = 13.680 mm

Brake End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	-	T1	-
2	Bearing width B1	-	T2	-
3	Bearing width B2	NACHI	T3	14.945
4	Bearing width B3	"	T4	14.945
5	Spacer cylindrical width B4	-	T5	-
6	Flange cover (2) spigot depth	C24-360-8123	T6	-
7	Housing Bearing depth	C37-153-2123	T7	43.510

V. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 153 618

T6 = 13.680 mm

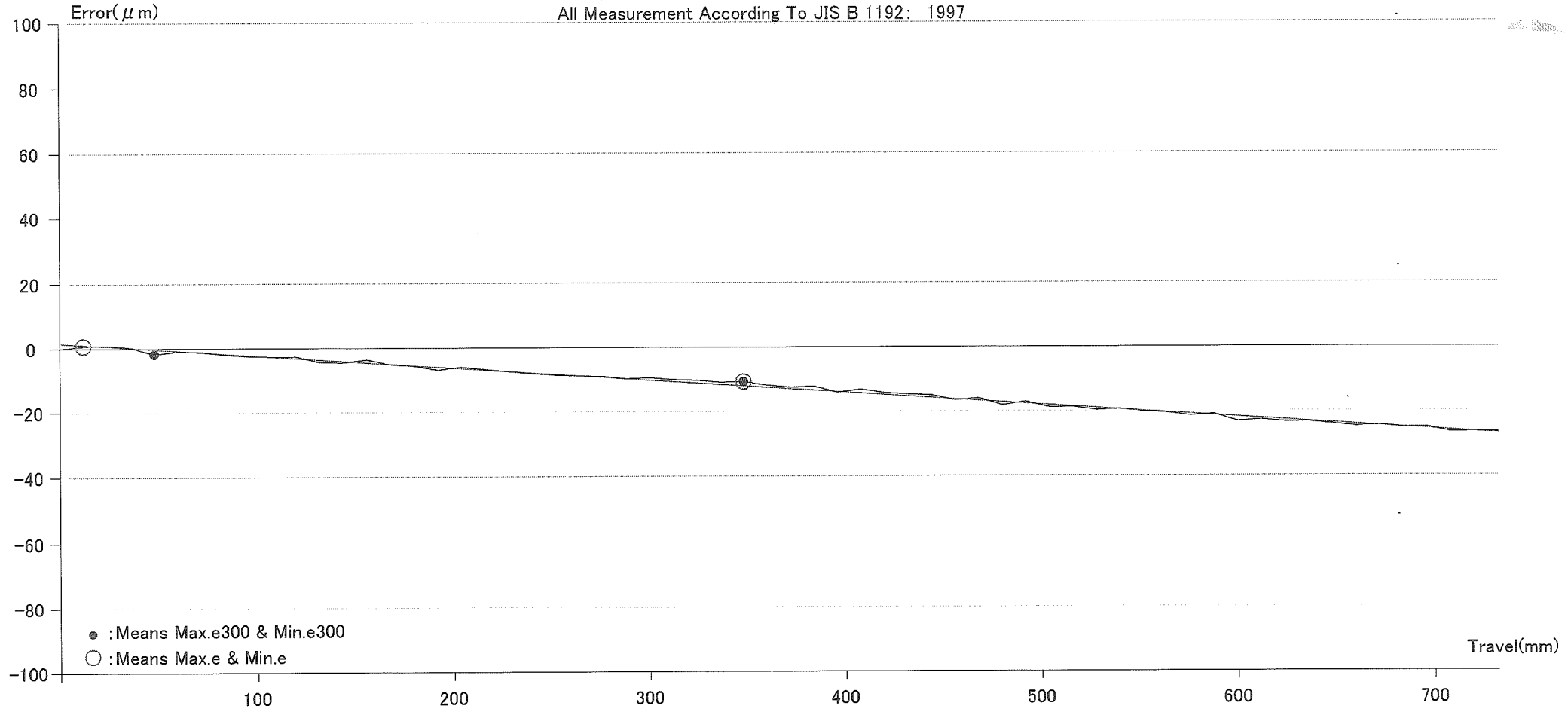
VI. Check points after assembly

	CHECK POINTS	Remark
1	Ensure the clearance between flange cover and bearing housing is 0.040 mm before tightening.	OK
2	Ensure Bearing housing flange concentricity is within 0.030 mm.	"
3	Ensure smooth rotation of ball screw assembly.	"
4	Ensure ball screw run out within 0.010 mm	"
5	Ensure tightness of all screws & marking with yellow marker	"
6	Ensure ball Screw report attached to Inspection Report.	"

Date:	Checked by	Approved by

PMI BALLSCREW INSPECTION CERTIFICATE

Inspected By Agilent Laser Measuring System
All Measurement According To JIS B 1192: 1997



Specifications:	
·Customer	:AMS
·Order No.	:S3-140011401030
·Drawing No.	:BA300510003-0
·Part No.	: *:S12-016-97
·Specification	:R40-12C1-FDWE-796-1060-0.008
·Nominal Diameter (mm)	:40
·Nominal Pitch (mm)	:12
·Travel Length (mm)	:732
·E(μm)	:±18
·T(μm)	:-24

Measuring Result:	Permissible Error	Actual Measuring Error
·T+E(μm)	-42~-6	-28.65
·e(μm)	13	2.87
·e300(μm)	8	2.77
·Tq(kgf - cm)	4.16~7.28	5.194-6.658
·Accuracy Grade	C3	C0

Manufactured By
PRECISION MOTION IND., INC.
www.pmi-amt.com

Measuring Environment:
·Refernce Temperatrue : 20℃ ± 1.0℃
·Relative Humidity : 50% ± 10%
Measuring Device:
·Air Bearing guided Measuring Machine
·Agilent Laser Interferometer Type 5527A

Inspector: ROCK

QC
2014/03/10
PASS

Y. Aki

2011-995

F-P010-01-04-E

1030

ACE MANUFACTURING SYSTEMS LTD

AMS
In Pursuit of Excellence

BALL SCREW ASSEMBLY REPORT

Machine Number : M11-995	Ball screw Sl No: 831400114010 30	Axis : Y

I. Ball screw Make & Specification : PMI [RM0 - 1204 - FDWC - 796 - 1060 - 0.008]

II. Ball Scerw Bearing Make & Specification : NACHI [35TAB06DB/GMP4]

Sl No	Description	Recommended	Motor End	Brake End
1	Ball screw rotational Torque (Forward / Reverse)	As per DS - PDN - MEC - SA - 02	1.3 N/m	1.4 N/m
2	Bearing housing Bore diameter	As per drawing	8 to 6 µ	5 to 4 µ
3	Bearing housing centre height difference	0.01 mm	0.01 mm	—
4	Bearing housing bore parallelism with respect to pads	0.01 mm	0.005 mm	0.005 mm

Motor End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	—	T1	—
2	Bearing width B1	—	T2	—
3	Bearing width B2	NACHI	T3	14.950
4	Bearing width B3	— " —	T4	14.945
5	Spacer cylindrical width B4	—	T5	—
6	Flange cover (2) spigot depth	C24-275-8173	T6	—
7	Housing Bearing depth	C37-103-8173	T7	43.515

IV. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 103 / 7

T6 = 13.680 mm

Brake End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	—	T1	—
2	Bearing width B1	—	T2	—
3	Bearing width B2	NACHI	T3	14.935
4	Bearing width B3	— " —	T4	14.935
5	Spacer cylindrical width B4	—	T5	—
6	Flange cover (2) spigot depth	C24-360-8173	T6	—
7	Housing Bearing depth	C37-153-8173	T7	43.490

V. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 153 / 7

T6 = 13.680 mm

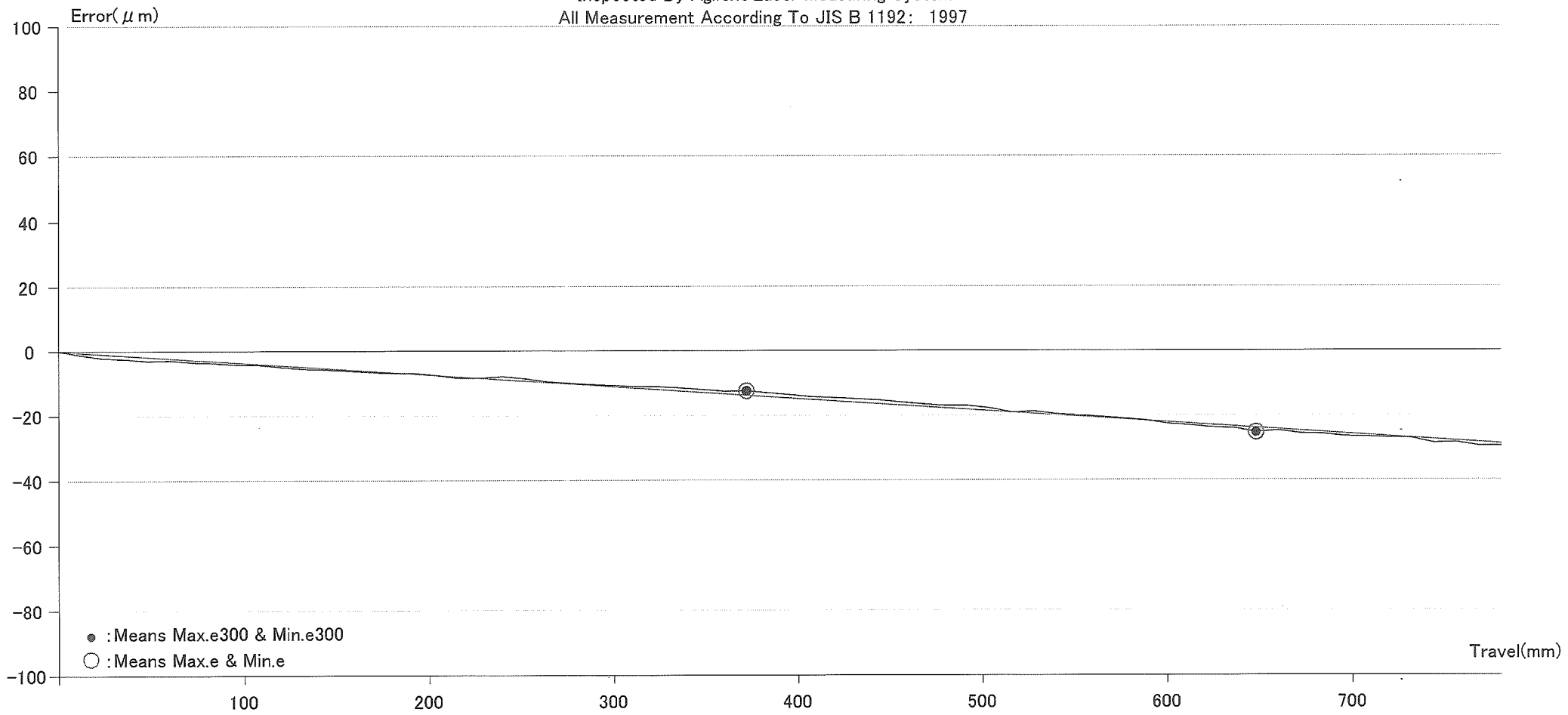
VI. Check points after assembly

	CHECK POINTS	Remark
1	Ensure the clearance between flange cover and bearing housing is 0.040 mm before tightening.	OK
2	Ensure Bearing housing flange concentricity is within 0.030 mm.	— " —
3	Ensure smooth rotation of ball screw assembly.	— " —
4	Ensure ball screw run out within 0.010 mm	— " —
5	Ensure tightness of all screws & marking with yellow marker	— " —
6	Ensure ball Screw report attached to Inspection Report	—

Date:	Checked by	Approved by

PMI BALLSCREWS — BALLSCREW INSPECTION CERTIFICATE

Inspected By Agilent Laser Measuring System
All Measurement According To JIS B 1192: 1997



Specifications:	
·Customer	:AMS
·Order No.	:S3-140011402011
·Drawing No.	:BA300510004-0
·Part No.	: *:S12-016-96
·Specification	:R40-12E1-FDWE-841-1105-0.008
·Nominal Diameter (mm)	:40
·Nominal Pitch (mm)	:12
·Travel Length (mm)	:780
·E(μm)	:±18
·T(μm)	:-25

Measuring Result:	Permissible Error	Actual Measuring Error
·T+E(μm)	-43～-7	-28.82
·e(μm)	13	2.69
·e300(μm)	8	2.69
·Tq(kgf－cm)	5.72～9.88	7.102 ~ 8.601
·Accuracy Grade	C3	C0

Manufactured By
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Measuring Environment:
·Refernce Temperatrue : 20°C ± 1.0°C
·Relative Humidity : 50% ± 10%
Measuring Device:
·Air Bearing guided Measuring Machine
·Agilent Laser Interferometer Type 5527A

Inspector: JAY R

QC
2014/03/07
PASS

2-RAIL
M11-995

F-P010-01-04-E

2011

ACE MANUFACTURING SYSTEMS LTD

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BALL SCREW ASSEMBLY REPORT

Machine Number : M11-995	Ball screw Sl No: S3140011402011	Axis : 2

I. Ball screw Make & Specification : PMI [RHO-12G1-FOWC-8H1-1105-0.008]

II. Ball Screw Bearing Make & Specification : NACHI (25TAB06DB/Pm)

Sl No	Description	Recommended	Motor End	Brake End
1	Ball screw rotational Torque (Forward / Reverse)	As per DS - PDN :- MEC - SA - 02	1.2 N/m	1.4 N/m
2	Bearing housing Bore diameter	As per drawing	9 to 8 µ	5 to 4 µ
3	Bearing housing centre height difference	0.01 mm	~	~
4	Bearing housing bore parallelism with respect to pads	0.01 mm	~	0.005mm

Motor End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	~	T1	~
2	Bearing width B1	~	T2	~
3	Bearing width B2	NACHI	T3	14.930
4	Bearing width B3	〃	T4	14.930
5	Spacer cylindrical width B4	~	T5	~
6	Flange cover (2) spigot depth	C24-1119-8123	T6	~
7	Housing Bearing depth	C37-122-7123	T7	43.480

IV. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 122

T6 = . mm

Brake End

Sl No	Description	Part Number	Thickness (mm)	Depth / width
1	Flange cover (1) spigot depth	~	T1	~
2	Bearing width B1	~	T2	~
3	Bearing width B2	NACHI	T3	14.945
4	Bearing width B3	〃	T4	14.940
5	Spacer cylindrical width B4	~	T5	~
6	Flange cover (2) spigot depth	C24-280-8123	T6	~
7	Housing Bearing depth		T7	43.505

V. Flange cover thickness T6 = T7 - (T1+T2+T3+T4+T5)+0.05 mm 154/20

T6 = 13.680 mm

VI. Check points after assembly

	CHECK POINTS	Remark
1	Ensure the clearance between flange cover and bearing housing is 0.040 mm before tightening.	OK
2	Ensure Bearing housing flange concentricity is within 0.030 mm.	〃
3	Ensure smooth rotation of ball screw assembly.	〃
4	Ensure ball screw run out within 0.010 mm	〃
5	Ensure tightness of all screws & marking with yellow marker	〃
6	Ensure ball Screw report attached to Inspection Report.	〃

Date:	Checked by	Approved by

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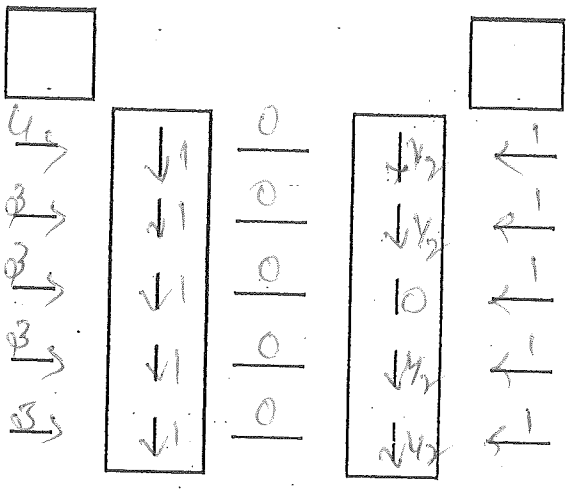
Sign: Date: 01/6/13

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INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

MACHINE NUMBER: ~~DVM 60~~ MH-995 DATE: 9-6-14

I. MACHINE BASE LEVELING: Least count of the Instrument: 1 div = 0.020mm / meter



SL NO	DESCRIPTION	RECOMMENDED (mm)	ACTUAL(mm)		
1	Leveling - Length wise (Longitudinal)	1 div / meter	1 div		
2	Leveling - Height variation between loc/non-loc guide ways	1 div / meter	0 div		
3	Leveling - Individual (Transverse)	3 div / meter	3 div		
4	Straightness of locating guide way	± 0.005	0.002		
5	Parallelism of Non locating guide way w.r.t.locating guide way	±0.005	0.003		
6	Ball screw Run out at Motor end	±0.010	0.010		
7	Ball screw Run out at Brake end	±0.010	0.010		
8	Ball screw alignment w.r.t. Top guide way	±0.020	0.010		
9	Ball screw alignment w.r.t. locating Side guide way	±0.020	0.010		
10	Ball screw Stretching	0.025 (±0.002 /meter)	0.025 N/m		
			M.E	CEN	B.E
11	Ball screw Torque	0.6 to 1.8 N-m	1.2 N/m	1.2 N/m	1.2 N/m
12	Slide & Jib Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
13	Slide, Jib & Single Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
14	Slide, Jib & Both Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
15	Final slide Torque – box guide way	2.5 to 3.5 N-m			
16	Final slide Torque – L M rail	1.6 to 2.5 N-m	1.5 N/m	1.5 N/m	1.5 N/m
17	Variation in torque for final slide	10%			
18	Remarks if any:				

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MANAGEMENT

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INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

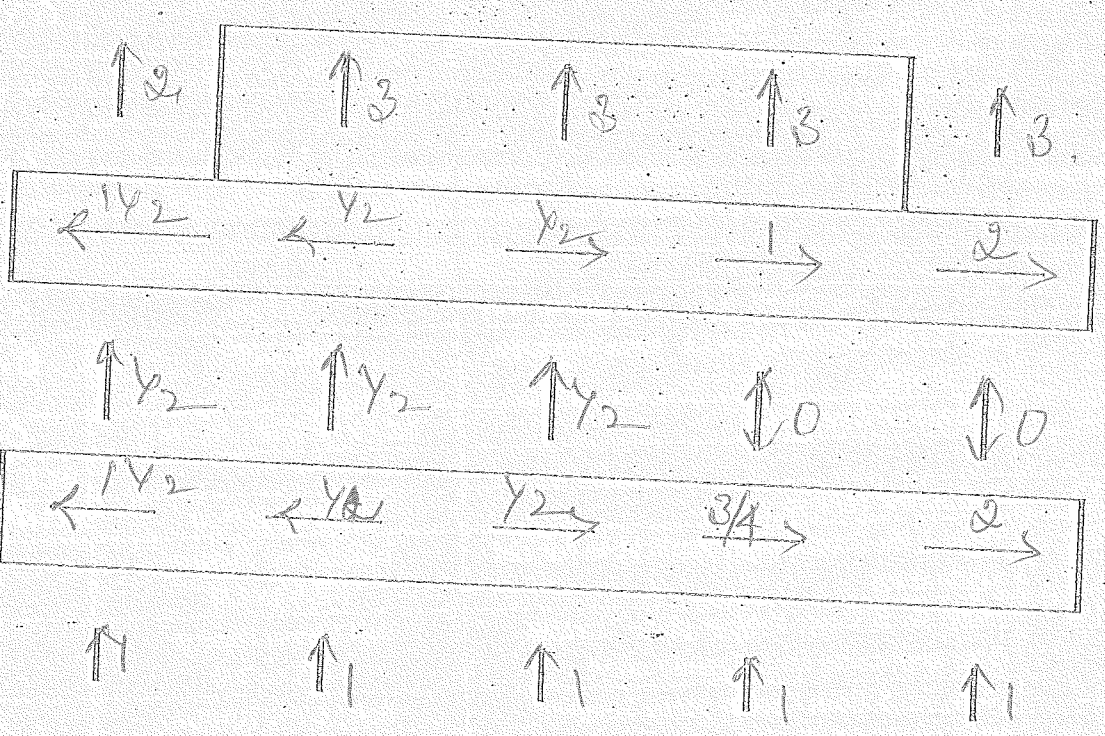
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MACHINE NUMBER: M11-995 DATE: 9.6.14.

SL NO	Check Points	Remark
1	Ensure all leveling screws & locknuts are fixed to the base.	✓
2	Ensure guide way butting face blue matching is minimum 70%	✓
3	Ensure locating & non locating guide way are fitted at respective position (L M m/c only)	✓
4	Ensure position of LM block ground surface butted to slide butting face.	✓
5	Ensure required height blocks are fitted to the base as per Work Order / Planning chart.	✓
6	Ensure plugs are fitted for L M guide way	✓

DATE: 9.6.14.

II. SADDLE LEVEL READING: Least count of the Instrument: 1 div = 0.020mm / meter



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INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

MACHINE NUMBER: MB-995 DATE: 9-6-14

SL NO	DESCRIPTION	RECOMMENDED (mm)	ACTUAL(mm)		
1	Leveling - Length wise (Longitudinal)	2 div / meter	2 div		
2	Leveling - Height variation between loc/non-loc guide ways	1 div / meter	½ div		
3	Leveling - Individual (Transverse)	3 div / meter	3 div		
4	Straightness of locating guide way	± 0.005	0.004		
5	Parallelism of Non locating guide way w.r.t.locating guide way	±0.005	0.003		
6	Ball screw Run out at Motor end	±0.010	0.010		
7	Ball screw Run out at Brake end	±0.010	0.010		
8	Ball screw alignment w.r.t. Top guide way	±0.020	0.010		
9	Ball screw alignment w.r.t. locating Side guide way	±0.020	0.010		
10	Ball screw Stretching	0.025 (±0.002 /meter)	0.025 mm/m		
			M.E	CEN	B.E
11	Ball screw Torque	0.6 to 1.8 N-m	1.2 N-m	1.2 N-m	1.2 N-m
12	Slide & Jib Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
13	Slide, Jib & Single Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
14	Slide, Jib & Both Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
15	Final slide Torque – box guide way	2.5 to 3.5 N-m			
16	Final slide Torque – L M rail	1.6 to 2.5 N-m	1.8 N-m	1.8 N-m	1.8 N-m
17	Variation in torque for final slide	10%			

Remarks if any :

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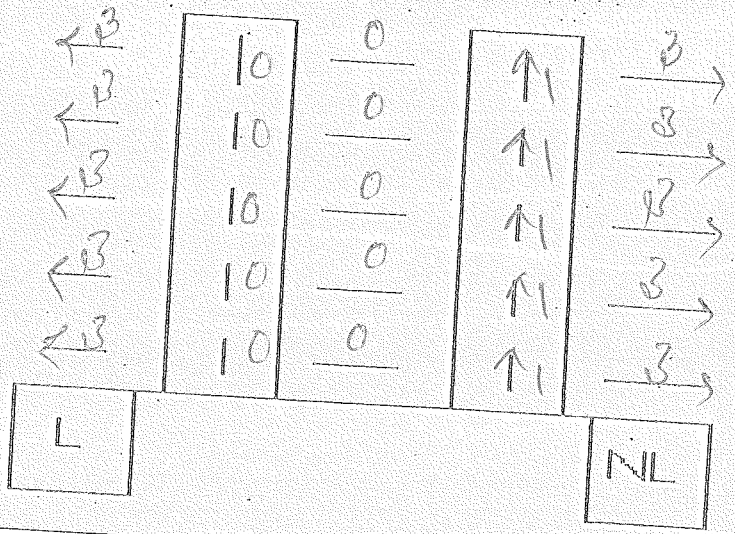
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MACHINE NUMBER: M11-995 DATE: 9-6-14

SL NO	Check Points	Remark
1	Ensure oil connecting holes are made clear by passing air & kerosene	✓
2	Ensure guide way butting face & saddle bottom blue matching is minimum 80%	✓
3	Ensure tightness of Eccentric pin locknuts. ((H G m/c only)	NA
4	Ensure proper taper matching of jib. ((H G m/c only)	NA
5	Ensure locating & non locating guide way are fitted at respective position (L M m/c only)	✓
6	Ensure position of LM block ground surface butted to slide butting face. (L M m/c only)	✓
7	Ensure Equal gap between ball nut & nut housing bore.	✓
8	Ensure Nut housing alignment is with in 0.010 mm/meter	✓
9	Ensure White Prussian matching between nut housing & ball nut butting face is80%	✓
10	Ensure bearing housing Dowelling and grub locking.	✓
11	Ensure Ball screw lock nut grub locking with lock tite.	✓
12	Ensure guide way wipers are fixed. (H G m/c only)	NA
13	Ensure Nylon stoppers are fixed as per specification.	✓
14	ensure plugs are fitted for lm guide way & grub screws are fixed to saddle mounting holes	✓

DATE: 9-6-14

III. COLUMN READING: Least count of the Instrument: 1 div = 0.020mm / meter



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INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

MACHINE NUMBER: MM-995 DATE: 9-6-14

SL NO	DESCRIPTION	RECOMMENDED (mm)	ACTUAL(mm)		
1	Leveling - Length wise (Longitudinal)	1 div / meter	1 div		
2	Leveling - Height variation between loc/non-loc guide ways	1 div / meter	0 div		
3	Leveling - Individual (Transverse)	3 div / meter	3 div		
4	Straightness of locating guide way	± 0.005	0.002		
5	Parallelism of Non locating guide way w.r.t.locating guide way	±0.005	0.002		
6	Ball screw Run out at Motor end	±0.010	0.010		
7	Ball screw Run out at Brake end	±0.010	0.010		
8	Ball screw alignment w.r.t. Top guide way	±0.020	0.010		
9	Ball screw alignment w.r.t. locating Side guide way	±0.020	0.010		
10	Ball screw Stretching	0.025 (±0.002 /meter)	0.025 N/m		
11	Mandrel radial run out	±0.010	0.010		
12	Mandrel reading at Top (parallelism wrt.head stock movement)	±0.010	0.010		
13	Mandrel reading at side (parallelism wrt.head stock movement)	±0.010	0.010		
			M.E	CEN	B.E
14	Ball screw Torque	0.6 to 1.8 N-m	1.2 N/m	1.2 N/m	1.2 N/m
15	Slide & Jib Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
16	Slide, Jib & Single Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
17	Slide, Jib & Both Keeper plate Torque (Not applicable for LM guide way)	Ball Screw Torque + 0.5 to 1.0			
18	Final slide Torque – box guide way	2.5 to 3.5 N-m			
19	Final slide Torque – LM rail	1.6 to 2.5 N-m	2 N/m	2 N/m	2 N/m
20	Variation in torque for final slide	10%			

Remarks if any :

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INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

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MACHINE NUMBER: M11-995 DATE: 9-6-14

SL NO	Check Points	Remark
1	Ensure oil connecting holes are made clear by passing air & kerosene.	✓
2	Ensure guide way butting face, head stock bottom face,& column bottom face blue matching is minimum 80%	✓
3	Ensure tightness of Eccentric pin locknuts. ((H G m/c only)	✓
4	Ensure proper matching of jib taper. ((H G m/c only)	NA
5	Ensure locating & non locating guide way are fitted at respective position (L M m/c only)	N.A
6	Ensure position of LM block ground surface butted to slide butting face. (L M m/c only)	✓
7	Ensure Equal gap between ball nut & nut housing bore.	✓
8	Ensure nut housing alignment is with in 0.010 mm	✓
9	Ensure White Prussian matching between nut housing & ball nut butting face is 80 %	✓
10	Ensure spindle cartridge butting face blue matching is 80%	✓
11	Ensure head stock is fixed as per requirement (ID, DD, and DD CTS).	✓
12	Ensure bearing housing Dowelling and grub locking.	✓
13	Ensure Ball screw lock nut grub locking with locktite.	✓
14	Ensure guide way wipers are fixed. (H G m/c only)	✓
15	Ensure Nylon stoppers are fixed as per specification.	NA
16	Ensure machine lifting tackles are fitted.	✓
17	Ensure plugs are fitted for lm guide way & grub screws are fixed to head stock mounting holes	✓

MACHINE NUMBER: DATE:

IV. SIDE ARM ALINGMENT READING: (Turret Type ATC only)

SL NO	DESCRIPTION	Recommended (mm)	ACTUAL (mm)
1	Shaft reading at Top (longitudinal)	±0.050	
2	Shaft reading at side (transverse)	±0.030	
3	Square ness of locating side	±0.010	
4	Distance between Z axis guide way top surface & side arm bore centre	REFER AS PER DRAWING NO	
5	Distance between sidearm front butting face on column & side arm bore centre	REFER AS PER DRAWING NO	

Remarks if any :

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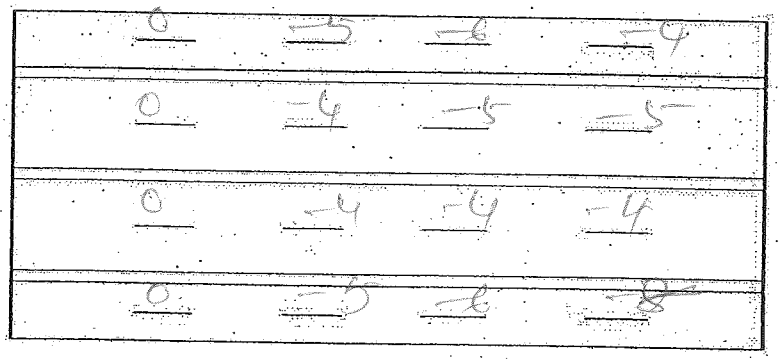
MACHINE NUMBER: M11-995 DATE: 13-6-14

RECOMMENDED TOLERANCE AS PER DS – DSN – TC – PROD – 01 – R0

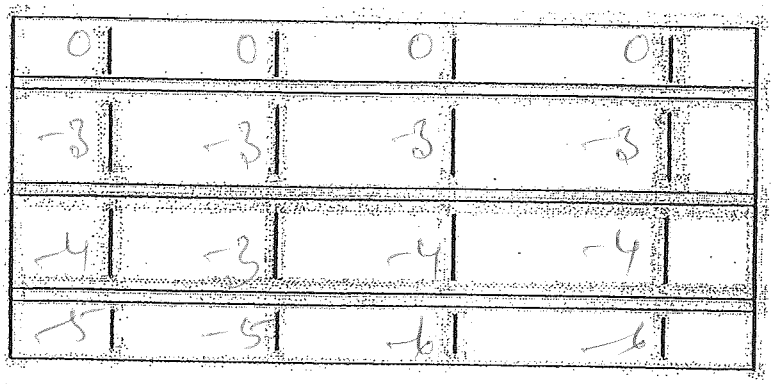
V. LEVELLING OF MACHINE: L.C of Instrument: 1 div = 0.020mm / m

Along X-axis			Along Y-axis		
Motor End	Middle	Brake End	Motor End	Middle	Brake End
¼ →	0	¾ →	↕ 0	↑ ½	↑ ¾
¼ →	0	¼ →	↑ ¼	↕ 0	↑ ¾
¼ →	0	¼ →	↕ 0	↕ 0	↕ 0

VI. TABLE TOP READING ALONG X- AXIS:



VII. TABLE TOP READING ALONG Y- AXIS:



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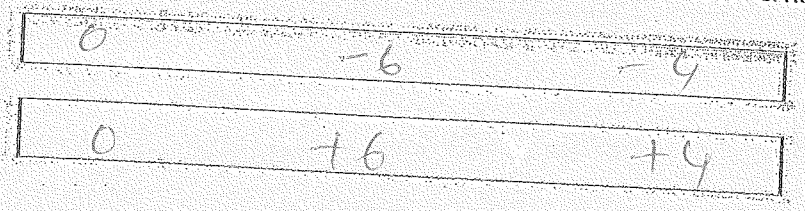
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INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

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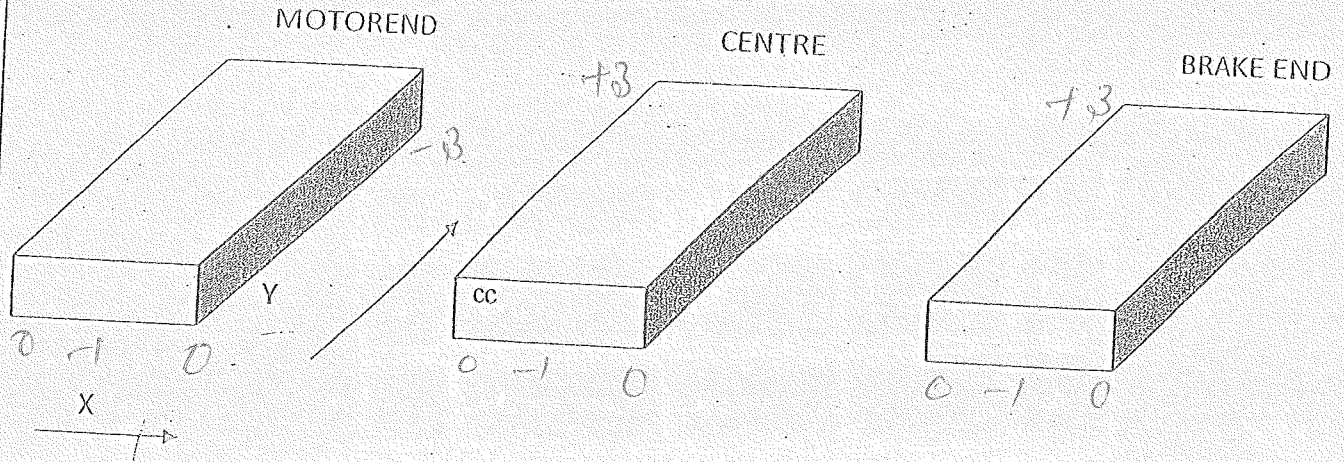
MACHINE NUMBER: M11-995 DATE:

VIII. PARALLELISM OF REFERENCE T-SLOT TO X-AXIS MOVEMENT AT FRONT & REAR SURFACE:



SL NO	Check Points	Remarl
1	Ensure oil connecting holes are made clear by passing air	✓
2	Ensure table bottom face blue matching IS minimum 70%	✓
3	Ensure tightness of Eccentric pin locknuts. ((H G m/c only)	N/A
4	Ensure proper matching of jib taper. ((H G m/c only)	N/A
5	Ensure position of LM block ground surface butted to slide butting face.	✓
6	Ensure Equal gap between ball nut & nut housing bore.	✓
7	Ensure nut housing parallelism is with in 0.010 mm	✓
8	Ensure Blue matching between nut housing & ball nut butting face is 70%	✓
9	Ensure Table Tenon slot size as per specification	✓

IX. SQUARENESS OF Y –AXIS MOVEMENT WITH RESPECT TO X – AXIS MOVEMENT:



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INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

MACHINE NUMBER: M11-995

DATE: 13-6-14

X. COLUMN READING (INITIAL):

XZ PLANE(SIDE)

500	+35
450	
400	
350	
300	
250	+15
200	
150	
100	
50	
0	0

YZ PLANE (FRONT)

500	+55
450	
400	
350	
300	
250	+25
200	
150	
100	
50	
0	0

COLUMN READING – AFTER GRINDING:

XZ PLANE(SIDE)

500	+15
450	
400	
350	
300	
250	+10
200	
150	
100	
50	
0	0

YZ PLANE (FRONT)

500	+25
450	
400	
350	
300	
250	+15
200	
150	
100	
50	
0	0

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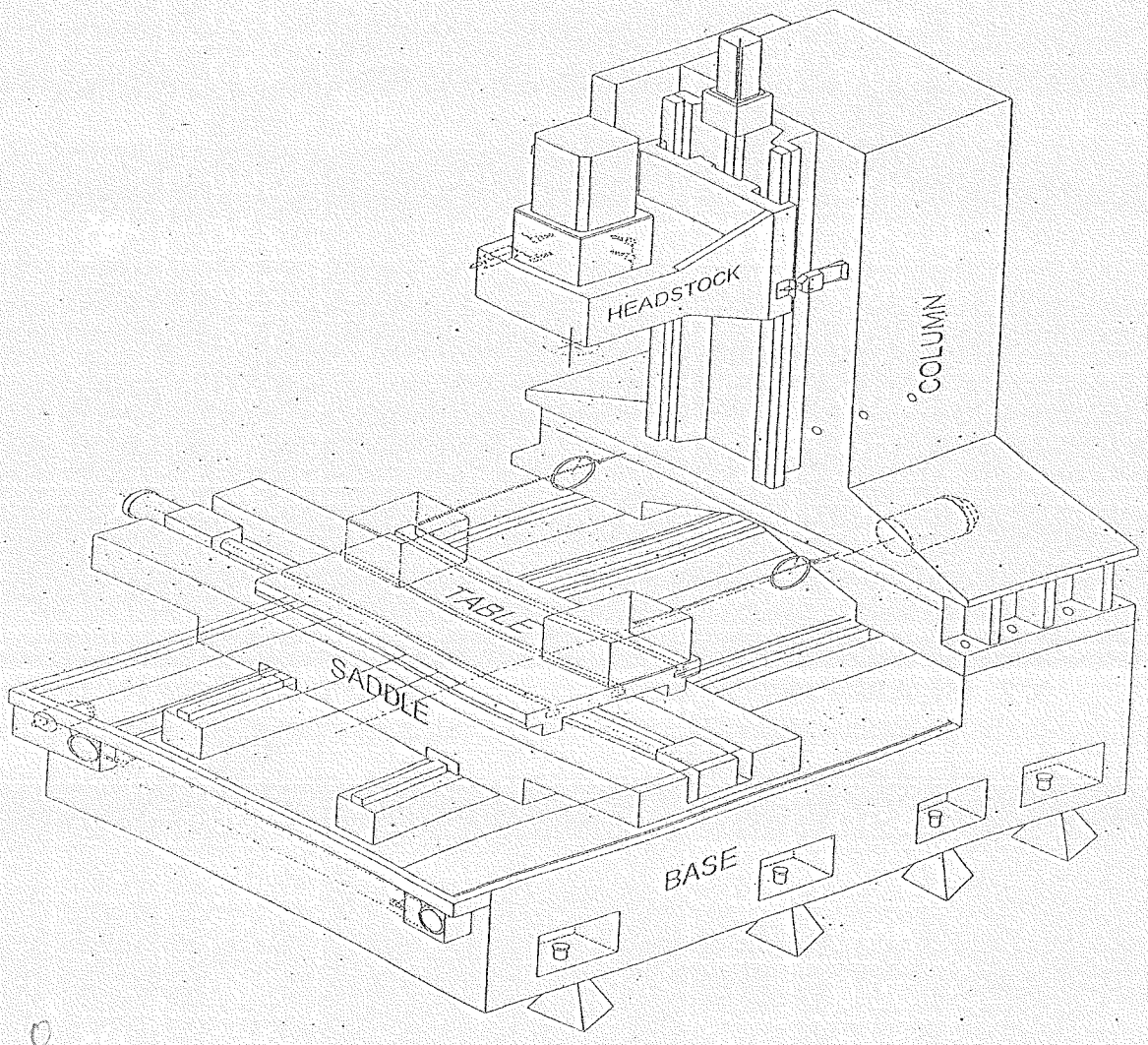
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INSPECTION REPORT-BASE BUILDING. 2 GUIDEWAYS

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MACHINE NUMBER: MM-995 DATE: 14/6/14

XI. COLUMN ALINGMENT READING WITH RESPECT TO X-AXIS MOVEMENT:
PERMISSIBLE VALUE = 0.30MM/METRE



0 0.080 mm

0	100	200	300	400	500	600	700	800	900	1000

Remarks if any :

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	PREPARED	CHECKED	APPROVED	10...10 PAGE NUMBER	F-QA-IPR-BBA-02-R0 FORMAT NUMBER

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LUBRICATION ASSEMBLY - INSPECTION REPORT- BBA

MACHINE NUMBER : M11- 995

I. Lubrication metering cartridge specification (cc of oil / shot) as per DS-PDN-MEC-LUB-01 :

Sl. No.	Axis	Ball screw	Box guideway top / LM block	Box guideway side	Jib	Keeper plate
1	X -Axis	0.25cc	0.10cc	–	–	–
2	Y -Axis	0.25cc	0.10cc	–	–	–
3	Z-Axis	0.25cc	0.10cc	–	–	–

Sl. No.	Check points:	Remark
1	Verify that metering cartridge are fixed as per DS-PDN-MEC-LUB-01.	OK
2	Ensure X,Y & Z guideways and ball screw are lubricating properly	OK
3	Ensure lubrication pressure set to 15 Bar.	OK
4	Ensure there is no leakage of oil from joints, cartridges & manifolds etc.	OK
5	Ensure that the lubrication hoses do not rub or foul with the stationary / moving parts during axis movement.	OK
6	Ensure that the lubrication hoses of appropriate length (as per DS-PDN-MEC-LUB-01) are used and properly clamped and routed.	OK

II Telescopic cover Assembly:

	Sl No	Axis	Extension block Parallelism w.r.t. axis guideway		Extn. Block Length Variation -	DRAWING NUMBER
			Top - Recommended 0.020 mm	Side- Recommended 0.030 mm	Recommended 0.050 mm	(S10 -)
	1	X - Motor End	0.020mm	0.020mm	0.040mm	S10-013-122-08
	2	X - Brake End	0.010mm	0.030mm	0.050mm	S10-013-122-08
	3	Y - Motor End	0.020mm	0.030mm	–	S10-013-524-00
	4	Y - Brake End	0.010mm	0.020mm	0.040mm	S10-013-525-01
	5	Z - Brake End	0.020mm	0.020mm	0.040mm	S10-013-277-02

III. Saddle cover alignment w.r.t. Y axis movement (If applicable) :

	Sl NO	Top - Recommended 0.050 mm	Side - Recommended 0.050 mm	
	1	–	–	–

IV. Check Points :

Sl NO	Check points:	Remark
1	Check telescopic cover sliding smoothly	OK
2	Ensure telescopic cover screws are tightened in box close condition	OK

Date: 20-6-14	PREPARED	CHECKED	APPROVED